Inequality, Markups, and the Phillips Curve

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August 2026

Draft version. Please do not cite.

Abstract

We develop a Two-Agent New Keynesian (TANK) model in which poor households have a higher price elasticity of demand than rich households, consistent with micro-level evidence from Swiss household data. Firms' desired markups then depend on the composition of demand across household types, so the consumption distribution enters the Phillips curve. When inequality rises, demand tilts toward the less elastic rich, desired markups rise, and cost-push shocks pass through more strongly to inflation. In US time series, markups rise after identified supply shocks and consumption inequality rises after an expansionary technology shock, as the model predicts. In a panel of OECD and other European economies, the markup response to an identified oil supply shock is larger in high-inequality countries—the sign the mechanism predicts—though this cross-country evidence is inconclusive.

JEL Classification: E31, E12, D43, D31, E32
Keywords: Inequality, Markups, Phillips curve, Heterogeneous price elasticities, Demand composition, Two-agent New Keynesian (TANK) model

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1 Introduction

We identify and quantify a channel through which income inequality affects the slope of the Phillips curve: when households differ in their price elasticities of demand, firms' desired markups depend on the composition of household demand. The question is motivated by two concurrent trends: income inequality has risen while the Phillips curve has flattened. In the United States, the pre-tax income share of the top 10 percent rose from about 34 percent in the early 1980s to about 45 percent by 2016, per the WID.world project (Alvaredo et al., 2017). Over the same period, a large literature documents a flattening of the Phillips curve: aggregate estimates of its reduced-form slope fell from about 0.7 in the 1970s to near zero before the pandemic (Blanchard, 2016; Hooper et al., 2020; Kuttner and Robinson, 2010), while estimates from regional cross-sections suggest a smaller decline from a slope that was already low in the early 1980s (Hazell et al., 2022). Our contribution is not a new estimate of the Phillips curve slope, but a mechanism through which income inequality affects it.

This paper is motivated by Auer et al. (2024), who use high-frequency Swiss household scanner data and the 2015 Swiss franc appreciation as a quasi-natural experiment whose baseline quantification implies low-income households are more than twice as price-sensitive as high-income households. We embed this fact in the Two-Agent New Keynesian (TANK) framework of Debortoli and Galí (2018, 2024) by letting the price elasticity of demand differ across household types. Poor households, who face binding liquidity constraints, have a high elasticity; rich households, with full financial market access, have a low elasticity. Firms set prices under monopolistic competition with Rotemberg adjustment costs, and their desired markup depends on the composition of demand across types, summarized by a heterogeneity index. When inequality rises, demand shifts toward the less elastic rich, and the desired markup rises.

The consumption distribution therefore enters the Phillips curve: the markup channel moves with the cycle, damping the price response to demand-driven output fluctuations and amplifying the pass-through of cost-push shocks. We call this economy the two-elasticity TANK, or TE-TANK. The channel is local: it operates at a given steady-state level of inequality and governs how shocks that move the consumption distribution affect inflation. We do not map the historical rise of inequality into model objects.

Section 2 develops the two-agent model and shows how inequality enters the Phillips curve through firms' desired markups. The model reduces to the familiar three-equation New Keynesian system. A positive demand shock compresses inequality, shifts demand toward the price-sensitive poor, and lowers desired markups, flattening the Phillips curve. By contrast, shocks that raise inequality run the mechanism in reverse. After a cost-push shock or an expansionary technology shock, demand tilts

toward the less elastic rich and desired markups rise, reinforcing the initial inflationary pressure or damping the fall in inflation. Because the pass-through of a cost-push wedge is decreasing in the slope of the Phillips curve, the flatter the curve, the more of the wedge reaches inflation. Redistribution therefore acquires a supply-side role, since compressing inequality lowers desired markups.

Closest in mechanism, Mongey and Waugh (2025) study how demand composition shifts markups in a model with oligopolistic competition and heterogeneous firms, using the elasticity estimates of Auer et al. (2024). Their aggregate policy experiment is a one-time fiscal transfer, and they abstract from nominal rigidities and monetary policy. We bring the same composition channel into a New Keynesian model and draw out its implications for the Phillips curve, the inflation–output trade-off, and the transmission of demand versus supply shocks.[4]

More broadly, desired markups that move with the income and wealth distribution are documented and modelled elsewhere: retail markups rise with household income and with local income dispersion (Sangani, 2024), local retail prices move with local house prices through markups rather than costs (Stroebel and Vavra, 2019), and non-homothetic demand or income dispersion can generate related price effects (Marto, 2023; Hu et al., 2025; Cavallari, 2020). Buyer composition moves markups in general equilibrium when shoppers differ in search intensity rather than in preferences, since a pool tilted toward the employed, who search less, weakens competition and raises prices (Kaplan and Menzio, 2016); that elasticity is a property of a frictional product market under flexible prices and never reaches a nominal rigidity. Variable markups also flatten the Phillips curve through firm entry (Fujiwara and Matsuyama, 2026) and market concentration (Wang and Werning, 2022); Jaravel (2019) supplies the micro evidence for the entry margin, where demand growth in the segments bought by high-income households draws in entrants who erode incumbents' market shares and markups, though there income groups differ in the varieties they buy rather than in their elasticity of substitution. A related composition channel runs through nominal rigidity rather than markups: the goods bought by high-income households adjust prices about a quarter less often, so shifting income toward them makes the aggregate price level stickier, though realistic changes in inequality leave the aggregate price and output responses essentially unchanged in that calibration (Cravino et al., 2020). In heterogeneous-agent models, however, the distribution typically shapes demand rather than price setting (Debortoli and Galí, 2024), and the empirical evidence suggests that even this demand-side channel leaves aggregate dynamics largely un-

[4] Enders (2020) also obtains endogenous countercyclical markups through a demand-elasticity channel, but there the elasticity mix moves with monetary redistribution rather than the consumption distribution. Variable desired markups also arise from firm-level demand curvature, through Kimball demand (Kimball, 1995; Klenow and Willis, 2016) or oligopolistic competition (Edmond et al., 2015).

changed (Bilbiie et al., 2025).[5] We place the composition term inside the Phillips curve.

In Section 3, we compare RANK, TANK, and TE-TANK under a common calibration and quantify the demand-composition channel shock by shock. At the baseline calibration, the channel removes about 20 percent of the Phillips-curve slope relative to the same economy with the feedback switched off. Most of that reduction is the standard TANK income feedback, which operates even when the two types share an elasticity: of the 19.48 percent, 15.82 percentage points come from that feedback and 3.66 percentage points from the elasticity gap itself. For demand shocks, the channel is weak and works in the opposite direction: TE-TANK inherits the amplification generated by hand-to-mouth households, but the elasticity gap slightly weakens it relative to mean-matched TANK and slightly dampens inflation. Technology shocks make the channel visible: they raise inequality, shift demand toward the less price-elastic rich, and dampen the fall in inflation. Cost-push shocks deliver the central quantitative result. Relative to a homogeneous-elasticity TANK matched to TE-TANK's effective elasticity and steady-state markup, cost-push inflation pass-through is 3.1 percent higher and the output contraction is 3.1 percent deeper.

This cost-push amplification also bears on the post-pandemic debate over markups and inflation. In our mechanism, the markup increase creates its own distributional consequence: higher markups temporarily raise profits, and the fiscal scheme rebates only part of the profit flow to hand-to-mouth households, so relative income shifts toward the rich. Measured inequality rises, demand tilts further toward these low-elasticity households, and desired markups rise further, all through demand composition alone. Consistent with the income shift in our mechanism, Broer et al. (2025) use 45 years of German administrative data and the oil-supply news shock of Känzig (2021) to show that a 10 percent oil-price increase lowers the earnings growth of low-income workers by about two percentage points after two years, while top incomes are largely unaffected. Their mechanism is different from ours, but the evidence points to the same distributional outcome: a shift in relative income toward high-income, low-elasticity households, which in our model raises desired markups.

Finally, in Section 4, we test the model's shock predictions in US quarterly time series and in a cross-country panel. In US time-series data, markups rise after both identified supply shocks, the expansionary technology shock and the contractionary oil supply news shock of Känzig (2021), and consumption inequality, the empirical counterpart of the model's heterogeneity index, rises after the technology shock, as the

[5] A large heterogeneous-agent literature studies monetary transmission and optimal policy (Kaplan et al., 2018; Auclert, 2019; Bilbiie, 2025; Debortoli and Galí, 2024; McKay and Wolf, 2023; Acharya et al., 2023), and a parallel line brings household heterogeneity into supply-shock transmission through incomplete markets (Guerrieri et al., 2022; Auclert et al., 2023) and unequal energy exposure (Chan et al., 2024).

model predicts.

The paper makes three contributions. The first is theoretical: we derive a distribution-dependent supply block in closed form, in which the demand-composition feedback flattens the Phillips curve and the flatter curve amplifies cost-push inflation. The second is quantitative: we compare TE-TANK with a homogeneous-elasticity economy that has the same steady-state markup, so the two economies differ only in elasticity dispersion. Dispersion alone raises cost-push pass-through and deepens the output contraction. The third is empirical: in US time-series data, markups rise after identified supply shocks and consumption inequality rises after an expansionary technology shock, as the model predicts, though the markup response is shared by any sticky-price model, RANK included. We also replicate the established finding of Nekarda and Ramey (2020) that markups rise after an expansionary monetary shock, whereas the sticky-price class, our model included, predicts a decline. In the cross-country panel, high-inequality countries show a larger markup response to the oil shock, as predicted, but the gap is not significant once inference accounts for the common shock, and its sign depends on the inequality sorter, so we treat the test as inconclusive.

2 A Two-Agent New Keynesian Model with Heterogeneous Elasticities

The model extends the two-agent New Keynesian (TANK) framework of Debortoli and Galí (2018, 2024) with one new assumption: hand-to-mouth households have a higher price elasticity of demand than households with asset market access, as Auer et al. (2024) document in Swiss micro-data. A fraction λ of households are hand-to-mouth (poor, denoted P); the remaining $1-\lambda$ have full asset market access (rich, denoted R); and $\varepsilon_P > \varepsilon_R$. This single change makes firms' desired markups depend on the composition of demand: when spending shifts toward rich households, firms face a less elastic customer base and raise desired markups. We call this economy the two-elasticity TANK, or TE-TANK.

The mechanism operates through the division of income between wages and profits. When the output gap is positive, markups and profit income fall relative to labor income, demand shifts toward the more price-sensitive poor households, and desired markups fall further; the reverse occurs when the output gap is negative. How strongly income shifts translate into demand shifts depends on fiscal policy: δ determines how much profit income the government collects and τ determines how it is redistributed across the two types. Redistribution therefore has supply-side effects in this model, by

changing the elasticity of demand firms face and hence their desired markups.

Standard TANK models introduce a fraction of hand-to-mouth households who consume their current income each period (Campbell and Mankiw, 1989; Galí et al., 2007; Bilbiie, 2008). This creates an indirect New Keynesian cross (Bilbiie, 2025): a lower real rate raises Ricardian consumption and labor demand, hand-to-mouth households spend the resulting wage income one for one, and aggregate demand rises further, amplifying the IS semi-elasticity from $1/\sigma$ to $1/[\sigma(1-\chi_y)]$, as shown in Section 2.5. With a common elasticity ε, however, the desired markup remains the constant $\varepsilon/(\varepsilon-1)$ regardless of the income distribution, so standard TANK cannot connect inequality to markups. The elasticity gap $\varepsilon_P > \varepsilon_R$ closes this loop by making the elasticity of demand firms face an expenditure-weighted average of the two types' elasticities, summarized by the heterogeneity index $H_t \equiv C_t^R/C_t$. Both benchmarks are nested: TANK is the case $\varepsilon_R = \varepsilon_P$, and RANK obtains when $\delta(1-\tau) = 1$, in which case the two-agent economy is observationally equivalent to a representative-agent economy.

2.1 Households

A unit mass of households is divided into a fraction λ of hand-to-mouth households (poor, P) and a fraction $1-\lambda$ of Ricardian households (rich, R) with full asset-market access. Both types share the same preferences over consumption and leisure,

$$U_t^h = \left(\frac{(C_t^h)^{1-\sigma}}{1-\sigma} - \frac{(N_t^h)^{1+\varphi}}{1+\varphi} \right) Z_t, \tag{1}$$

where C_t^h is consumption, N_t^h is labor supply, σ is the inverse intertemporal elasticity of substitution, φ is the inverse Frisch elasticity, and Z_t is a preference shock with $\log Z_t = \rho_z \log Z_{t-1} + \varepsilon_t^z$. Each type aggregates consumption over varieties with its own CES elasticity of substitution ε_h, which implies type-specific demand for variety i, $C_t^h(i) = (P_t(i)/P_t^h)^{-\varepsilon_h} C_t^h$ (Appendix D.2). Under symmetric pricing, the type-specific price indices coincide, so $P_t^R = P_t^P = P_t$ (Section 2.2). We set $\varepsilon_P = 6.6$ for poor households and $\varepsilon_R = 3.0$ for rich households, following the baseline quantification in Auer et al. (2024), whose Swiss micro-data identify the income gradient of the elasticity; the level is pinned by choosing the elasticity at a top income (Section 3).

Rich households have full asset-market access; utility maximization subject to a standard budget constraint yields the familiar Euler equation, labor-supply condition, and equity-pricing equation (Appendix D.1). Poor households cannot save or borrow and consume all labor income plus transfers each period. We assume common hours across types, $N_t^P = N_t^R \equiv N_t$, at a wage that satisfies the Ricardian intratemporal condition. This is a modelling convention rather than an implication of the house-

hold problem: under separable preferences and $C_t^P < C_t^R$, hand-to-mouth households would choose higher hours at the common wage.

2.2 Firms, Demand Composition, and Markups

A continuum of monopolistically competitive firms produces differentiated goods using the technology

$$Y_t(i) = A_t N_t(i)^{1-\alpha}, \tag{2}$$

where A_t follows $\log A_t = \rho_a \log A_{t-1} + \varepsilon_t^a$ and $\alpha \in (0,1)$ governs decreasing returns to labor. Firms face Rotemberg quadratic price adjustment costs: changing the price from $P_{t-1}(i)$ to $P_t(i)$ costs $\frac{\zeta}{2}\left(\frac{P_t(i)}{P_{t-1}(i)} - 1\right)^2 Y_t$.

Aggregate demand for variety i reflects the two household types:

$$Y_t(i) = \sum_h s_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h, \tag{3}$$

where $s_h \in \{1-\lambda, \lambda\}$ is the population share of type h (Appendix D.2). Define the rich household's demand relative to the average as $H_t^R \equiv Y_t^R/Y_t = C_t^R/C_t \equiv H_t$, so that rich households account for the expenditure share $(1-\lambda)H_t$. Market clearing implies

$$H_t^P = \frac{1-(1-\lambda)H_t}{\lambda}.$$

Appendix D.4.1 derives this identity and the effective elasticity in (D.18). Thus the effective (expenditure-weighted) elasticity faced by a firm in a symmetric allocation is

$$\varepsilon_{\text{eff},t} = (1-\lambda)H_t\varepsilon_R + \lambda H_t^P \varepsilon_P = H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P.$$

This object summarizes the demand-composition channel: a higher H_t shifts demand toward rich households, who are less price-sensitive. The firm chooses $P_t(i)$ to maximize the present value of profits subject to (3). In a symmetric equilibrium, all firms face the same technology, adjustment cost, and aggregate demand schedule, so they choose the same price, $P_t(i) = P_t$ for all i. Because firms post one price per variety and do not price discriminate across household types, both types face the same price level, $P_t^R = P_t^P = P_t$. The full firm problem is in Appendix D.4.

We treat ε_R and ε_P as fixed within type, as implied by homothetic within-type preferences, and calibrate their between-type gap using Auer et al. (2024). The fixed gap is a deliberate simplification: it stands in for a demand system in which the elasticity is derived from preferences rather than imposed—non-homothetic demand as in the class of Comin et al. (2021), variable-elasticity aggregators as surveyed by Matsuyama

(2023), or, closest to our setting, Mongey and Waugh (2025), where elasticities vary endogenously with wealth. Imposing the gap instead of deriving it keeps $\varepsilon_{\text{eff},t}$ moving only with the composition of demand H_t. Under non-homothetic within-type preferences the levels of the elasticity-consumption gradients cancel from the dispersion-pure comparison, and only a difference in gradients across types matters: it scales the channel by $1-\gamma$, where γ measures the curvature of the elasticity in log consumption. Under the constant-proportional-gradient form calibrated to the same cross-section, $\gamma = 0.44$, leaving the channel at 56 percent of its baseline strength with an unchanged sign. Appendix D.12 derives the condition.

Optimal markup. Under flexible prices ($\zeta = 0$), the first-order condition yields a markup that depends on the heterogeneity index H_t:

$$\mathcal{M}_t = \frac{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P}{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}. \tag{4}$$

Equivalently, $\mathcal{M}_t = \varepsilon_{\text{eff},t}/(\varepsilon_{\text{eff},t} - 1)$. When rich households consume more relative to the average (higher H_t) and are less price-sensitive ($\varepsilon_R < \varepsilon_P$), firms optimally charge higher markups. When $\varepsilon_R = \varepsilon_P = \varepsilon$, the markup collapses to the standard constant $\varepsilon/(\varepsilon - 1)$. The derivation is in Appendix D.4.1.

2.3 Fiscal and Monetary Policy

Fiscal policy. The government collects a fraction δ of firm profits and redistributes the revenue across household types according to the transfer-tilt parameter $\tau \in [0, 1]$ (with $\tau = 0$ the most progressive scheme, equal per-capita transfers, and higher τ tilting transfers toward rich households). Appendix D.1 provides the household budget constraints and the details of the fiscal scheme. The government budget balances each period. Although the fiscal block has two parameters, only one combination of them matters for equilibrium outcomes.

Lemma 1 (Fiscal Sufficient Statistic)**.** *Fix the non-fiscal parameters $(\beta, \alpha, \varphi, \sigma, \lambda, \zeta, \varepsilon_R, \varepsilon_P)$ and the monetary-policy rule (ϕ_π, ϕ_y). After imposing symmetric equilibrium and eliminating transfers and equity values from the household budget constraints, the redistribution share δ and the transfer tilt τ enter every steady-state allocation object and every coefficient of the first-order equilibrium system for $(\tilde{y}_t, \pi_t, \hat{h}_t, \hat{\imath}_t)$ only through the scalar*

$$R \equiv \delta(1-\tau), \qquad 1 + \delta(\tau - 1) = 1 - R. \tag{5}$$

In particular, $(\bar{H}, \bar{\mathcal{M}}, \delta_2, \kappa_y, \kappa_h, \Gamma, \chi_y, \kappa)$, defined in Sections 2.4 and 2.5, are all functions of

R alone given the non-fiscal parameters. Hence fiscal pairs with the same R generate identical steady-state allocations and first-order dynamics.

Proof in Appendix D.9.1.

Although transfer flows and equity valuations depend separately on δ and τ, equity transactions cancel and transfers combine with after-tax dividends, leaving allocations dependent only on R. The parameter R measures how much profit income is rebated according to population shares. At $R = 0$, poor households receive none; at $R = 1$, profits are fully collected and returned as equal lump-sum transfers, reproducing the RANK allocation. The baseline calibration ($\delta = 1, \tau = 0.7$) gives $R = 0.3$.

Monetary policy. The central bank follows a Taylor rule:

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \hat{y}_t + v_t, \tag{6}$$

where ρ is the steady-state real rate, ϕ_π and ϕ_y are the policy response coefficients, and v_t is a monetary policy shock.

2.4 Steady State

The steady state has a recursive structure. Define the composite parameters

$$\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P), \qquad \mathsf{B} = \varepsilon_P, \qquad \mathsf{C} = \frac{1-\lambda R}{1-\lambda}, \qquad \mathsf{D} = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}.$$

First, the heterogeneity index and the flex-price markup are jointly determined by

$$\bar{H} = \mathsf{C} - \frac{\mathsf{D}}{\bar{\mathcal{M}}}, \tag{7}$$

$$\bar{\mathcal{M}} = \frac{\mathsf{A}\bar{H} + \mathsf{B}}{\mathsf{A}\bar{H} + \mathsf{B} - 1}. \tag{8}$$

The first equation follows from the rich household's budget constraint. The composite C captures the concentration of profit-related income among rich households, while $\mathsf{D}/\bar{\mathcal{M}}$ captures the equalizing effect of common labor income; equation (7) therefore expresses inequality as the balance between profit concentration and wage equalization. The second is equation (4) in steady state: consumption heterogeneity changes the effective demand elasticity faced by firms and therefore their desired markup. The elasticity gap enters through A, while fiscal policy enters only through R, via C and D (Lemma 1).

The fixed point captures a feedback between redistribution, inequality, and market power. A decline in R shifts profit income toward rich households and raises $\bar{H}$. Be-

cause rich households are less price-sensitive, this shift lowers the effective demand elasticity and raises the desired markup. The higher markup shifts income from wages toward profits and further increases inequality.

This two-equation fixed point has a closed-form characterization. Substituting (8) into (7) reduces it to a single quadratic in $\bar{H}$,

$$\mathsf{A}\,\bar{H}^2 + \big[\mathsf{B} - \mathsf{A}(\mathsf{C} - \mathsf{D})\big]\bar{H} - \big[\mathsf{B}(\mathsf{C} - \mathsf{D}) + \mathsf{D}\big] = 0, \tag{9}$$

which determines $\bar{H}$ and, through (8), $\bar{\mathcal{M}}$. Once these two objects are known, output follows recursively from labor supply and production:

$$\bar{Y} = \left(\frac{1-\alpha}{\bar{\mathcal{M}}\bar{H}^{\sigma}}\right)^{\frac{1-\alpha}{\alpha+\varphi+\sigma(1-\alpha)}}. \tag{10}$$

Holding primitives fixed, both endogenous objects lower steady-state output: a higher markup depresses labor demand, while a higher $\bar{H}$ raises rich consumption and hence depresses labor supply through the Ricardian intratemporal condition. Appendices D.3 and D.5 derive these equations; Appendix D.9.2 establishes existence and uniqueness and selects the economically admissible root.

Proposition 1 (Existence, Uniqueness, and Fiscal Comparative Statics)**.**

(i) ***Existence and uniqueness.*** *For every fiscal pair $(\delta, \tau) \in [0,1]^2$, the quadratic (9) has exactly one root consistent with strictly positive consumption for both household types. If $R < 1$, this root satisfies $\bar{H} \in (1, \frac{1}{1-\lambda})$ and $\bar{\mathcal{M}} > 1$; if $R = 1$, it satisfies $\bar{H} = 1$.*

(ii) ***Fiscal comparative statics.*** *Along this branch, $\bar{H}$ is strictly decreasing in R. Because $\varepsilon_P > \varepsilon_R$, the steady-state markup is strictly increasing in $\bar{H}$ and therefore strictly decreasing in R. A higher transfer share to poor households thus lowers both steady-state inequality and the desired markup.*

Proof in Appendix D.9.2.

The model nests both benchmark economies. At $R = 1$, the steady state coincides with RANK at the population-weighted elasticity $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$; when $\varepsilon_R = \varepsilon_P = \varepsilon$, the model reduces to standard TANK with fixed desired markup $\varepsilon/(\varepsilon - 1)$.[6]

2.5 Log-Linear Equilibrium

Lowercase letters denote logarithms, $\hat{x}_t$ denotes the percentage deviation of x_t from its steady-state value, and $\tilde{y}_t \equiv \log(Y_t/Y_t^n)$ is the output gap relative to flexible-price

[6] On the fiscal square, $R = 1$ requires $(\delta, \tau) = (1, 0)$ and gives $\bar{H} = 1$. Standard TANK can instead retain consumption inequality when $R < 1$, but its desired markup does not respond to it. Appendix D.9.2 provides the algebra for both nesting cases.

output Y_t^n.

The log-linear equilibrium consists of four relations: an auxiliary distributional relation, a dynamic IS curve, a New Keynesian Phillips curve, and the Taylor rule (6). The distributional relation ties the heterogeneity index $\hat{h}_t$ to the output gap period by period, so substituting it out returns the familiar three-equation system. The system differs from the textbook model in two ways: hand-to-mouth households create a general equilibrium feedback through the labor market, and the heterogeneity index moves endogenously with the output gap, with a strength that depends on fiscal policy through R.

Distribution and aggregate demand. Two composites summarize the distributional feedback:

$$\Gamma = \frac{1-\alpha}{\bar{\mathcal{M}}\bar{H}} \cdot \frac{\lambda\big(1+\delta(\tau-1)\big)}{1-\lambda} = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}}, \qquad \chi_y = \frac{\Gamma\left(\dfrac{\varphi+\alpha}{1-\alpha}+\sigma\right)}{1+\Gamma\sigma}.$$

Γ scales the feedback and vanishes in the RANK limit $R = 1$; χ_y turns it into the loading of inequality on the output gap. The heterogeneity index follows from the rich household's budget constraint normalized by aggregate output: it rises with the markup and falls with the share of profits retained by the government. Log-linearizing around the steady state yields the distributional relation

$$\hat{h}_t = -\chi_y\, \tilde{y}_t. \tag{11}$$

Appendix D.3 provides the exact derivation. When $\tilde{y}_t$ is positive, labor income rises relative to profit income, compressing inequality; when $\tilde{y}_t$ is negative, the reverse occurs. At the baseline calibration (Table 1), $\Gamma = 0.188$ and $\chi_y = 0.4746$: a one percent positive output gap lowers the heterogeneity index by about half a percent (Section 3).

Dynamic IS curve. The log-linearized Euler equation of rich households, combined with market clearing and the hand-to-mouth constraint, gives:

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] - \frac{1}{\sigma(1-\chi_y)}\left(\hat{\imath}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n\right), \tag{12}$$

where $r_t^n = (1-\rho_z)z_t + \sigma\mathbb{E}_t[\Delta\hat{y}_{t+1}^n]$ is the natural real rate. χ_y captures the New Keynesian cross generated by hand-to-mouth households: real-rate changes move Ricardian consumption and labor demand, which changes hand-to-mouth disposable income and, through fiscal redistribution, dividend income as well. The same coefficient therefore governs the inequality dynamics in (11) and the demand amplification in (12),

linking both channels to fiscal policy through R. In the RANK limit ($R = 1$), $\Gamma = 0$, so $\chi_y = 0$ and (12) collapses to the textbook IS curve. The full derivation, including the pre-substitution form in which (12) contains $\mathbb{E}_t[\Delta\hat{h}_{t+1}]$ explicitly, is in Appendix D.6.

The RANK nesting is now immediate:

$$\bar{H} = 1 \iff R = 1 \iff \Gamma = 0 \iff \chi_y = 0 \iff \hat{h}_t \equiv 0 \quad \text{along every first-order equilibrium path.} \tag{13}$$

The final equivalence concerns the heterogeneity loading, not any single realized path: since $\hat{h}_t = -\chi_y \tilde{y}_t$, a path with a closed output gap has $\hat{h}_t \equiv 0$ even when $\chi_y \neq 0$. When $\chi_y = 0$, the loading itself vanishes identically across all admissible disturbances. Under these equivalent conditions, TE-TANK coincides in steady state and in all first-order dynamics with RANK at the population-weighted elasticity $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$. Thus elasticity dispersion has aggregate effects only when consumption is unequal.

The distributional relation ties inequality to the output gap for any disturbance. Since $\chi_y > 0$ whenever $R < 1$, inequality moves opposite to the output gap. Shock-specific cyclicality then follows from the equilibrium sign of the gap: demand shocks move output and the gap in the same direction, so inequality is counter-cyclical; a cost-push shock lowers both, so inequality rises as output falls. For a technology shock the sign of the gap depends on the policy rule: under the calibrated rule of Section 3, a positive shock raises output but opens a negative gap, so inequality rises with output.

New Keynesian Phillips curve. Log-linearizing the firm's first-order condition around the zero-inflation steady state yields the NKPC (full derivation in Appendix D.7):

$$\pi_t = \beta\, \mathbb{E}_t[\pi_{t+1}] + \kappa_h\, \hat{h}_t + \kappa_y\, \tilde{y}_t + u_t, \tag{14}$$

with

$$\kappa_y = \delta_2 \left(\frac{\varphi + \alpha}{1-\alpha} + \sigma \right),$$

$$\kappa_h = \frac{(1-\lambda)\bar{H}}{\zeta} \left(1 - \overline{MC}\right)(\varepsilon_P - \varepsilon_R) + \delta_2 \sigma,$$

$$\delta_2 = \frac{1}{\zeta}[\bar{H}(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1] = \frac{\varepsilon_{\text{eff}}(\bar{H}) - 1}{\zeta},$$

where $\overline{MC} = 1/\bar{\mathcal{M}}$ is steady-state real marginal cost, δ_2 is the marginal-cost loading in the Phillips curve, and u_t is an exogenous cost-push disturbance, specified in Sec-

tion 2.6.[7] The coefficient κ_y is the output-gap loading with the heterogeneity feedback switched off: the conventional marginal-cost block, evaluated at the TE-TANK steady state. The coefficient κ_h carries two routes: a marginal-cost route through Ricardian consumption, $\delta_2\sigma$, which is present even when both types share one elasticity, and a desired-markup route, which is present only when $\varepsilon_P \neq \varepsilon_R$. The marginal-cost route arises because the wage is priced by the Ricardian household's marginal rate of substitution at its own consumption; Broer et al. (2020) stress the same mechanism in a worker–capitalist economy with flexible wages, where marginal cost tracks workers' consumption rather than aggregate output. Since $1 - \overline{MC} = 1/\varepsilon_{\text{eff}}(\bar{H})$, the second route is itself an elasticity:

$$\eta_\mu \equiv \left.\frac{\partial \log \mathcal{M}_t}{\partial \log H_t}\right|_{\bar{H}} = \frac{(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)}{\varepsilon_{\text{eff}}(\bar{H})\left[\varepsilon_{\text{eff}}(\bar{H}) - 1\right]} > 0, \qquad \kappa_h = \delta_2\big(\sigma + \eta_\mu\big). \tag{15}$$

Here η_μ is the steady-state elasticity of the desired markup with respect to the heterogeneity index, so the second equality splits κ_h into its two routes. Substituting the distributional relation (11) gives the reduced-form NKPC:

$$\pi_t = \beta\,\mathbb{E}_t[\pi_{t+1}] + \underbrace{(\kappa_y - \kappa_h\chi_y)}_{\kappa}\,\tilde{y}_t + u_t. \tag{16}$$

The inequality channel enters through $-\kappa_h\chi_y$: a positive output gap compresses inequality ($\hat{h}_t < 0$ when $\tilde{y}_t > 0$), which lowers firms' desired markups and puts downward pressure on inflation. Writing $\chi_y = \Gamma G/(1+\Gamma\sigma)$ with $G \equiv \frac{\varphi+\alpha}{1-\alpha} + \sigma$, the terms in $\Gamma\sigma$ cancel and the slope factors exactly:

$$\kappa = \delta_2 G\,\frac{1-\Gamma\eta_\mu}{1+\Gamma\sigma}, \qquad \delta_2 G = \kappa_y. \tag{17}$$

The feedback-off slope κ_y is multiplied by two factors. The first, $1/(1+\Gamma\sigma)$, is the standard TANK income feedback, which survives under a common elasticity. The second, $1-\Gamma\eta_\mu$, is the desired-markup correction, which is absent under a common elasticity. Section 2.6 uses this factorization to build the benchmark comparisons.

The factorization yields two immediate implications under the baseline common-hours closure. First, $\text{sign}(\chi_y) = \text{sign}(1-R)$, so inequality moves opposite to the output gap whenever $R < 1$. Second, $\kappa_h > 0$ and

$$\kappa < \kappa_y \iff \chi_y > 0 \iff R < 1.$$

[7] Unlike its Calvo counterpart, δ_2 carries no decreasing-returns correction $(1-\alpha)/(1-\alpha+\alpha\varepsilon)$: that factor arises because a resetting Calvo firm's own marginal cost differs from the average, whereas under Rotemberg pricing all firms stay symmetric and this wedge never opens (Ascari and Rossi, 2012).

The endogenous heterogeneity feedback therefore flattens the Phillips curve relative to the same steady state with $\hat{h}_t \equiv 0$. This comparison holds the steady-state objects fixed and removes only the demand-composition feedback; it is not a comparison with a homogeneous-elasticity economy.

Model closure. Together with the Taylor rule (6), equations (12), (11), and (16) form the complete first-order equilibrium system in $(\tilde{y}_t, \pi_t, \hat{h}_t, \hat{i}_t)$. Because (11) determines $\hat{h}_t$ from the output gap, the system reduces to the standard three equations in $(\tilde{y}_t, \pi_t, \hat{i}_t)$.

2.6 Benchmark Comparisons and Cost-Push Transmission

The results in this subsection compare TE-TANK with three benchmarks. The *feedback-off* benchmark is TE-TANK itself with the steady-state objects $(\bar{H}, \bar{\mathcal{M}}, \delta_2)$ held fixed and the distributional relation switched off, $\hat{h}_t \equiv 0$; its Phillips-curve slope is κ_y. The *mean-matched* TANK is a homogeneous-elasticity economy at the population-weighted mean $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$, with the same fiscal parameters. It preserves the mean of the primitive elasticities but has its own, lower, steady-state markup, so a comparison with it mixes the dispersion effect with a markup-level effect. The *markup-matched* TANK(ε^*) is a homogeneous-elasticity economy at the elasticity ε^* whose constant markup $\varepsilon^*/(\varepsilon^*-1)$ equals TE-TANK's steady-state markup $\bar{\mathcal{M}}$. With common fiscal parameters, matching the markup also matches $\bar{H}$, δ_2, Γ, and χ_y, so the two economies differ only in the loading κ_h; this benchmark isolates the dispersion-pure part of the mechanism. Because the markup-matched economy has $\eta_\mu = 0$, its reduced-form slope is $\kappa^{\text{TANK}^*} = \delta_2 G/(1+\Gamma\sigma)$, and by (17),

$$\kappa^{\text{TE}} = \kappa^{\text{TANK}^*}\left(1 - \Gamma\eta_\mu\right),$$

so the entire dispersion-pure difference between the two slopes is the factor $1-\Gamma\eta_\mu$. Appendix D.10 compares the structural objects across the three economies and RANK, and reports their joint sensitivity to the elasticity gap and level along a fixed-ratio path.

IS amplification. On the demand side, the interest-rate semi-elasticity in (12) is $1/[\sigma(1-\chi_y)]$. When $R < 1$ and $\chi_y \in (0,1)$, hand-to-mouth behavior amplifies the response relative to RANK by the finite factor $1/(1-\chi_y) > 1$. By Lemma 1, the fiscal configuration enters this comparison only through R. The comparison with the homogeneous-elasticity benchmarks is the content of the next proposition.

Proposition 2 (Elasticity Dispersion and IS Amplification)**.** *Fix the population-weighted mean elasticity* $\bar{\varepsilon} \equiv (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$ *and fiscal parameters with* $R < 1$. *Relative to*

homogeneous-elasticity TANK with $\varepsilon = \bar{\varepsilon}$, TE-TANK has

$$\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}, \qquad \bar{\mathcal{M}}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}.$$

Consequently,

$$\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}, \qquad 1 < \frac{1}{1-\chi_y^{\text{TE}}} < \frac{1}{1-\chi_y^{\text{TANK}}}.$$

Thus TE-TANK's IS curve is more rate-sensitive than RANK's but less rate-sensitive than that of mean-matched TANK.

Proof in Appendix D.9.3.

This ordering is a markup-level result: elasticity dispersion raises $\bar{H}$ and $\bar{\mathcal{M}}$, increasing the product $\bar{\mathcal{M}}\bar{H}$ in the denominator of Γ and thereby lowering χ_y. Under markup matching, TANK(ε^*) instead shares TE-TANK's $\bar{\mathcal{M}}\bar{H}$ and hence its χ_y, so the two amplification factors coincide. The contribution of the elasticity gap is therefore not stronger demand amplification; it is the endogenous markup and flattening described above, together with the cost-push amplification in Proposition 3 below. Section 3 reports the amplification factors at the baseline calibration.

Cost-push transmission. The disturbance u_t in (14) is an AR(1), $u_t = \rho_u u_{t-1} + \varepsilon_t^u$ with $\rho_u \in [0,1)$: a reduced-form stand-in for wage-markup or energy-cost shocks that move marginal cost independently of the output gap.[8] For the cost-push results below, maintain $\beta \in (0,1)$, $\sigma > 0$, $\kappa > 0$, $\phi_\pi > 1$, $\phi_y \geq 0$, $\rho_u \in [0,1)$, and $0 \leq \chi_y < 1$. When two economies are compared, these restrictions hold in both. They imply $\tilde{\sigma} \equiv \sigma(1-\chi_y) > 0$ and standard determinacy.

Proposition 3 (Cost-Push Pass-Through)**.** *Consider the cost-push wedge u_t in isolation, so $z_t = a_t = v_t = 0$, $r_t^n = 0$, and $\hat{y}_t = \tilde{y}_t$. Under the maintained cost-push conditions in the preceding paragraph, the unique bounded solution is $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$, where*

$$\Psi_\pi = \frac{1}{D_\pi} > 0, \qquad \Psi_y = -\Theta\Psi_\pi < 0,$$

with

$$D_\pi = (1-\beta\rho_u) + \kappa\Theta > 0, \qquad \Theta = \frac{\phi_\pi - \rho_u}{\tilde{\sigma}(1-\rho_u) + \phi_y} > 0.$$

Pass-through satisfies

$$\frac{\partial\Psi_\pi}{\partial\kappa} = -\Theta\Psi_\pi^2 < 0, \qquad \frac{\partial\Psi_\pi}{\partial\tilde{\sigma}} = \frac{(1-\rho_u)\kappa(\phi_\pi-\rho_u)}{\left[\tilde{\sigma}(1-\rho_u)+\phi_y\right]^2}\Psi_\pi^2 > 0.$$

[8] The calibration sets $\rho_u = 0.9$, $\sigma_u = 0.05$; the paper's headline results are TE-TANK/TANK ratios of impact responses to the same unit shock, which are invariant to σ_u.

Holding all non-elasticity primitives, policy coefficients, and the wedge process fixed, TE-TANK has greater pass-through than homogeneous-elasticity TANK if and only if

$$\frac{\Psi_\pi^{\text{TE}}}{\Psi_\pi^{\text{TANK}}} > 1 \iff \kappa^{\text{TE}}\Theta^{\text{TE}} < \kappa^{\text{TANK}}\Theta^{\text{TANK}}.$$

Proof in Appendix D.9.4.

A sufficient condition for greater pass-through is $\kappa^{\text{TE}} \leq \kappa^{\text{TANK}}$ and $\chi_y^{\text{TE}} \leq \chi_y^{\text{TANK}}$, with at least one inequality strict. Under the common-hours closure, the condition holds structurally against the markup-matched benchmark whenever $R < 1$: the two economies share χ_y, and $\kappa^{\text{TE}} = \kappa^{\text{TANK}^*}(1 - \Gamma\eta_\mu) < \kappa^{\text{TANK}^*}$. Against the mean-matched benchmark, Proposition 2 delivers $\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}$; the slope comparison is quantitative, and Section 3 verifies it at the baseline calibration. A positive cost-push wedge also generates stagflation: inflation rises and the output gap falls. Across the two economies,

$$\frac{|\Psi_y^{\text{TE}}|}{|\Psi_y^{\text{TANK}}|} = \frac{\Theta^{\text{TE}}}{\Theta^{\text{TANK}}} \frac{\Psi_\pi^{\text{TE}}}{\Psi_\pi^{\text{TANK}}},$$

so the output contraction is larger in TE-TANK whenever the pass-through amplification dominates any offsetting difference in the demand block.

The mechanism operates through demand composition. A cost-push recession opens a negative output gap and raises inequality. Demand shifts toward less price-sensitive rich households, inducing firms to raise desired markups and reinforcing the initial cost pressure. The flatter Phillips curve implies that a given output contraction produces less disinflation, while a weaker New Keynesian cross further raises pass-through. Demand and technology shocks produce the opposite inflation effect because their output movements do not reinforce the initial cost pressure.

Divine coincidence. Absent the exogenous wedge ($u_t = 0$), setting $\pi_t = \tilde{y}_t = 0$ satisfies (16) at all times: stabilizing inflation also closes the output gap. For demand and technology shocks, the flexible-price allocation leaves H_t—and hence the flexible-price desired markup—at its steady-state value, so heterogeneous elasticities do not create an additional cost-push wedge; the TE channel changes only the *pass-through* of the exogenous wedge u_t, through κ (Proposition 3).[9]

[9]The divine coincidence is a flexible-wage property. Under nominal wage rigidity it breaks for disturbances that move the flexible-wage marginal rate of substitution—in this model's shock set, only the technology shock.

3 Quantitative Analysis

3.1 Calibration

Table 1 reports the quarterly calibration. Common parameters follow standard New Keynesian practice (Debortoli and Galí, 2018, 2024): $\beta = 0.995$ (a 2% annual real rate), $\sigma = \varphi = 1$, $\alpha = 1/3$, and $\zeta = 105.63$. The adjustment cost reproduces the marginal-cost slope of a Calvo economy with decreasing returns at a price-stickiness parameter of 0.753, corresponding to an average price duration of four quarters.[10] Following Debortoli and Galí (2018), we set the hand-to-mouth share to $\lambda = 0.40$, close to evidence placing it near one-third (Kaplan et al., 2014). We impose full dividend taxation, $\delta = 1$.

The key parameters are the demand elasticities. Following the baseline quantification of Auer et al. (2024), we set $\varepsilon_P = 6.6$ for poor households and $\varepsilon_R = 3.0$ for rich households, corresponding to annual incomes of 20,000 and 120,000 Swiss francs; hence $\varepsilon_P/\varepsilon_R = 2.2$.[11] These estimates identify expenditure switching between imported and domestic goods. Applying them to substitution across domestic varieties assumes that the rich–poor gradient in price sensitivity transports across margins. This assumption is weaker for their second approach, which estimates substitution between any pair of products within a sector regardless of import status, on a sample that includes domestic goods, and recovers essentially the same gradient as the first; restricting that sample to imports leaves the estimates largely unchanged. Three caveats remain: the data cover groceries, the identifying price variation is an import-price shock in both approaches, and the nesting imposes one elasticity per sector. Appendix D.10 reports how the Phillips-curve slope and the steady-state markup move with the elasticity level along a fixed-ratio path.

The baseline TANK and RANK columns use the homogeneous elasticity $\varepsilon^* = 3.933$, which reproduces TE-TANK's steady-state markup $\bar{\mathcal{M}} = 1.341$; RANK additionally sets $\tau = 0$. Appendix C.4 treats the CEX consumption ratio, which lies outside the fiscal block's feasible range, as an untargeted validation moment. Appendix C.3 reports the full parameterization and implied steady states.

[10] The mapping equates the Rotemberg slope $(\varepsilon^* - 1)/\zeta$ with the Calvo slope $\frac{(1-\theta)(1-\beta\theta)}{\theta} \cdot \frac{1-\alpha}{1-\alpha+\alpha\varepsilon^*}$, whose second factor corrects for decreasing returns; at $\varepsilon^* = 3.933$ and $\zeta = 105.63$ it holds at $\theta = 0.753$.

[11] Auer et al. (2024) estimate the slope of the elasticity in log income at 2.19 (SE 0.55) in their first approach and 1.93 (SE 0.87) in their second, and calibrate it to 2 elasticity points per log income point, which gives a gap of 3.6 points between annual incomes of 20,000 and 120,000 francs. Neither baseline design identifies the level; they choose the elasticity at 120,000 francs to be 3, so the poor elasticity is 6.6. They also recover the level under stronger assumptions and obtain 4.92 and 1.76 at 20,000 francs in the two approaches, but do not use these values because the approaches disagree. Their own sensitivity exercise varies the top-income elasticity from 1.5 to 5 while holding the elasticity differences across households fixed, which implies a ratio between 1.72 and 3.39.

TABLE 1: Baseline Calibration

Description	Symbol	TE-TANK	TANK	RANK
Common parameters				
Discount factor	β		0.995	
Risk aversion	σ		1.00	
Inverse Frisch elasticity	φ		1.00	
Capital share	α		1/3	
Price adjustment cost	ζ		105.63	
Model-specific parameters				
Share of hand-to-mouth	λ	0.40	0.40	0.00
Share of dividends	δ	1.00	1.00	1.00
Transfer tilt toward rich	τ	0.70	0.70	0.00
Rich elasticity	ε_R	3.00	3.93	3.93
Poor elasticity	ε_P	6.60	3.93	3.93

Notes: Common parameters are shared across all three economies. TANK and RANK use the homogeneous elasticity $\varepsilon^* = 3.933$, at which $\varepsilon/(\varepsilon-1)$ equals TE-TANK's steady-state markup $\bar{\mathcal{M}} = 1.341$; they are therefore markup-matched to TE-TANK. RANK is the representative-agent limit $\lambda = 0$. By Lemma 1 fiscal policy enters only through $R = \delta(1-\tau)$, and at $R = 1$ profits are rebated equally so that steady-state inequality vanishes (Proposition 1); the parameterization with $\lambda = 0.40$ and $R = 1$ is therefore the same economy. Appendix C.3 (Tables C.2–C.3) reports the full parameterization, including the Taylor rule and shock-persistence parameters not shown here.

At the baseline calibration, the composite Phillips-curve slope is 0.0833 in RANK, 0.0701 in markup-matched TANK, and 0.0671 in TE-TANK. Within TE-TANK, the demand-composition feedback lowers the slope from its feedback-off value $\kappa_y = 0.083$ by 19.48 percent. The standard TANK income feedback accounts for 15.82 percentage points of this reduction, while the desired-markup channel accounts for 3.66 percentage points.

Because the demand elasticities are calibrated externally to Auer et al. (2024), the steady-state markup $\bar{\mathcal{M}} = 1.341$ is an outcome rather than a target. The elasticity level is a free calibration parameter in Auer et al. (2024), who sweep it themselves, and here it pins the steady-state markup $\varepsilon/(\varepsilon-1)$, so the main sensitivity path holds the elasticity gap fixed and varies the level. That fixed-gap path is the first of three one-dimensional paths through the plane of the mean elasticity and the elasticity gap: the second holds the ratio fixed and varies the level, and the third holds the mean fixed and varies the gap. The first is primary because the gap comes from the estimated gradient; the other two show what the level or the ratio would have to do to change the answer. Over the implied markup range 1.21 to 1.61, the dispersion-pure cost-push gap moves from

1.6 to 5.7 percent around the baseline 3.1 percent, and the elasticity ratio drifts from 1.75 to 3.01. Holding the ratio at 2.2 instead narrows the range to 2.4 to 3.8 percent over the same markup span; the ratio depends on the chosen level, the gap does not. Appendix C.4 reports the first two paths, and the bounds below, in Table C.6.

A 10 percent dispersion-pure gap requires $\Delta\varepsilon \approx 8.0$, or $(\varepsilon_R, \varepsilon_P) = (1.23, 9.26)$ and a ratio of 7.5, at the baseline λ and R. That calculation runs along the fixed-mean path, which holds the mean elasticity at its calibrated $\bar{\varepsilon} = 4.44$ and varies $\Delta\varepsilon$ alone. The implied markup there is 1.45, inside the 1.21 to 1.61 span, so what rules out this parameterization is not the markup but $\varepsilon_R = 1.23$, near-unit-elastic demand from rich households, and a ratio more than three times the calibrated one. Holding $\varepsilon_P/\varepsilon_R = 2.2$, the gap cannot exceed 4.0 percent at any elasticity level, but that ceiling is reached at $\bar{\mathcal{M}} = 1.93$, above the published markup span, so the binding value within the span is the 3.8 percent reported above. The distributional parameters cannot substitute either: at $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ the gap is strictly decreasing in R and strictly increasing in λ, so over $\lambda \in (0,1)$ and $R \in [0,1]$ subject to the strict admissibility constraint $\chi_y < 1$ it has a supremum of 7.0 percent, approached at $R = 0.0$ as $\lambda \to 0.656$ and $\chi_y \to 1$ but attained at no admissible point. The size of the mechanism is set by the estimated rich–poor gradient, not by the elasticity level or the fiscal block.

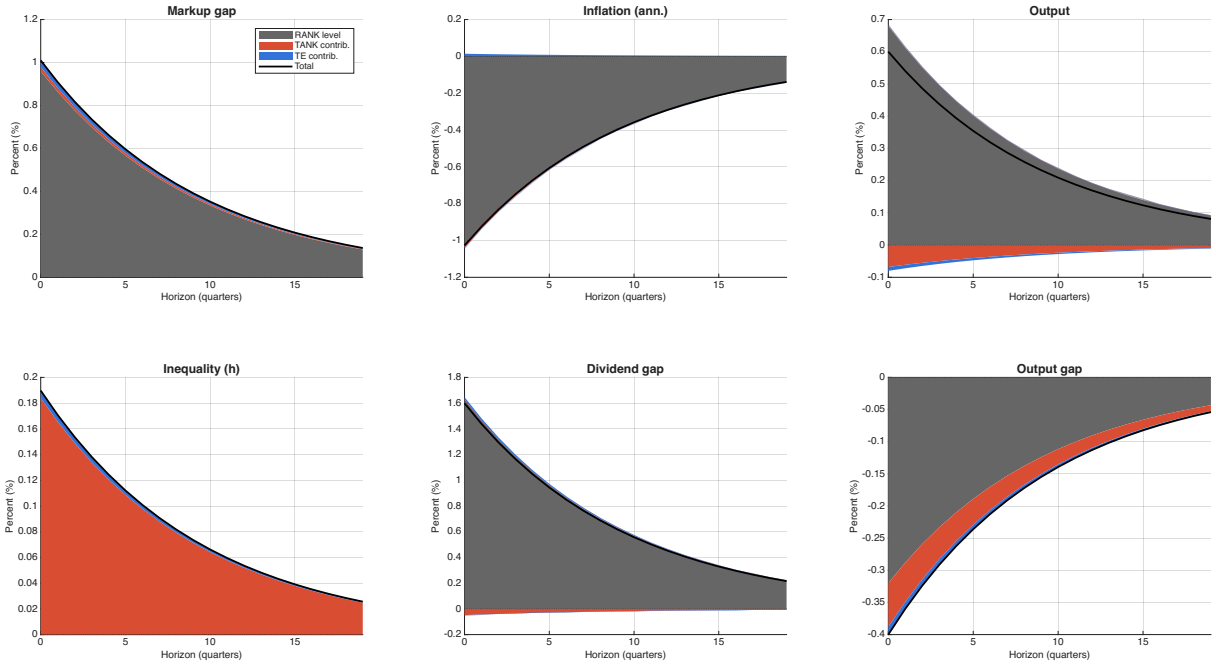
3.2 Impulse Response Functions

We solve the log-linearized model by first-order perturbation.[12] The impulse-response figures share one design. Each band decomposition builds the TE-TANK response ($\varepsilon_R = 3.0$, $\varepsilon_P = 6.6$; **solid black line**) from three additive bands along the accounting path RANK $\to$ TANK $\to$ TE-TANK: **gray** is the RANK response, **red** is the contribution of hand-to-mouth aggregation (TANK minus RANK), and **blue** is the contribution of the elasticity gap (TE-TANK minus TANK). The accounting is sequential and path-dependent along that order, not a partition into independent channels. Both homogeneous corners are evaluated at the markup-matched elasticity $\varepsilon^* = 3.933$, which reproduces TE-TANK's steady-state markup $\bar{\mathcal{M}} = 1.341$, so the TANK corner shares TE-TANK's $\bar{H}$, $\bar{\mathcal{M}}$, and χ_y and differs only in κ_h; the blue band is therefore the dispersion-pure contribution of elasticity dispersion. The panel layout is the same throughout: markup gap, inflation, and output in the top row; inequality ($\hat{h}_t$), the dividend gap, and the output gap in the bottom row. Inflation is annualized; all other variables are percentage deviations from the steady state at a quarterly frequency.

Figure 1 applies this decomposition to a positive technology shock. The gray and red bands reproduce the familiar New Keynesian and TANK shapes, with hand-to-

[12] We use Dynare 6.4.

FIGURE 1: Technology Shock: Channel Decomposition



Notes: Impulse responses to a one-standard-deviation positive technology shock; bands, homogeneous corners, panel layout, and units as described in the text. The shock lowers marginal cost, and price adjustment costs keep prices from falling one for one, so the markup gap and the dividend gap rise while inflation falls. The red band mutes the output gain relative to RANK and carries the rise in inequality, which is zero in RANK by construction: dividends rise while hours fall, shifting income toward rich households. The blue band is positive in the inflation panel: the rise in inequality shifts demand toward the less price-elastic rich households, desired markups rise, firms become more reluctant to cut prices, and the elasticity gap damps the fall in inflation.

mouth aggregation muting the output gain, while the blue elasticity-gap band damps the inflation response. The corresponding decomposition for an expansionary monetary policy shock, where hand-to-mouth aggregation instead amplifies the response and the elasticity gap again damps inflation, is Figure D.3 in Appendix D.11. We focus the main text on the cost-push shock, which produces the paper's central quantitative result.

Cost-Push Shock. A cost-push shock reverses the sign of the TE channel: heterogeneous elasticities now amplify inflation and deepen the output loss, whereas they damp the inflation response to the demand and technology shocks. The shock is the exogenous wedge u_t in the NKPC (14), an AR(1) with persistence $\rho_u = 0.9$ that raises marginal cost independently of the output gap. Figure C.1 in Appendix C.5 plots the TE-TANK response to a one-standard-deviation wedge (solid black) directly against TANK($\varepsilon^* = 3.933$) (dashed blue); in the inflation panel the gap between the two economies is $+0.018$ of a total $+0.604$ percentage points on impact.

On impact, the wedge raises annualized TE-TANK inflation by 0.60 percentage

points and lowers output by 0.51 percent. Relative to TANK(ε^*), inflation pass-through is 3.1 percent higher and the output contraction 3.1 percent deeper: Ψ_π rises from 2.928 to 3.019 and $|\Psi_y|$ from 9.896 to 10.204. The two gaps coincide because the two economies share $\bar{H}$, $\bar{\mathcal{M}}$ and χ_y, so the entire first-order difference runs through the Phillips-curve slope and none of it through the demand block. These impact responses are the closed-form pass-throughs Ψ_π and Ψ_y of Proposition 3; the Dynare responses match them to machine precision. Both sufficient conditions of Section 2.6 hold at the calibration: $\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}$ against the mean-matched benchmark (Proposition 2), together with $\kappa^{\text{TE}} < \kappa^{\text{TANK}(\bar{\varepsilon})}$ (Table C.7), and $\kappa^{\text{TE}} = 0.0671 < 0.0701 = \kappa^{\text{TANK}}$ with equal χ_y against the markup-matched benchmark.

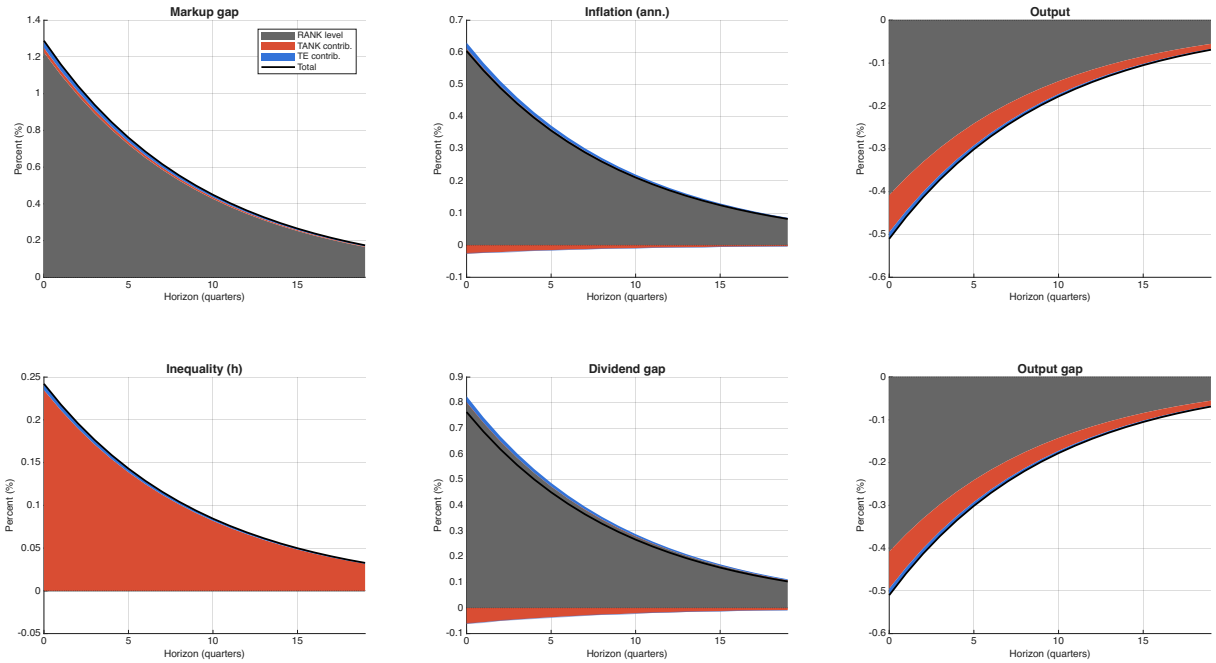
We quote headline magnitudes as the dispersion-pure gap against TANK($\varepsilon^* = 3.933$). At this matched steady state the two economies differ in only one coefficient of the analytical first-order policy block, κ_h, and therefore in the slope κ. The primary estimand is thus the effect of elasticity dispersion conditional on the same steady-state markup and distribution. The population-mean-preserving benchmark answers a different question: TANK($\bar{\varepsilon} = 4.44$) holds the population-weighted mean of the primitive elasticities fixed, so the gap against it also includes the equilibrium markup-level consequence of introducing dispersion. Against this benchmark, Taylor pass-through is 15.74 percent higher and the output contraction is 14.53 percent deeper (Appendix C.5, Table C.7). The table maps between four homogeneous-elasticity conventions and decomposes the population-mean-preserving gap into dispersion-pure and markup-level components.

The three economies are not ordered under the calibrated Taylor rule. RANK has the highest pass-through, $\Psi_\pi = 3.061$, ahead of TE-TANK's 3.019 and TANK's 2.928. Hand-to-mouth aggregation makes the IS curve more rate-sensitive, $\tilde{\sigma} = \sigma(1 - \chi_y) = 0.53$ against 1 in RANK, so the same rule delivers a larger output contraction per unit of inflation and buys more disinflation. The IS amplification factors $1/(1 - \chi_y)$ of Proposition 2 are 1.000 in RANK, 1.903 in TE-TANK—and, by construction, in markup-matched TANK—and 1.972 in mean-matched TANK. Figure 2 reports the sequential accounting from RANK through hand-to-mouth aggregation to the elasticity gap.

The mechanism is the demand-composition channel running in reverse. The wedge opens a negative output gap, and inequality rises as real wages fall and profits rise ($\hat{h}_t = -\chi_y \tilde{y}_t > 0$), as the inequality panel of Figure C.1 in Appendix C.5 shows. Demand tilts toward the less price-elastic rich households, desired markups rise, and firms add inflation on top of the exogenous wedge; the distance between the two lines in that figure's markup-gap panel is this endogenous markup response. Markups and profits thus rise during a cost-driven inflation with no change in conduct—the pattern at the center of the "sellers' inflation" debate (Weber and Wasner, 2023; Lorenzoni

and Werning, 2023; Bilbiie and Känzig, 2024)—here through static profit maximization under a shift in customer composition.

FIGURE 2: Cost-Push Shock: Sequential Channel Decomposition at the Markup-Matched Benchmark



Notes: Impulse responses to a one-standard-deviation cost-push shock, the AR(1) wedge u_t in equation (14) ($\rho_u = 0.9$, $\sigma_u = 0.05$); bands, homogeneous corners, panel layout, and units as described in the text. The red band is negative in the inflation panel, a consequence of the pass-through ordering discussed above. The paper's dispersion-pure comparison is the final TE-minus-TANK step alone, the blue band, not the difference between TE-TANK and RANK.

4 Empirical Evidence

The evidence falls into two tiers. Most of what is tested here—that markups rise after supply shocks, that they fall after demand shocks, and that consumption inequality rises in a technology-driven expansion—follows from the sticky-price core this model shares with RANK and with a homogeneous-elasticity TANK. Those are not tests of elasticity heterogeneity. We run them because a mechanism built on that core is not worth taking seriously if the core itself fails in the data. One prediction is specific to the elasticity gap: the markup response to a cost-push shock rises with steady-state inequality. The aggregate data discipline the signs in both tiers. They do not discipline the magnitude in the second. The gap the calibrated model implies at the observed cross-country inequality spread is an order of magnitude below the smallest gap this design could have detected, so the cross-country test carries almost no information about the size of the channel (Appendix B.6).

The model's testable content is shock-conditional. Conditional on supply shocks, markups should rise and consumption inequality should rise with them, as described in Sections 2.5 and 2.6. Conditional on demand shocks, the sticky-price mechanism predicts falling markups and a TE channel too weak to matter (Section 3). Along the calibrated fiscal branch, the markup response to a cost-push shock increases with steady-state inequality (Corollary 4); Proposition 3 links this result to inflation pass-through. This section tests these predictions with local projections on identified shocks. Section 4.1 lays out the design, centered on the oil supply-news shock; Section 4.2 summarizes the unconditional markup and consumption-inequality responses to the technology, oil, and demand shocks, with the full proxy-by-proxy record in Appendix B.2; Section 4.3 tests the amplification prediction across countries, with the complementary US time-series design in Appendix B.3 and the cross-country details in Appendix B.4; and Section 4.4 collects the signs.

4.1 Empirical Strategy

For each horizon $h = 0, 1, \ldots, 12$ we estimate the local projection of Jordà (2005),

$$\mu_{t+h} - \mu_{t-1} = a_h + \beta_h s_t + \sum_{l=1}^{4} \gamma_{l,h} s_{t-l} + \sum_{l=1}^{4} d_{l,h} \Delta\mu_{t-l} + e_{t+h}, \tag{18}$$

where μ_t is the log markup *level* and s_t an identified shock, so that β_h is the cumulative markup response at horizon h, measured from $t-1$ and normalized to a one-standard-deviation shock. The projection is lag-augmented, so ordinary least squares with Eicker–Huber–White (HC3) standard errors delivers valid inference without any HAC correction (Montiel Olea and Plagborg-Møller, 2021). We estimate one regression per shock–outcome pair and assess joint significance across horizons with a horizon-stacked block-wild bootstrap that also delivers sup-t simultaneous confidence bands, with the identified shocks entering as given regressors (the SVAR first stage of the technology shock is not re-estimated within the resampling; Appendix B.2 discusses); where the sample is shorter—the cross-country panel and the consumption-inequality outcome—we truncate at $h \leq 8$.

This section's focus is the oil supply-news shock, the published series of Känzig (2021), identified from high-frequency oil-futures reactions to OPEC announcements; we aggregate it to the quarter and standardize it, keeping the published sign, so $+1$ SD is contractionary (future oil supply falls and the oil price rises). Oil is the natural cost-push laboratory: oil-price movements are a leading source of variation in US headline inflation relative to core. It is a cost-push shock that also redistributes income toward the top (Broer et al., 2025): in the model both the exogenous cost pressure and

the income shift toward the inelastic rich raise desired markups, so the markup response tested below is a joint test of distribution-dependent pricing rather than a pure pass-through test. Using the same instrument, Broer et al. (2025) find textbook supply-shock responses of German inflation and activity, with a strong first stage, corroborating its identification outside the United States. Their policy counterfactuals further indicate that the shock's aggregate and distributional effects are largely invariant to the systematic monetary response, which mitigates the concern that cross-country differences in policy reactions, rather than in demand composition, drive the heterogeneity documented below. Two further shocks discipline the unconditional predictions of Section 4.2: a *technology* shock, identified exactly as in Nekarda and Ramey (2020) from a quarterly five-variable Cholesky SVAR with utilization-adjusted TFP (Fernald, 2012) ordered first; and a *demand* shock, the high-frequency monetary policy series of Jarociński and Karadi (2020) for the markup outcome and the narrative shock of Romer and Romer (2004) for the shorter consumption-inequality sample. Full identification detail for both is in Appendix B.2.

The outcomes are markups and consumption inequality. Markups are measured by the four national-accounts proxies of Nekarda and Ramey (2020)—the inverse labor share under Cobb–Douglas technology (the baseline), a Cobb–Douglas variant with overhead labor, and CES labor- and capital-margin measures, estimated over 1975Q1–2017Q4 ($N = 172$) for the oil shock and over 1949Q1–2017Q4 for the technology shock (1964Q1 onward for the overhead-labor variant)—and a fifth price-over-unit-cost measure, estimated over 1985Q1–2025Q2 ($N = 162$); definitions and the full proxy battery are in Appendix B (Table B.1). Consumption inequality is the P90/P10 ratio of log consumption from Coibion et al. (2017), constructed from the BLS Consumer Expenditure Survey (quarterly, 1981Q4–2008Q4). This outcome is the model-matched object: the heterogeneity index $\hat{h}_t$ is a statement about *consumption* inequality, not about top income shares. With two types, the model's exact analogue of a percentile ratio is the rich-to-poor ratio of mean consumption, which is monotone in H_t; the P90/P10 ratio indexes the same dispersion without being a literal counterpart, so we read only the sign and timing of its response.

4.2 Markups and Inequality after Supply and Demand Shocks

Markups rising after supply shocks and falling after demand shocks are properties of any standard sticky-price New Keynesian model, RANK included. Markups do rise after both supply shocks and across all four proxies: the technology-shock response replicates Nekarda and Ramey (2020), significantly so on impact and imprecisely at longer horizons, and the oil-shock response survives an energy-inclusive marginal-cost

measure (Appendix B.2, Figures B.1 and B.2). The demand shock disagrees: markups rise after an expansionary monetary policy shock (Nekarda and Ramey, 2020), whereas the whole sticky-price class predicts a decline (Figure B.3).[13] Because the TE channel is negligible for demand shocks (Section 3), the disagreement concerns the shared core rather than the elasticity gap, and the mechanism tests in Section 4.3 condition on supply and cost-push shocks.

Consumption inequality, the mechanism's second input, rises significantly after an expansionary technology shock (sup-t joint $p = 0.018$), matching the cyclicality predicted in Section 2.5, while the oil and demand shocks leave it an honest null (joint $p = 0.66$ and 0.95; Appendix B.2, Figure B.4). The model predicts a rise for the oil shock and a fall for the demand shock, with a negligible incremental TE effect for the demand shock and a small one for the oil shock; neither estimate is precise enough to detect the predicted response, so both nulls are uninformative rather than evidence for or against the model.

4.3 Zooming in on the Oil Shock

The sharpest prediction that separates TE-TANK from standard TANK is cross-sectional. After the same oil shock—a cost-push disturbance bundled with redistribution toward the top (Section 4.1)—the calibrated model predicts a larger markup response at higher steady-state inequality (Corollary 4, proved in Appendix D.10). The sign is what the model delivers: along the admissible fiscal branch the impact markup response is strictly increasing in $\bar{H}$ under two conditions on primitives that hold strictly at the calibration. The magnitude is not: mapping the observed cross-country inequality spread onto $\bar{H}$ requires an extrapolation off the model grid and a unit elasticity between income and consumption dispersion, so the model-implied gap of Appendix B.6 is an illustrative bound rather than a derivation. Carrying the corollary across countries requires one more assumption. The model has a single lever on steady-state inequality, the fiscal parameter $R = \delta(1 - \tau)$, which measures how much profit income is rebated equally across households; the corollary raises $\bar{H}$ by lowering R, holding the demand elasticities, the hand-to-mouth share λ, and all other primitives fixed. Reading the corollary across countries therefore treats observed inequality differences as if they were generated by that lever alone, and countries differ in much

[13] TE-TANK inherits this failure. Matching the sign requires machinery we deliberately do not adopt: a cost channel, under which the fall in the policy rate lowers firms' financing costs and marginal cost (Suveg, 2026); Kimball demand curvature or strategic complementarities, which tie desired markups to relative prices and damp the predicted decline (Kimball, 1995; Klenow and Willis, 2016; Edmond et al., 2015); and the composition of measured marginal cost, which decides the sign of the estimate itself (Bils et al., 2018). We treat the sign as a known limitation of the core, not as something the elasticity gap can repair. The demand-shock markup evidence also cannot separate TE-TANK from its benchmarks, since the elasticity gap barely moves that response.

more than their redistribution schemes; beyond the power problem documented below, this is a further reason to read the cross-country test as suggestive rather than decisive. A complementary US time-series design, which exploits within-country fluctuations in inequality over time rather than its cross-country dispersion, is directionally positive for both the technology and the oil shock but underpowered on the available US sample (Appendix B.3, Figure B.5).

Cross-country evidence. The cross-country test replaces time-series variation in US inequality with the much larger dispersion of inequality levels across countries. For each country i we estimate a separate local projection on the common global oil shock,

$$\begin{aligned} \mu_{i,t+h} - \mu_{i,t-1} &= a_{i,h} + \beta_{i,h}\, s_t + \sum_{l=1}^{L+1} \rho_{i,l,h}\, \Delta\mu_{i,t-l} + \sum_{l=1}^{L+1} \omega_{i,l,h}\, s_{t-l} + e_{i,t+h}, \\ \hat{\beta}_h^{\mathrm{MG}} &= \frac{1}{N} \sum_i \hat{\beta}_{i,h}, \end{aligned} \tag{19}$$

and average the country coefficients with the Mean Group estimator of Pesaran and Smith (1995), taking standard errors from the cross-country dispersion; this is consistent under the slope heterogeneity a cross-country panel almost surely exhibits, unlike a pooled estimator. In equation (19), L is a common control-lag length chosen by a panel AIC rule, and carrying $L+1$ lags of both the outcome difference and the shock is the lag augmentation of Montiel Olea and Plagborg-Møller (2021), so ordinary least squares with Eicker–Huber–White (HC3) standard errors again delivers valid inference without any HAC correction. The shock is Känzig (2021)'s published global oil supply-news series, common to all countries and not re-estimated, so identification comes entirely from how the response $\hat{\beta}_{i,h}$ differs across countries. The markup outcome is a factor-share-based (national-accounts) proxy, the inverse labor share $\mu_{i,t} = \log(\text{GDP}/\text{compensation of employees})$; countries are split at the median time-averaged WID top-10% share. This top-10% income share is the cross-country availability sorter, not the model-matched consumption-dispersion object; Appendix C.4 distinguishes the two objects. The sample is 37 OECD and other European economies at quarterly frequency through 2019Q4—29 OECD members plus eight further European economies covered by the Eurostat national accounts, of which Bulgaria, Croatia, and Romania are OECD accession candidates (Table B.4)—excluding the net oil exporters Norway, Mexico, and Canada, for whom an oil supply-news shock is a positive terms-of-trade shock with the opposite incidence and whose oil rents mechanically depress the labor share.

The high-minus-low inequality gap in the cross-country markup response is positive at every horizon, $+0.0019$ at $h = 4$ and $+0.0026$ at $h = 8$ (Figure 3). Because the

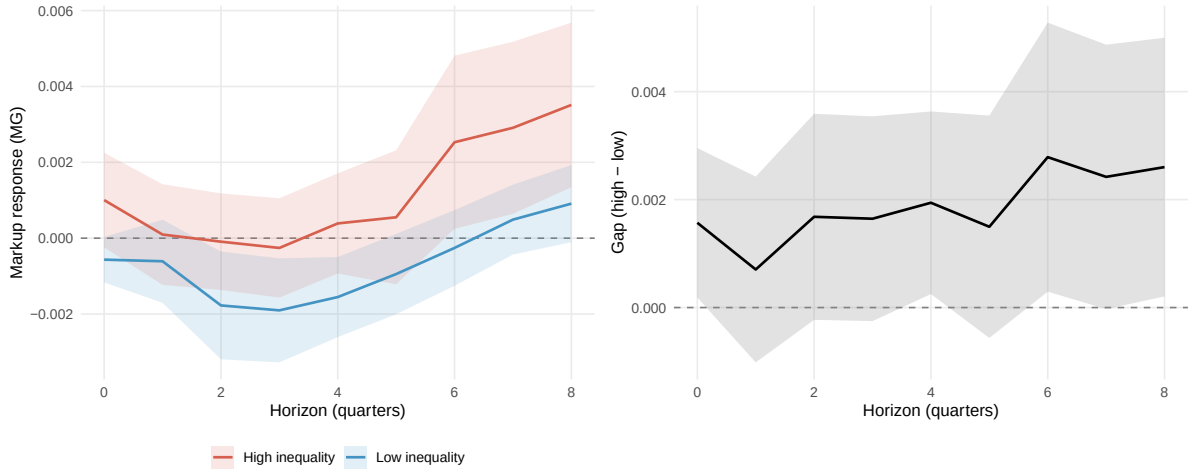
oil shock is common to all countries, we lead with common-shock-robust inference: a moving-block bootstrap in the time dimension, which resamples the shared quarterly axis and shrinks the effective number of independent draws to roughly the number of oil episodes (Appendix B.4). Under it the gap is insignificant at $h = 4$ ($p = 0.254$) and only marginal at $h = 8$ ($p = 0.103$), both at the pre-specified primary block length of 10 quarters ($B = 4999$); block lengths of eight and twelve give similar results. The cross-country dispersion standard errors, which underlie the bands in Figure 3, are less demanding: they place both headline horizons at the 10 percent level, and in the balanced subsample of countries with at least 80 usable quarters ($N = 35$), which holds the Mean Group sample fixed across horizons, the $h = 8$ gap sharpens to $+0.0029$ (5 percent level). Under either cut the gap is significant pointwise but not simultaneously across horizons. The Mean Group markup response itself is flat to slightly negative over the first year and builds to $+0.0022$ (SE 0.0007, significant at the 1 percent level) by $h = 8$, so the average response, like the gap, is a medium-horizon phenomenon. The gap survives a swap to the gross-operating-surplus profit-share proxy and the omission of any single country, but it does not extend to broad-dispersion sorters: sorting by the SWIID disposable-income Gini (Solt, 2020) instead of the pre-tax top-10% share attenuates it to insignificance, and sorting by the market Gini flips its sign (Table B.8). The sign of the gap is robust to the identifying instrument, though its magnitude and significance are not: replacing the Känzig (2021) oil-news series with the noisier single-maturity proxy of Kilian (2024), a variant of the same OPEC-announcement design, halves the gap and removes its significance on the matched sample (Table B.8). We therefore read the test of the amplification prediction as inconclusive: the gap has the predicted sign at every horizon,[14] but its significance depends on the inference method, and its sign on the breadth of the inequality sorter.[15] Driscoll–Kraay standard errors on the pooled interacted panel leave its implied gap significant, and the pooled interaction survives seven confounders jointly (Table B.2); that pooled estimand differs from the Mean Group gap (Appendix B.5), and these estimates do not overturn the inconclusive Mean Group result under common-shock inference.

Power and model-implied magnitudes. The cross-country design could not have detected the amplification the calibrated model implies. Two probabilities describe

[14] The qualitative amplification prediction is positive at every horizon. The calibrated model's mapped gap is essentially zero and slightly negative at $h = 4$ (-0.000118) because it rescales the estimated unconditional response, which is negative at that horizon, by the model's positive amplification ratio; the sign is inherited from the data, not predicted by the model (Appendix B.6).

[15] A sharper, pre-specified version of the same test—regressing the country coefficients $\hat{\beta}_{i,h}$ on the structural traits the model's closed form signs—returns a comprehensive null under the same common-shock logic (Appendix B.7, Table B.7). That exercise runs on all 40 countries for which the first stage delivers a coefficient, and treats the 37-country net-exporter exclusion used here as a separate pre-specified robustness family.

FIGURE 3: Cross-Country Markup Response by Inequality Group (Mean Group, Oil Shock)



Notes: Mean Group markup responses to a +1 SD Känzig (2021) oil supply-news shock, high- versus low-inequality countries (37 OECD and other European economies, split at the median WID top-10% share; net oil exporters NOR, MEX, and CAN excluded). Right panel: the high-minus-low gap. Shaded bands are 90% intervals from the cross-country dispersion of the country coefficients, which treats the countries as independent draws from the common shock; the common-shock-robust alternative, a moving-block bootstrap in the time dimension, is in Appendix B.4. Inference on the gap uses the standard error of the difference; the overlap of the left panel's bands is therefore not informative about it (right panel).

any test: its size is how often it rejects when there is nothing to find, and its power is how often it rejects when the effect is exactly the one the model predicts. Here the two nearly coincide—power is 5.6 percent against a nominal size of five percent—so failing to reject conveys almost nothing about the mechanism. Figure B.7 in Appendix B.6 draws the arithmetic: it places the estimated gap, the model's prediction, and the range this design could have detected on one scale, and beneath it the two sampling distributions—under no amplification and under the model's prediction—against the rejection region. At $h = 8$ the smallest gap this design finds with eighty percent power is 0.00506, while the model, mapped into the units of the test at the observed cross-country inequality spread, predicts 0.000423: a factor of 12.0. Closing that distance means shrinking the standard error by the same factor, hence an effective sample about 144 times larger, roughly 6500 years of quarterly data. More countries do not substitute. The oil shock is common to all of them, so the effective sample is the time dimension, which the bootstrap credits with about 18 independent blocks. The estimated gap is itself several times larger than the model predicts, 0.0026 against 0.000423, so the positive point estimates cannot be attributed to the elasticity-gap channel either. Appendix B.6 reports the mapping and the same calculation for the US design.

Limitations. Three further caveats qualify the amplification result beyond the sorter dependence noted above. First, both cross-country outcomes—the inverse labor share and the gross-operating-surplus profit share—are factor-share-based (national-accounts) markup proxies, which can move with wage incidence, employment composition, energy costs, the terms of trade, and sectoral reallocation while desired markups are unchanged. Oil shocks compress the bottom of the wage bill directly, with selective exit of low-wage workers (Broer et al., 2025), so part of the measured labor-share decline can reflect labor-market incidence and composition rather than a rise in desired markups. The gross-operating-surplus proxy checks this from the profit side, reading the markup off operating surplus rather than the labor margin the incidence channel operates on, and the amplification gap survives under it (Appendix B.8, Table B.8). Neither this check nor the energy-inclusive marginal-cost measure (Appendix B.2) rules the concern out entirely.

Second, the production-approach markup does not corroborate the oil result. The country-year markup levels of De Loecker and Eeckhout (2018), which do not use the factor-share identity, deliver a significantly *negative* average response at short horizons on the 19-country annual subsample (1980–2016) and a null amplification gap at every horizon. The two measures disagree, and the oil-shock evidence reported here is a statement about the factor-share-based proxies, not about production-approach markups.

Third, the distributional evidence behind the mechanism predates the episode that motivates it: the Coibion et al. (2017) consumption-inequality series ends 2008Q4 and the DLEU production-approach markups end 2016, so the 2021–2023 inflation surge lies outside the window on which the mechanism is estimated. Extending the oil-shock design through 2023Q4 leaves the average markup response and the amplification gap positive and significant at the 10 percent level or better ($+0.0015$ at $h = 8$ and $+0.0020$ at $h = 4$, respectively; Appendix B.4).

4.4 Signs: Model versus Evidence

The technology and oil shocks match on the markup sign; the demand shock does not. Table B.9 in Appendix B collects the full record of model-predicted and estimated signs for the markup and consumption-inequality responses, shock by shock. The TE amplification prediction is directionally positive but imprecise in the US state-dependent design; in the cross-country oil-shock gap it is positive at every horizon—the sign the mechanism predicts at the medium horizons—but is not significant under the inference we lead with. One further fact sits outside the shock-conditional design: average markups are higher in countries with higher average top-10% income shares, with a

between-country slope of 0.93 (SE 0.20) across 138 countries. That relationship is consistent with the model's steady-state channel but does not identify it, and it concerns long-run levels rather than within-country dynamics (Appendix A).

Robustness and the regression alternative. Appendix B collects the supporting battery behind this section. We do not pursue the complementary direct approach of regressing the Phillips curve slope on inequality. Aggregate time-series estimates of the New Keynesian Phillips curve are weakly identified and highly sensitive to specification, not least to the measure of inflation expectations (Mavroeidis et al., 2014; Coibion and Gorodnichenko, 2015), and when policy accommodates cost-push shocks the aggregate regression traces the targeting rule rather than the structural curve (McLeay and Tenreyro, 2020). The remedies that literature prescribes are to identify the slope from cross-sectional variation (Hazell et al., 2022; Beschin et al., 2025) or from identified shocks (Barnichon and Mesters, 2020). We take the second route, which is why the evidence in this section is shock-conditional throughout.

5 Conclusion

Heterogeneous price elasticities place the consumption distribution inside the Phillips curve. We develop this mechanism in an otherwise standard two-agent New Keynesian model, calibrated to the baseline quantification of Auer et al. (2024): desired markups depend on the composition of demand across household types, so they move whenever a shock moves inequality.

The mechanism has one sign for demand and technology shocks and the opposite sign for cost-push shocks. It flattens the reduced-form Phillips curve by about 20 percent relative to the same economy with the feedback switched off—mostly through the standard TANK income feedback, with 3.66 percentage points from the elasticity gap itself—and damps the inflation response to demand and technology shocks, while a cost-push recession raises inequality, tilts demand toward the inelastic rich, and raises desired markups on top of the shock. Against TANK($\varepsilon^* = 3.933$), the homogeneous economy at TE-TANK's own effective elasticity, cost-push inflation is 3.1 percent higher under a Taylor rule—the dispersion-pure effect of elasticity heterogeneity alone. Table C.7 in the appendix maps this gap into the other homogeneous-elasticity conventions and adds the markup-level component that the elasticity composition itself induces. A flatter Phillips curve is not benign in this model: the same feedback that flattens the curve raises the pass-through of cost shocks. The model does not decompose the measured decline of the slope since the 1980s (Kuttner and Robinson, 2010; Blanchard, 2016; Hooper et al., 2020); it states what a flatter curve costs when a cost

shock arrives.

The empirical record is shock-conditional. The supply-shock findings are sign-consistent: markups rise after identified supply shocks, and consumption inequality rises after an expansionary technology shock, as the model predicts. In the cross-country panel, the markup response to a common oil supply news shock is larger in high-inequality countries, the sign the mechanism predicts. That gap is positive at every horizon and survives the gross-operating-surplus proxy and the omission of any single country, but it is significant only pointwise, not simultaneously across horizons and not once common-shock-robust inference shrinks the effective sample to a few oil episodes. It attenuates under an alternative inequality sorter, and the production-approach markup gives a negative average response and a null gap. We read the test as inconclusive rather than as support. The estimates also reproduce an established result: markups rise after expansionary demand shocks (Nekarda and Ramey, 2020), whereas standard sticky-price models predict a decline.

The calibration also misses a moment it does not target: the CEX consumption ratio lies outside the fiscal block's feasible range, so the baseline understates consumption dispersion (Appendix C.4). A natural next step is a structural estimation of the model, so that the elasticity gap and the fiscal parameters are disciplined jointly by the data rather than calibrated.

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Appendix

Inequality, Markups, and the Phillips Curve

Rahel Braun Sarah M. Lein Florian Stich

August 2026

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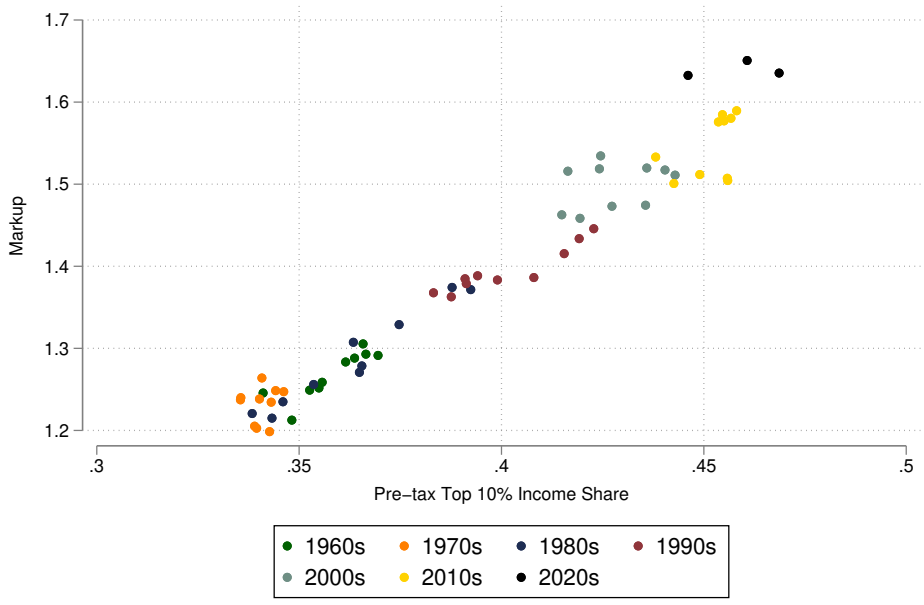
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A Markup and Inequality: Additional Evidence

Higher average top-10% income shares are associated with higher average markups across countries, but this unconditional relationship does not identify the model's mechanism. This appendix collects the level fact quoted in Section 4.4, the US time series, alternative markup proxies, and the within-country estimates.

Rising markups alongside inequality. Between 1980 and 2016, the US pre-tax income share of the top 10% rose from about 34% to 45%, while the aggregate sales-weighted markup rose from approximately 1.2 to 1.6 (Figure A.1). The figure reports annual observations from 1960 to 2023. The two series rise together, but this unconditional co-movement does not identify a causal relationship.[16] This sales-weighted aggregate markup and the model's calibrated steady-state markup ($\bar{\mathcal{M}} \approx 1.34$) are different concepts at different levels: Figure A.1 aggregates firm-level production-function markup estimates across the whole economy with sales weights, while the model targets a single representative pricing block calibrated to the Auer et al. (2024) elasticity estimates. The model's steady-state comparative statics share the sign of the *co-movement* documented here; the model is calibrated neither to the level of either series nor to the historical trend.

FIGURE A.1: The Co-Movement of Market Power and Income Concentration

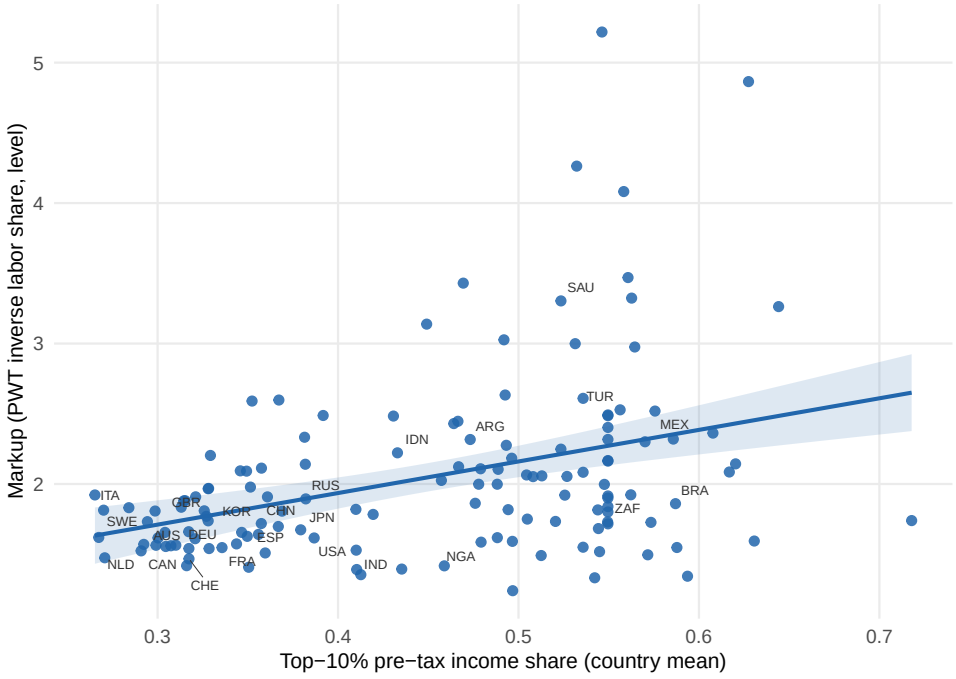


Notes: Aggregate sales-weighted markups versus pre-tax income share of top 10%. Each point: annual observation, colored by decade.
Sources: WID.world and Compustat.

[16] Markups are estimated using the production approach; see Appendix C.2 for details.

Markup levels rise with inequality levels across countries. The time-series co-movement documented above spans a single country. Its cross-sectional counterpart—the level analogue of the steady-state mechanism formalized in Section 2—holds across countries: economies with higher long-run income concentration operate at higher markup levels. Figure A.2 plots country-mean markup levels, measured as the inverse labor share from the Penn World Table, against country-mean WID pre-tax top-10% income shares. PWT corrects labor shares for self-employment income where mixed-income data are available, but coverage is incomplete and residual measurement error may correlate with informality and income concentration. The between-country slope is 0.93 (SE 0.20, $p < 0.001$, $N = 138$ countries). The relationship is not specific to this proxy: the between-country slope is 0.56 ($p < 0.01$, $N = 41$) under the production-approach markups of De Loecker and Eeckhout (2018) and 0.96 ($p < 0.001$, $N = 185$) under the ILO labour income share. Because the test compares long-run averages, no shock identification is required. Table A.1 reports the underlying level regressions for all three sources, together with the within-country fixed-effects and long-difference estimates discussed below.

FIGURE A.2: Markup Levels and Inequality Levels Across Countries



Notes: Each point is one of 138 countries: the time-mean PWT inverse labor share from 1951 to 2019 against the time-mean WID pre-tax top-10% income share. The line is the OLS fit with its 95% confidence band. PWT adjusts labor shares for the labor income of the self-employed (Gollin, 2002). All countries are included; selected countries are labelled.
Sources: Penn World Table 10.x and WID.world.

TABLE A.1: The Between-Country Markup-Inequality Slope: Proxy and Estimator Robustness

	Between	Within FE	Long-diff.
Inverse LS: PWT `labsh`	0.935^{***}	0.202	0.044
N=138	(0.197)	(0.142)	(0.209)
DLEU production-approach	0.564^{***}	0.285	0.625
N=41	(0.189)	(0.279)	(0.665)
Inverse LS: ILO labour income share	0.964^{***}	0.055	−0.413
N=185	(0.181)	(0.192)	(0.254)

Notes: Dependent variable: log markup; regressor: WID pre-tax top-10% income share; standard errors in parentheses. Estimator definitions in the accompanying text. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The three samples differ in coverage rather than in the underlying relationship: the Penn World Table has the broadest country and time coverage, the production-approach markups of De Loecker and Eeckhout (2018) are limited to countries and years with sufficiently detailed firm financial-statement data, and the ILO labour-share series has its own reporting coverage; the intersection of all three sources is too thin to support a separate common-sample estimate, so we report the coverage difference rather than compute one. Two caveats keep the reading honest. First, this is a steady-state *level* fact, not a time-series claim: within-country fixed-effects and long-difference estimators are statistically insignificant in all three sources, so the data speak to differences in long-run levels across countries, not to within-country trends. Second, the level association is consistent with the model's steady-state channel from $\bar{H}$ to $\bar{\mathcal{M}}$ but does not identify it; the shock-conditional tests in Section 4 do the identification work.

B Local Projections

This appendix collects the estimation detail and robustness behind the local-projection evidence of Section 4: the markup-measure menu, the unconditional responses to the technology, oil, and demand shocks, the consumption-inequality responses, the US state-dependent analysis, and the cross-country robustness. Since markup cyclicality is sensitive to how the markup is measured (Nekarda and Ramey, 2020), we evaluate the unconditional prediction across five distinct measures before turning to the state-dependent tests. Throughout this appendix standard errors are the HC3 (leverage-corrected) variant of Eicker–Huber–White heteroskedasticity-robust standard errors. HC3 is more conservative in small samples than the HC1 formula employed in the

Montiel Olea–Plagborg-Møller lag-augmented asymptotic theory (Montiel Olea and Plagborg-Møller, 2021), so inference is conservative where the per-regime sample is modest. Robustness checks reported in summary form below cover smooth and bias-corrected local projections, a threshold BVAR, and alternative sample restrictions, together with a set of cross-country checks: an alternative oil-shock instrument, a secondary US-identified technology shock, alternative markup proxies, alternative inequality sorters, and sample-composition checks. This appendix reports the point estimates and conclusions.

B.1 Markup Measures

Table B.1 lists the five measures. They span the main methodological choices in the literature: (i) the inverse labor share under Cobb–Douglas technology, the Nekarda and Ramey (2020) baseline, which we also use in the cross-country panel below; (ii) a Cobb–Douglas specification that adds overhead labor following Galí and Gertler (1999); (iii) a CES labor-margin measure; (iv) a CES capital-margin measure constructed under constant utilization; and (v) the price-over-unit-cost markup for non-financial corporations from Enders (2020). Measures (i)–(iv) use quarterly US national-accounts data; measure (v) uses the NFC price-cost ratio. Spanning several production-function assumptions and input margins helps separate genuine cyclical patterns from measurement artifacts.

TABLE B.1: Markup measures

Markup measure	Definition	Source / parameter
Inverse labor share (CD)	$\mu^{CD} = -\ln s_L$	BLS labor share s_L
CD + overhead labor	$\mu^{CD,OH} = \ln(\alpha / s'_L)$	s'_L = variable-labor share
CES, labor margin	$\mu^{CES}_L = \mu^{CD} + \ln\left[\alpha_L \, (Y/ZL)^{1/\sigma - 1}\right]$	BLS, $\sigma = 0.5$
CES, capital margin	$\mu^{CES}_K = \mu^{CD} + \ln\left[1 - \alpha_K \, (Y/uK)^{1/\sigma - 1}\right]$	BLS + BEA capital, const. u
Price / unit cost (Enders)	$\mu = \ln(P/UC)$	BLS NFC Prod. & Costs

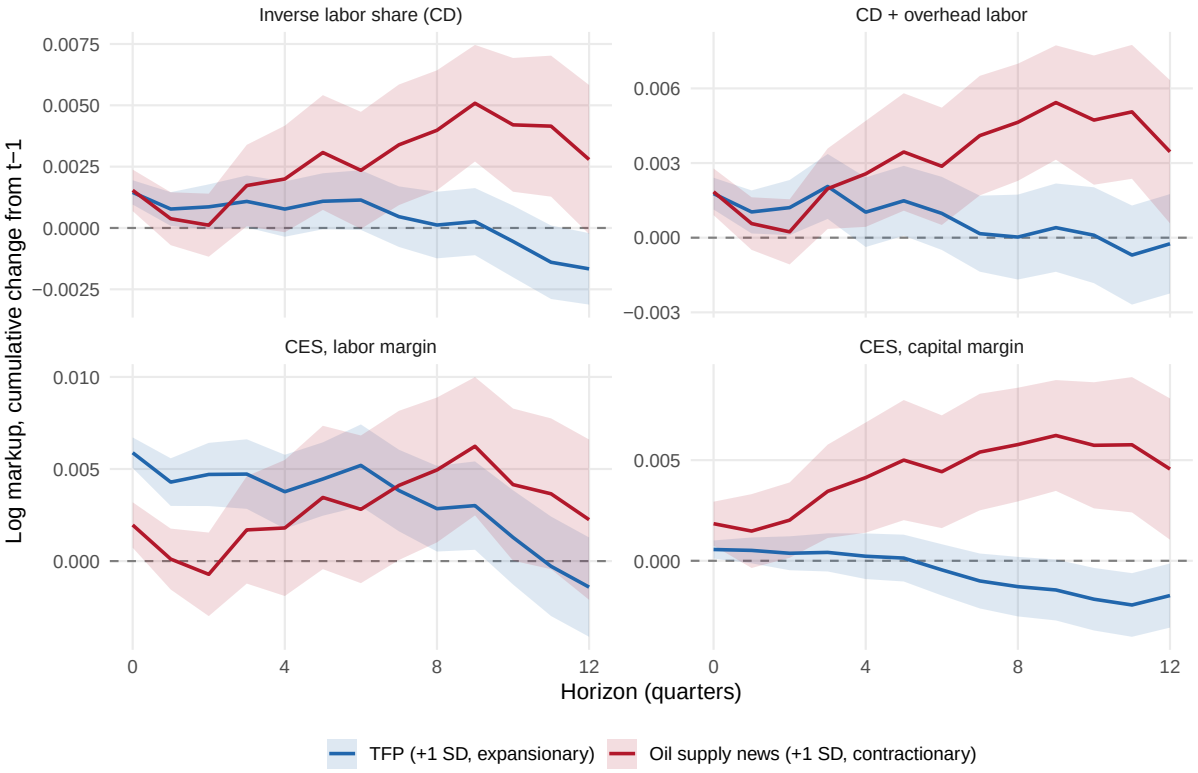
Notes: The five markup measures used in the local-projection analysis. Measures (i)–(iv) are constructed from quarterly US national-accounts data; measure (v) is the price-over-unit-cost ratio for non-financial corporations. Measure (i) is also the variable used in the cross-country panel in Section B.4.

B.2 Unconditional Responses to Identified Shocks

Markups across both supply shocks: the headline picture. Figure B.1 reports the markup responses to the technology and oil shocks across the four proxies of Nekarda and Ramey (2020), estimated directly from equation (18). Markups rise after both. The technology shock is the clean test of the model's mechanism: marginal cost falls and, with sticky prices, the ratio of price to marginal cost rises—the response replicates

Nekarda and Ramey (2020) and is evidence consistent with New Keynesian pricing in general, not only with our extension. Two qualifications discipline the reading. First, the US markup response to the technology shock is concentrated in the pre-1975 sample, so the oil shock is the reliable driver in modern data. Second, the inverse-labor-share proxy used here measures the labor margin, while energy costs enter marginal cost directly, so part of its rise after the contractionary oil shock could in principle be the mechanical wedge between the two; the energy-inclusive marginal-cost check below (Figure B.2) addresses this directly.

FIGURE B.1: Markup Responses to Supply Shocks: Unconditional Local Projections



Notes: Cumulative log-markup responses from equation (18) on US quarterly data, one panel per markup proxy. Blue: identified TFP shock, +1 SD expansionary (SVAR innovation). Red: Känzig (2021) oil supply news, +1 SD contractionary (published sign)—the two shocks move the economy in opposite directions by construction. Shaded bands are ±1 standard error (approximately 68% pointwise; EHW/HC3).

The technology shock $\varepsilon_t^{\text{TFP}}$ is identified in a *first stage*, exactly as in Nekarda and Ramey (2020): we estimate a quarterly five-variable Cholesky SVAR—utilization-adjusted TFP (Fernald, 2012) in log levels ordered first, followed by log real GDP per capita, the log GDP deflator, the three-month Treasury-bill rate, and the log markup, with four lags and a quadratic trend—and take the reduced-form residual of the TFP equation, which enters equation (18) as the shock s_t for horizons $h = 0, \ldots, 12$. Be-

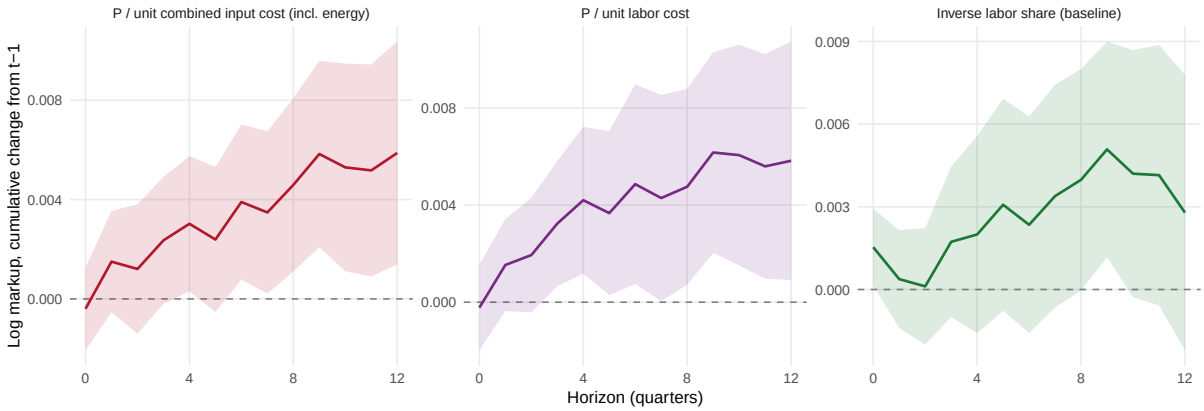
cause the projection is lag-augmented, ordinary least squares with Eicker–Huber–White (HC3) standard errors delivers valid inference without any HAC correction (Montiel Olea and Plagborg-Møller, 2021). The identified shock is essentially unchanged when the markup is dropped from the SVAR (correlation 0.995) and across ordering, trend, and sample choices, so the concern that the outcome enters the system is immaterial. The local projections and their bootstrap treat this identified shock as a given regressor; first-stage estimation uncertainty is not propagated into the bands. The invariance of the shock series just documented limits but does not eliminate this concern; the cross-country oil design is unaffected, since the oil supply-news series is a published external instrument with no first stage estimated here.

The unconditional markup response is the sharper of the two LP findings in this appendix: a joint bootstrap test across all horizons ($h = 0, \ldots, 12$) rejects the null that the entire path is zero ($p = 0.041$), and the sup-t simultaneous confidence band excludes zero at the impact horizon. The rejection is driven by that impact response: the path is positive but imprecise thereafter, and the mild negative excursion beyond two years is indistinguishable from zero at every horizon, pointwise and jointly. The markup rises on impact, and the sign is robust across measures: four of the five are positive and significant at impact (only the CES capital-margin measure is insignificant), and the response holds on the Enders price-over-unit-cost measure, which does not reduce to the labor-share identity, becoming *larger*, not smaller, when contemporaneous hours are added as controls, so it is not a labor-hoarding artifact. This matches Nekarda and Ramey (2020): the nature of the shock, not its sign, governs markup cyclicality. A smooth (penalized) local projection (Barnichon and Brownlees, 2019; Li et al., 2024) sharpens rather than overturns this result (joint p falls from 0.041 to 0.017 at the GCV-optimal penalty), and a bias-corrected local projection (Herbst and Johannsen, 2024) leaves it essentially unchanged (finite-sample bias below 2% of the standard error at every horizon).

Marginal-cost-based markup: does the oil-shock rise survive putting energy in the cost base? The headline inverse-labor-share proxy omits energy from measured marginal cost, so part of its response to the contractionary oil shock could in principle be the mechanical wedge between the labor margin and the true marginal cost. We test this directly with the Enders (2020) price-over-unit-cost markup, in two variants: price over unit labor cost alone, and price over the *combined* cost of labor and energy inputs, which puts energy directly in the denominator. Both rise significantly after the Känzig (2021) oil supply-news shock over 1985Q1–2025Q2 ($N = 162$): the energy-inclusive measure by $+0.30$ percent at $h = 4$ (10 percent level) and $+0.46$ percent at $h = 8$ (5 percent level); the labor-cost-only measure by $+0.42$ percent at $h = 4$ (5 percent level)

and +0.48 percent at $h = 8$ (10 percent level). Over the longer 1975Q1–2017Q4 window used for the national-accounts inverse-labor-share baseline ($N = 172$), that baseline's own response is directionally the same but smaller and not significant at $h = 4$ (+0.20 percent) or $h = 8$ (+0.40 percent). The oil-shock markup rise therefore survives, and is if anything sharper under, a cost measure that puts energy directly in the denominator; it is not an artifact of the labor-share proxy omitting energy from marginal cost.

FIGURE B.2: Marginal-cost-based markup response to the oil shock



Notes: Cumulative log-markup responses to a +1 SD Känzig (2021) oil supply-news shock, lag-augmented local projection with EHW standard errors. Left and middle panels: the Enders (2020) price-over-unit-cost markup, with energy included in (left) and excluded from (middle) the cost base; 1985Q1–2025Q2 ($N = 162$). Right panel: the national-accounts inverse-labor-share baseline; 1975Q1–2017Q4 ($N = 172$), for comparison.

Demand shock: the sign that disagrees with the model. Markups rise after an expansionary monetary policy shock, reproducing the established result in Nekarda and Ramey (2020). We use the Jarociński and Karadi (2020) high-frequency series purged of central-bank information effects (1990Q2–2017Q4), sign-flipped so that +1 SD is expansionary. All four proxies in Figure B.3 are positive from $h = 1$ onward and significant at the 10 percent level at several horizons; the overhead-labor measure reaches +0.78 percent at $h = 12$ ($t = 2.1$). A sticky-price New Keynesian model—RANK, TANK, and TE-TANK alike—predicts a decline because expansionary demand raises marginal cost against sticky prices. The disagreement concerns the common sticky-price core rather than the elasticity-gap mechanism, which is quantitatively negligible for demand shocks (Section 3).

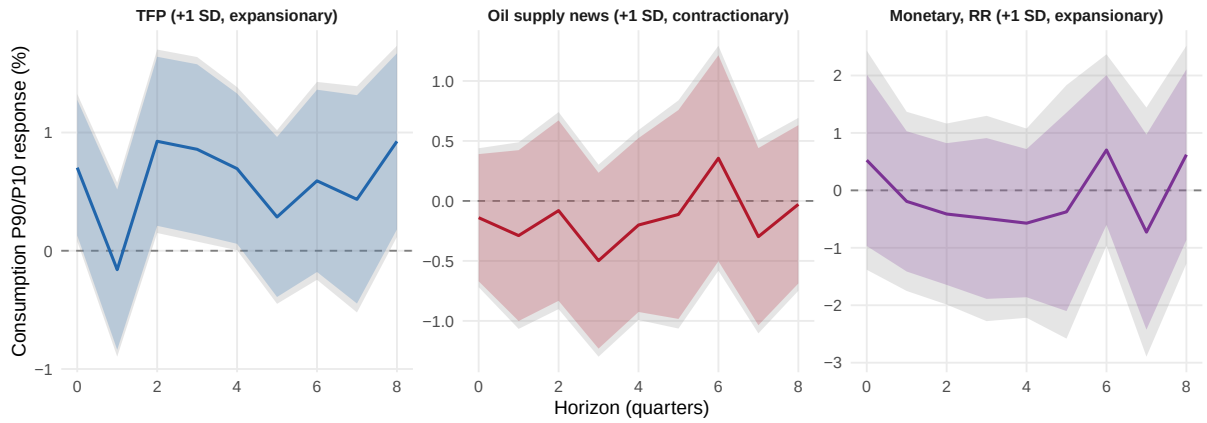
FIGURE B.3: Markup Response to an Expansionary Demand Shock



Notes: Same design as Figure B.1 (cumulative log markup from $t-1$; lag-augmented; EHW/HC3 standard errors). Shock: Jarociński and Karadi (2020) monetary policy shock, $+1$ SD *expansionary* (sign-flipped from the published convention to match the model's impulse responses); sample 1990Q2–2017Q4 ($N = 111$). Shaded bands are 90% pointwise intervals (± 1.645 SE).

Consumption inequality after three identified shocks. Figure B.4 turns to the second input of the mechanism: consumption inequality, measured by the Coibion et al. (2017) P90/P10 ratio. After an expansionary technology shock, the consumption P90/P10 ratio rises—inequality is pro-cyclical conditional on technology, as predicted in Section 2.5—and the response is jointly significant (sup-t joint $p = 0.018$ over $h = 0$–8). The oil and demand panels are honest nulls: neither shock moves consumption inequality significantly in either direction (joint $p = 0.66$ and 0.95). Neither null should be read as support for the model; a null is at best uninformative, never confirmatory. For the demand shock the model's level prediction is a fall in consumption inequality: an expansionary demand shock opens a positive output gap, and inequality moves against the gap ($\hat{h}_t = -\chi_y \tilde{y}_t$ with $\chi_y = 0.475$; Section 2.5). The elasticity gap barely changes this response, so the incremental TE effect is approximately zero (Section 3); the null is too imprecise to detect the predicted fall, so the result is uninformative rather than evidence for or against the model. For the oil shock the model predicts a rise, and the null is likewise uninformative: the consumption-inequality sample is short ($N = 110$), and the bands are wide. Broer et al. (2025) detect distributional effects of the same oil shock only in monthly administrative pre-tax earnings, using roughly 200 million person-month observations to identify an effect on earnings growth of about two percentage points over two years; against that benchmark, an undetectable response in a quarterly, survey-based consumption ratio is expected rather than puzzling.

FIGURE B.4: Consumption Inequality after Three Identified Shocks



Notes: Cumulative responses of the Coibion et al. (2017) consumption P90/P10 ratio (BLS Consumer Expenditure Survey, quarterly), in percent per +1 SD shock; same lag-augmented design as the markup projections. Estimation samples: 1981Q4–2008Q4, $N = 109$ (TFP, monetary); 1981Q3–2008Q4, $N = 110$ (oil; AIC selects one fewer lag). Left: TFP shock (+1 SD expansionary). Middle: oil supply news (+1 SD contractionary). Right: Romer and Romer (2004) monetary shock (+1 SD expansionary). Colored bands: pointwise 90% intervals; grey bands: 90% sup-t simultaneous bands. Joint p-values (sup-t, $h = 0$–8): TFP 0.018, oil 0.66, monetary 0.95.

B.3 State-Dependent Response: US Time Series

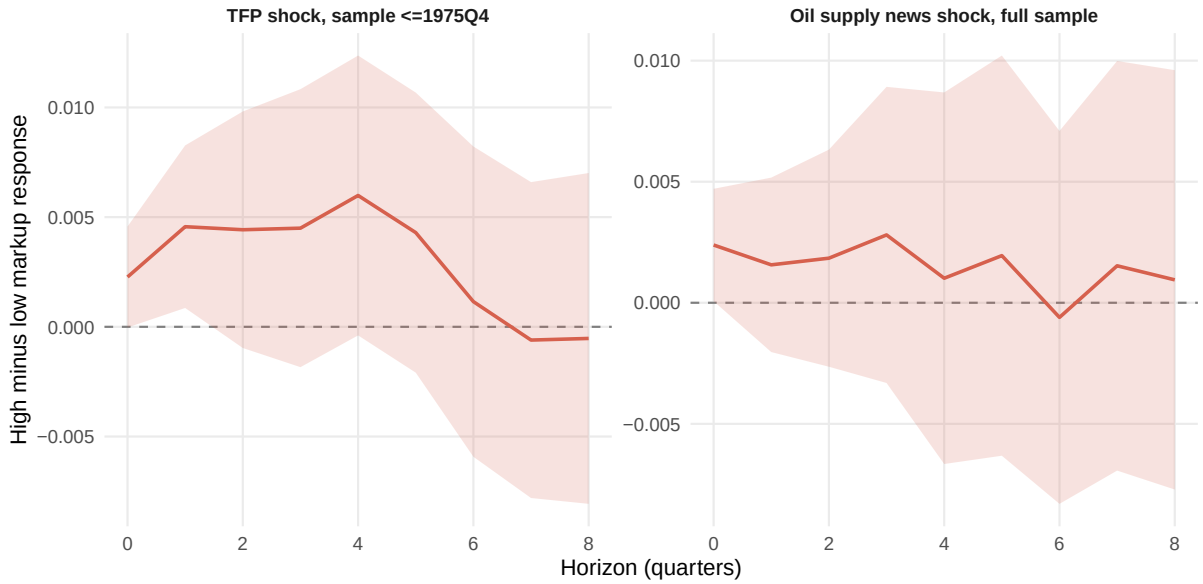
To test whether the markup response is amplified under high inequality, we augment equation (18) with an indicator for high-inequality periods. Let $D^H_{t-1} = \mathbf{1}\{\widetilde{H}_{t-1} > 0\}$, where $\widetilde{H}_t$ is the *linearly detrended* WID pre-tax top-10% income share (detrended because the raw share is nearly collinear with time, correlation 0.98, so a raw-level split would conflate "high inequality" with "late sample"), lagged one quarter to keep the state predetermined relative to the shock (Jorda and Taylor, 2025). The state-dependent specification is

$$y_{t+h} - y_{t-1} = a_h + \beta^L_h \varepsilon^{\text{TFP}}_t + \Delta_h \left(D^H_{t-1} \cdot \varepsilon^{\text{TFP}}_t\right) + \text{controls} + \sum_k \rho_{k,h} \left(D^H_{t-1} \cdot X_{k,t}\right) + u_{t+h}, \tag{B.1}$$

where β^L_h is the response in low-inequality periods and $\Delta_h = \beta^H_h - \beta^L_h$ is the differential. Coefficients are reported with lag-augmented EHW (HC3) standard errors, and we assess significance with a horizon-stacked joint block-bootstrap test and a sup-t simultaneous confidence band (following Jorda and Taylor 2025, Section 7). On the full sample ($N_{\text{high}} = 81$, $N_{\text{low}} = 110$), the differential is positive at every horizon and across all five markup measures—directionally consistent with the demand-composition prediction—but imprecisely estimated: the headline (inverse-labor-share) measure gives $\Delta_4 = +0.0021$ (HC3 s.e. 0.0031); the joint bootstrap p-value for $H_0\colon \Delta_h = 0$ across all h is 0.33, and the design is underpowered at this sample size.

A pre-1975/oil-shock variant. Restricting the technology shock to the pre-1975 sample, where its unconditional response is concentrated, and adding the oil shock as a second column sharpens the picture. Figure B.5 reports the differential Δ_h over $h \leq 8$ for the technology shock on this pre-1975 sample ($N = 101$) and for the oil shock on the full sample ($N = 165$). Both differentials are directionally positive: for the technology shock, $\Delta_4 = +0.0060$ ($t = 1.5$; joint $p = 0.13$), rising to $\Delta_4 = +0.012$ (joint $p = 0.014$) on the CES labor-margin measure; for the oil shock, $\Delta_4 = +0.0010$ ($t = 0.2$; joint $p = 0.10$). The estimates are positive but generally not conventionally significant.

FIGURE B.5: State-Dependent Markup Responses: High-Minus-Low Inequality Differential



Notes: High-minus-low differential Δ_h from the state-dependent local projection (equation (B.1)), for the inverse-labor-share markup. The state D_{t-1} equals one when the linearly detrended WID pre-tax top-10% share is above trend, lagged one quarter (predetermined); the state series is extended back to 1949 from the raw annual WID data. Left: TFP shock (+1 SD expansionary), sample through 1975Q4 ($N = 101$). Right: Känzig (2021) oil supply news (+1 SD contractionary), full sample ($N = 165$). Shaded bands are 90% pointwise intervals (EHW/HC3).

Further robustness. Three checks confirm that the imprecision of the US differential is intrinsic to the sample rather than an estimator artifact. A threshold Minnesota BVAR estimated on the same five-variable system gives a posterior median high-minus-low difference of the same sign (+0.0008 at $h = 4$) but with credible intervals roughly 50% wider than the LP's, spanning zero at every horizon; this widening is expected because the LP conditions on a pre-identified external shock while a full-system BVAR re-estimates the shock jointly with the dynamics. A smooth local projection (Barnichon and Brownlees, 2019; Li et al., 2024) pulls the point estimate down to +0.0008 at $h = 4$—roughly 40 percent of the standard-LP value—while the joint Wald p-value rises from 0.33 to 0.81; the t-statistic is stable across a four-point smoothing-penalty grid, so under-smoothing is not manufacturing the imprecision, but smoothing does not resolve it either—if anything it moves the estimate further from significance, the opposite of what smoothing does for the unconditional result above, where it sharpens the joint test. A bias-corrected LP (Herbst and Johannsen, 2024), by contrast, leaves both the point estimate and precision essentially unchanged ($\Delta_4 = +0.0022$, joint $p = 0.33$), matching the standard-LP baseline almost exactly. The cross-country counterpart—the Mean Group oil-shock design—is in Section 4.3 of the main text; Section B.4 collects its robustness.

B.4 Cross-Country Panel: Design Detail and Robustness

The headline Mean Group estimates appear in Section 4.3. The sample contains 37 OECD and other European economies through 2019Q4 and excludes the net oil exporters Norway, Mexico, and Canada. Each country has a separate local projection on the common Känzig (2021) oil shock, and the country coefficients are averaged following Pesaran and Smith (1995). The oil shock also redistributes income toward the top, so the exercise is a joint test of distribution-dependent pricing rather than a pure pass-through test.

The pooled comparison is

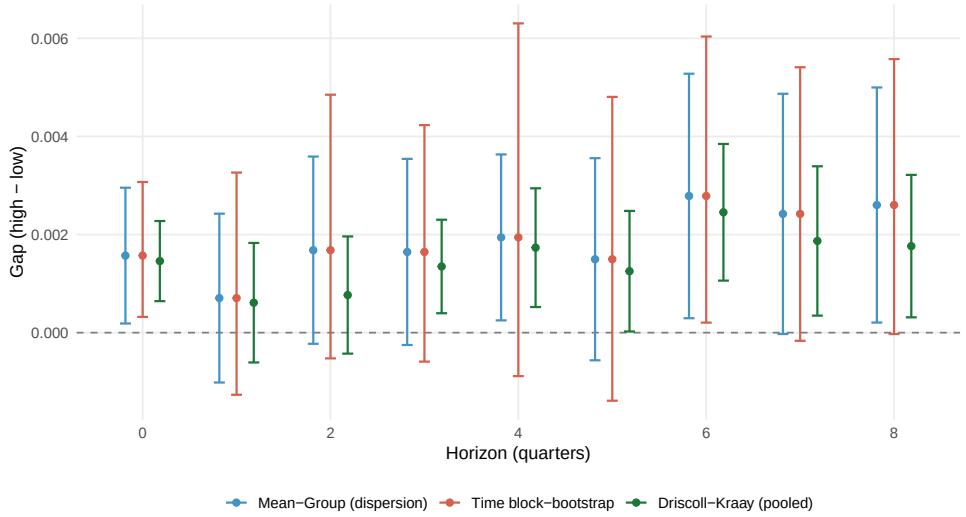
$$\mu_{i,t+h} - \mu_{i,t-1} = a_i + \beta_h s_t + \gamma_h \left(s_t \times \tilde{H}_i\right) + \sum_k \theta_{k,h} \left(s_t \times X_{k,i}\right) + \text{lags} + u_{i,t+h}, \qquad \text{(B.2)}$$

where $\tilde{H}_i$ is the demeaned country-mean top-10% income share and $X_{k,i}$ contains time-invariant country controls. Time fixed effects are omitted because they would absorb the common shock. The pooled estimator imposes a common slope. Mean Group is therefore the primary estimator, while the pooled specification supplies Driscoll–Kraay inference and confounder checks.

Common-shock inference weakens the headline gap. The moving-block bootstrap resamples the shared quarterly axis and applies the same quarters to every country ($B = 4999$; block lengths 8, 10, and 12). The gap remains positive but has $p = 0.254$ at $h = 4$ and $p = 0.103$ at $h = 8$ under the primary ten-quarter block. Driscoll–Kraay standard errors on equation (B.2) instead give $t = 2.35$ and 2.00. Figure B.6 compares the three methods. The blue and red bands share the Mean Group estimate; the green band uses the pooled implied gap, which is 31 percent smaller at $h = 8$.

The main robustness checks preserve the positive point estimate but not a uniform inference claim. The pooled interaction survives seven confounders jointly ($t = 2.21$ at $h = 4$ and 1.89 at $h = 8$), although the restricted sample and correlated traits reduce precision. Table B.2 names the seven traits and reports the horse race specification by specification, and Table B.4 lists the sample country by country. Extending the Mean Group sample through 2023Q4 gives an average markup response of $+0.0015$ at $h = 8$ and a high-minus-low gap of $+0.0020$ at $h = 4$, both significant at the 10 percent level or better. The consolidated proxy, sorter, and sample checks appear in Table B.8.

FIGURE B.6: Common-shock-robust inference on the high-minus-low gap



Notes: High-minus-low markup gap after the oil shock. Blue: Mean Group estimate with country-dispersion standard errors. Red: the same estimate with the shared-time moving-block bootstrap ($B = 4999$, block length 10). Green: the pooled implied gap from equation (B.2) with Driscoll–Kraay standard errors. Bands are 90 percent intervals.

TABLE B.2: Confounder Horse Race for the Pooled Interaction

	$h = 4$		$h = 8$	
	$\hat{\gamma}_h$ (DK se; t)	n	$\hat{\gamma}_h$ (DK se; t)	n
Baseline (no confounder controls)	0.0230** (0.0098); $t = +2.35$	3711	0.0234** (0.0117); $t = +2.00$	3563
Individual confounders (added one at a time)				
+ Imports / GDP	0.0217** (0.0093); $t = +2.34$	3711	0.0254** (0.0100); $t = +2.54$	3563
+ Capital-account openness (Chinn–Ito)	0.0230** (0.0097); $t = +2.38$	3387	0.0221* (0.0134); $t = +1.65$	3255
+ Energy intensity of production	0.0230** (0.0099); $t = +2.31$	3711	0.0238** (0.0118); $t = +2.01$	3563
+ Capital share	0.0229*** (0.0086); $t = +2.66$	3663	0.0190* (0.0101); $t = +1.87$	3519
+ Employment protection legislation (coverage 33/37)	0.0210** (0.0090); $t = +2.32$	3343	0.0191* (0.0110); $t = +1.73$	3211
+ Inflation-targeting regime share (coverage 37/37)	0.0223** (0.0093); $t = +2.39$	3711	0.0214* (0.0113); $t = +1.90$	3563
+ Net energy imports (coverage 37/37)	0.0232** (0.0100); $t = +2.33$	3711	0.0236** (0.0120); $t = +1.97$	3563
Joint specifications				
Joint — first four confounders	0.0210** (0.0090); $t = +2.34$	3387	0.0271** (0.0126); $t = +2.16$	3255
Joint — all seven confounders	0.0216** (0.0098); $t = +2.21$	3111	0.0270* (0.0143); $t = +1.89$	2991

Notes: $\hat{\gamma}_h$ is the coefficient on the interaction $s_t \times \tilde{H}_i$ in equation (B.2), with Driscoll–Kraay standard errors in parentheses and the implied t-statistic beside them. Each row of the middle block adds one time-invariant country trait, interacted with the shock, to the baseline specification. "Joint — first four confounders" adds imports/GDP, capital-account openness, energy intensity of production, and the capital share together; "Joint — all seven confounders" adds employment protection legislation, the inflation-targeting regime share, and net energy imports as well. Table B.3 opens up that seven-confounder regression. Sample size varies with trait coverage, as the n column records: imports, energy intensity, the inflation-targeting regime share, and net energy imports are available for all 37 countries; employment protection for 33; capital-account openness and the capital share for fewer than 37, the latter because the capital share is missing for one country throughout. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

TABLE B.3: Trait Loadings inside the Seven-Confounder Specification

	$h = 4$		$h = 8$	
	$\hat{\gamma}_h$ (DK se; t)	n	$\hat{\gamma}_h$ (DK se; t)	n
$s_t \times \tilde{H}_i$ (interaction of interest)	0.0216^{**} (0.0098); $t = +2.21$		0.0270^{*} (0.0143); $t = +1.89$	
$s_t \times$ Imports / GDP	-0.0002 (0.0010); $t = -0.26$		0.0014 (0.0012); $t = +1.21$	
$s_t \times$ Capital-account openness (Chinn–Ito)	0.0005 (0.0006); $t = +0.71$		0.0010 (0.0009); $t = +1.09$	
$s_t \times$ Energy intensity of production	-0.0003 (0.0006); $t = -0.42$		-0.0001 (0.0006); $t = -0.09$	
$s_t \times$ Capital share	0.0003 (0.0009); $t = +0.34$		0.0009 (0.0009); $t = +0.97$	
$s_t \times$ Employment protection legislation	0.0004 (0.0008); $t = +0.50$		0.0011^{*} (0.0006); $t = +1.84$	
$s_t \times$ Inflation-targeting regime share	-0.0009 (0.0008); $t = -1.16$		-0.0016^{**} (0.0007); $t = -2.41$	
$s_t \times$ Net energy imports	-0.0012 (0.0008); $t = -1.47$		-0.0013 (0.0011); $t = -1.19$	

Notes: Every interaction coefficient inside the joint seven-confounder regression reported in the final row of Table B.2, so the trait loadings can be read against the loading of interest. The first row is the interaction of interest and reproduces that row exactly. Driscoll–Kraay standard errors in parentheses with the implied t-statistic beside them; $n = 3111$ at $h = 4$ and $n = 2991$ at $h = 8$. Only two traits carry a non-trivial own loading, both at $h = 8$, which is what makes the interaction of interest hard to attribute to a correlated institutional characteristic. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

TABLE B.4: Cross-Country Sample: Coverage and Inequality Split

ISO3	First	Last	N	H_i (%)	Group	ISO3	First	Last	N	H_i (%)	Group
SVK	1995Q1	2019Q4	100	28.4	Low	DEU	1991Q1	2019Q4	116	34.3	High
NLD	1995Q1	2019Q4	100	28.6	Low	ESP	1995Q1	2019Q4	100	34.8	High
CZE	1995Q1	2019Q4	100	29.3	Low	MNE	2006Q1	2019Q4	56	34.9	High
SVN	1995Q1	2019Q4	100	29.3	Low	GRC	1995Q1	2019Q4	100	35.2	High
ITA	1995Q1	2019Q4	100	30.3	Low	*CAN*	1981Q1	2019Q4	156	35.3	*excl. (oil)*
DNK	1995Q1	2019Q4	100	30.5	Low	CYP	1995Q1	2019Q4	100	35.7	High
SWE	1993Q1	2019Q4	108	30.5	Low	LTU	1995Q1	2019Q4	100	35.8	High
NOR	1995Q1	2019Q4	100	30.6	*excl. (oil)*	GBR	1955Q1	2019Q4	260	36.2	High
HUN	1995Q1	2019Q4	100	30.7	Low	BGR	1995Q1	2019Q4	100	36.8	High
FIN	1990Q1	2019Q4	120	31.4	Low	LVA	1995Q1	2019Q4	100	36.8	High
MLT	1995Q1	2019Q4	100	31.8	Low	PRT	1995Q1	2019Q4	100	37.2	High
AUT	1995Q1	2019Q4	100	31.8	Low	LUX	1995Q1	2019Q4	100	37.4	High
CHE	1990Q1	2019Q4	120	32.0	Low	SRB	1995Q1	2019Q4	100	37.7	High
BEL	1995Q1	2019Q4	100	32.2	Low	EST	1995Q1	2019Q4	100	39.9	High
AUS	1959Q3	2019Q4	242	32.6	Low	ROU	1995Q1	2019Q4	100	40.3	High
FRA	1980Q1	2019Q4	160	33.1	Low	JPN	1994Q1	2019Q4	104	41.5	High
HRV	1995Q1	2019Q4	100	33.2	Low	USA	1947Q1	2019Q4	292	42.7	High
BIH	2000Q1	2019Q4	80	33.8	Low	ISR	1995Q1	2019Q4	100	50.9	High
IRL	1995Q1	2019Q4	100	34.2	Low	TUR	1995Q1	2019Q4	100	53.8	High
POL	1995Q1	2019Q4	100	34.2	Low	*MEX*	2008Q1	2019Q4	48	58.9	*excl. (oil)*

Notes: All forty countries for which the first stage delivers a coefficient, sorted by the WID top-ten-percent income share H_i. "First" and "Last" are the first and last quarters with a non-missing markup observation at or before 2019Q4; N counts those quarters. The inclusion rule is at least forty usable quarters. These counts measure panel coverage and are not the regression samples: the balanced subsample of Section 4.3 thresholds the horizon-maximum estimation sample, which is smaller than panel coverage wherever the shock series or the lag structure costs quarters, and two countries fall below eighty on that measure while only one does here. Group assigns each country to the low or high half at the median H_i of the baseline sample, giving eighteen high and nineteen low. The three net oil exporters are set in italics and marked "excl. (oil)"; they are excluded from the 37-country baseline and enter only the all-country comparison reported in Section 4.3.

B.5 Common-Shock Inference: Two Estimands, Three Standard Errors

The three inference methods apply to two estimands. Country-dispersion standard errors and the moving-block bootstrap use the Mean Group gap $\hat{\Delta}_h = \bar{\beta}_{\text{hi},h} - \bar{\beta}_{\text{lo},h}$, equal to $+0.0019$ at $h = 4$ and $+0.0026$ at $h = 8$. Driscoll–Kraay standard errors use the pooled implied gap $\hat{\gamma}_h(\bar{H}_{\text{hi}} - \bar{H}_{\text{lo}})$ from equation (B.2), equal to $+0.0017$ and $+0.0018$. The estimands differ because the first averages heterogeneous country slopes before comparing groups, while the second fits one slope to the continuous sorter.

The country-dispersion standard error treats 37 countries as independent draws even though each country receives the same shock series. The moving-block bootstrap instead resamples the common 1975Q1–2019Q4 time axis and re-estimates every country on the same resampled quarters. With $B = 4999$ and a primary block length of ten

quarters, it yields $p = 0.254$ at $h = 4$ and $p = 0.103$ at $h = 8$; block lengths of eight and twelve give similar results. Driscoll–Kraay inference treats time periods as the independent units and allows unrestricted cross-sectional dependence within each period. The time bootstrap is the most conservative method because it credits the sample with roughly the number of oil episodes rather than the number of countries or quarters.

B.6 Power and Model-Implied Magnitudes

The inference methods above say how precise the estimates are. This subsection asks the prior question: how large would the cross-country gap have to be for this design to detect it, and how large does the calibrated model say it is? We report the minimum detectable effect $\text{MDE}_h = (z_{0.975} + z_{0.80})\,\hat{\sigma}_h = 2.80\,\hat{\sigma}_h$, the conventional threshold at a two-sided five percent size and eighty percent power. We take $\hat{\sigma}_h$ from the moving-block bootstrap rather than from country dispersion, because the bootstrap is the primary inference of Section 4.3 and the most conservative of the three methods.

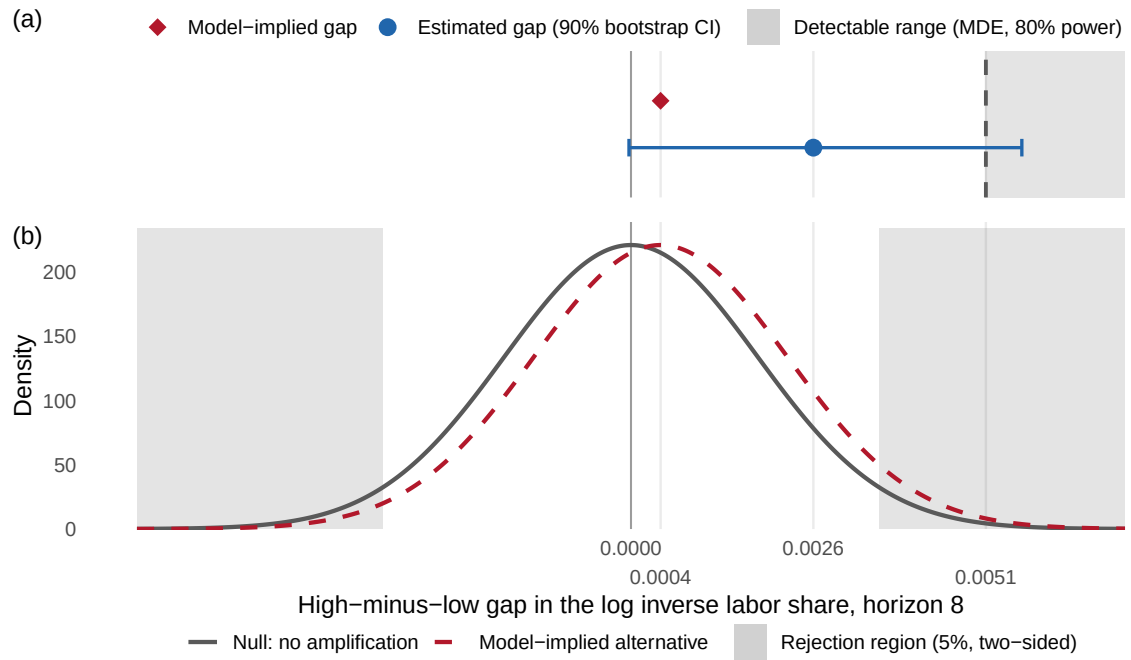
The model-implied gap is $\Delta_h^{\text{model}} = A(g_{xc})\,\hat{\theta}_h$, where $\hat{\theta}_h$ is the estimated unconditional Mean Group response and $A(g_{xc})$ is the model's amplification of the cost-push markup response between two economies whose steady-state inequality differs by the observed cross-country sorter gap. Scaling the model ratio by the *estimated* unconditional response makes the comparison immune to proportional measurement attenuation in the labor-share proxy, which affects the gap and the average alike. At the calibrated fiscal branch, $A = 0.194$ at the observed gap of 21.4 percent. Two caveats attach. First, that gap exceeds the largest gap the model's fiscal block can generate, 16.8 percent, so A is a linear extrapolation off the top of the model grid; a quadratic fit changes it in the fourth decimal. Second, mapping the top-ten-percent income share into the model's inequality index assumes a unit elasticity between the two, and Appendix C.4 is explicit that income and consumption dispersion are not interchangeable. The exhibit is therefore an illustrative bound rather than a derivation. Its conclusion is what survives the mapping. A quadratic rather than linear extrapolation gives $A = 0.195$. Refusing to extrapolate at all—taking A at the largest gap the fiscal block can generate—gives 0.152, and with it a predicted gap of 0.000331, below the detectable one by a factor of 15.3. The verdict does not depend on how A is carried from the model grid to the observed spread.

The sign of Δ_h^{model} at each horizon is inherited from the data, not from the model: A is a positive number, so $\Delta_h^{\text{model}} = A\,\hat{\theta}_h$ carries whatever sign the estimated response $\hat{\theta}_h$ carries. The model's own sign prediction does not vary with the horizon: along the admissible fiscal branch the impact markup response to a cost-push shock is positive (Corollary 4), and the response decays geometrically at the shock's persistence $\rho_u = 0.9$

without crossing zero, so the model predicts a positive gap at every horizon. Dividing the mapped gap by A recovers the estimate that anchors it: $0.000423/0.194 = 0.0022$ at $h = 8$, the Mean Group response of Section 4.3, and $-0.000118/0.194 = -0.0006$ at $h = 4$, where that response is slightly negative. A negative Δ_4^{model} is therefore a property of the rescaling convention, not a prediction of the model.

Table B.5 reports the comparison. At $h = 8$ the design detects gaps of 0.00506 and the model predicts 0.000423, a factor of 12.0. The power of the existing test against the model's own prediction is 5.6 percent, against a nominal size of five percent. Reaching eighty percent power would require shrinking the standard error by a factor of 12.0, hence an effective sample about 144 times larger—roughly 6500 years of quarterly data at the current cross-section. Adding countries does not substitute: under a common shock the effective sample is the time dimension, which the bootstrap credits with about 18 independent blocks at the primary block length. At $h = 4$ the estimated unconditional response is negative, so the mapped gap Δ_4^{model} is negative as well, about one-fiftieth of the detectable magnitude in absolute value; the model's own predicted sign is positive there, as at every horizon.

FIGURE B.7: What the Cross-Country Design Could Have Detected



Notes: All quantities are high-minus-low gaps in the log inverse labor share at horizon eight, the units of Figure 3. Panel (a): the estimated gap $\hat{\Delta}_8 = 0.0026$ with its 90% moving-block bootstrap interval $[-0.00003,\ 0.00558]$, the model-implied gap $\Delta_8^{\text{model}} = 0.000423$, and the detectable range, which begins at the minimum detectable effect $\text{MDE}_8 = 0.00506$ (Table B.5). Panel (b): the sampling distribution of $\hat{\Delta}_8$ under no amplification and under the model's prediction, both with the bootstrap standard deviation 0.0018, with the two-sided five percent rejection region shaded; the probability of rejecting under the model's own prediction is 5.6 percent against a nominal size of five percent.

TABLE B.5: Detectable Effects and Model-Implied Gaps: Cross-Country Design

	$\hat{\Delta}_h$	Bootstrap s.d.	MDE_h	Δ_h^{model}	$\Delta_h^{\text{model}}/\text{MDE}_h$	Power
h=4	0.0019	0.0022	0.00611	−0.000118	−0.019	0.050
h=8	0.0026	0.0018	0.00506	0.000423	0.083	0.056

Notes: $\hat{\Delta}_h$ is the Mean Group high-minus-low gap in the log inverse labor share after the oil supply-news shock; the bootstrap standard deviation is from the moving-block time bootstrap with $B = 4999$ and a block length of 10 quarters. $\text{MDE}_h = 2.80\,\hat{\sigma}_h$ is the smallest gap detectable at a two-sided five percent size with eighty percent power. Δ_h^{model} is the calibrated model's predicted gap at the observed cross-country inequality spread. Power is that of the two-sided five percent test evaluated at Δ_h^{model}; it is bounded below by the size of the test, so values near 0.05 mean the test carries essentially no information about the model's prediction.

The same calculation for the US state-dependent design is even more lopsided. Table B.6 compares the estimated differential with its detectable threshold and with the model-implied differential at the empirical regime gap g^* and at a counterfactual ten percent gap. The predicted differential is at least two orders of magnitude below the MDE at both horizons. No refinement of the US design—a longer window, a better markup proxy, a one-sided rather than two-sided detrending of the inequality state—closes a gap of that size. The US time series is not a marginal test of the mechanism's magnitude; it is not a test of it at all.

TABLE B.6: Detectable Effects and Model-Implied Differentials: US State-Dependent Design

	$\hat{\Delta}_h$	(s.e.)	90% CI	MDE_h	Model Δ_h (g^*)	Model Δ_h (g=10%)
h=4	0.0021	(0.0031)	[−0.0030, 0.0071]	0.0086	0.000025	0.000070
h=8	0.0011	(0.0038)	[−0.0051, 0.0073]	0.0105	0.000004	0.000011

Notes: $\hat{\Delta}_h$ is the high-minus-low differential in the markup response to the TFP shock across inequality regimes, from equation (B.1), with heteroskedasticity-robust standard errors. MDE_h uses the same convention as Table B.5. The model columns evaluate the calibrated differential at the empirical regime gap g^* and at a counterfactual gap of ten percent.

B.7 Structural Sorters: A Pre-Specified Second Stage

The pre-specified structural-sorter test is a null. The second-stage specification—sample rule, trait definitions, primary horizons, estimator designations, block-bootstrap procedure, and multiplicity families—was frozen on 5 August 2026, one day before the second stage itself was re-run; the first-stage Mean Group coefficients that this second stage regresses were already computed and stored by 8 July 2026, so the freeze fixed how those coefficients would be analyzed rather than what they showed.

The second stage regresses each country's oil-shock coefficient $\hat{\beta}_{i,h}$ on five standardized traits: the steady-state markup, weak redistribution, their product, an ordinal hand-to-mouth ranking, and the top-10% income share as an unsigned comparison. The primary horizons are $h = 4$ and $h = 8$. The four structural traits carry pre-specified positive signs; the top-10% income share is an unsigned comparison sorter. The primary sample contains up to 40 countries.

Inference re-estimates both stages in every shared-time block-bootstrap draw ($B = 4999$, primary block length 10 quarters) and applies Holm adjustment separately to the primary and net-exporter-exclusion families. No cell survives the adjustment: the smallest adjusted p-value in Table B.7 is 0.6976. The markup, redistribution, and composite slopes have the predicted sign at both horizons but are imprecise. The ordinal hand-to-mouth trait is wrong-signed. The structural ranking is therefore not supported by the second stage.

TABLE B.7: The Structural-Sorter Second Stage: Pre-Specified Traits and Bootstrap Inference

		Primary (all 40)		X1 (excl. NOR/MEX/CAN)		
Trait	Statistic	$\hat{\theta}$	N	$\hat{\theta}$	N	Sign
Panel A: primary horizon $h = 4$						
S_M	OLS slope	$+0.000777$ $p = 0.4066$, $p_H = 1.0000$	39	$+0.000645$ $p = 0.4622$, $p_H = 1.0000$	36	✓
S_R	OLS slope	$+0.000746$ $p = 0.4586$, $p_H = 1.0000$	36	$+0.001328$ $p = 0.2242$, $p_H = 1.0000$	33	✓
S_Θ	OLS slope	$+0.000717$ $p = 0.5048$, $p_H = 1.0000$	36	$+0.000917$ $p = 0.4180$, $p_H = 1.0000$	33	✓
S_λ	Spearman ρ	-0.183 $p = 0.2528$, $p_H = 1.0000$	18	-0.183 $p = 0.2528$, $p_H =$ n/a (X1-inv.)	18	×
S_H	OLS slope	$+0.000811$ $p = 0.3304$, $p_H = 1.0000$	40	$+0.001303$ $p = 0.0880$, $p_H = 0.6976$	37	n/a
Panel B: primary horizon $h = 8$						
S_M	OLS slope	$+0.000243$ $p = 0.7834$, $p_H = 1.0000$	39	$+0.001016$ $p = 0.2564$, $p_H = 1.0000$	36	✓
S_R	OLS slope	$+0.000346$ $p = 0.7364$, $p_H = 1.0000$	36	$+0.001629$ $p = 0.1624$, $p_H = 0.9744$	33	✓
S_Θ	OLS slope	$+0.000136$ $p = 0.8982$, $p_H = 1.0000$	36	$+0.001440$ $p = 0.2128$, $p_H = 1.0000$	33	✓
S_λ	Spearman ρ	-0.084 $p = 0.5666$, $p_H = 1.0000$	18	-0.084 $p = 0.5666$, $p_H =$ n/a (X1-inv.)	18	×
S_H	OLS slope	$+0.000344$ $p = 0.6938$, $p_H = 1.0000$	40	$+0.001461$ $p = 0.0872$, $p_H = 0.6976$	37	n/a

Notes: Second-stage coefficients at the two pre-specified horizons. Cardinal traits use OLS slopes on standardized values; the ordinal hand-to-mouth trait uses Spearman's ρ. Bootstrap p-values re-estimate both stages on the shared resampled quarters ($B = 4999$, block length 10). p_H applies Holm adjustment within the primary and net-exporter-exclusion families. S_R rises when redistribution falls, and $S_\Theta = S_M S_R$. No adjusted test is significant.

B.8 Cross-Country Robustness

The robustness results separate a stable sign from unstable magnitude and inference. Table B.8 reports the baseline gap of $+0.0019$ at $h = 4$ and $+0.0026$ at $h = 8$, then varies the instrument, markup proxy, inequality sorter, and sample. The alternative oil instrument preserves the positive sign but removes significance. The GOS proxy gives $+0.0031$ and $+0.0023$, while the annual production-approach proxy is an imprecise null. Top-income and top-wealth sorters give positive gaps, but the market Gini reverses the sign. Every single-country omission preserves the positive gap, whereas the simultaneous band contains zero.

TABLE B.8: Cross-Country Robustness of the High-Minus-Low Gap

	Gap (high−low)		N
	$h = 4$	$h = 8$	
Panel A: identifying shock and instrument			
Känzig (2021), headline (≤2019Q4)	0.0019* (0.0010)	0.0026* (0.0015)	37
Kilian (2024) m12, matched window (1989Q2–2017Q2)	0.0007 (0.0014)	0.0011 (0.0016)	37
US TFP SVAR shock (secondary)	−0.0010 (0.0019)	−0.0009 (0.0029)	39
Panel B: markup proxy			
GOS profit-share proxy (μ_{gos})	0.0031*** (0.0009)	0.0023** (0.0011)	36
Baseline proxy, restricted to μ_{gos} country-quarters	0.0020* (0.0011)	0.0028* (0.0015)	36
DLEU annual production approach (annual frequency)	0.0110 (0.0099) at $h = 4$ years		19
Panel C: inequality sorter			
WID top-10% income share (baseline)	0.0019* (0.0010)	0.0026* (0.0015)	37
WID top-10% wealth share	0.0010 (0.0011)	0.0030** (0.0014)	37
Capital share	0.0007 (0.0011)	0.0009 (0.0015)	36
SWIID market Gini	−0.0016 (0.0011)	−0.0017 (0.0015)	37
SWIID disposable Gini	0.0004 (0.0011)	0.0013 (0.0015)	37
Panel D: sample composition and inference			
Leave-one-out range (min–max across 37 single-country drops)	[+0.0015, +0.0023]	[+0.0018, +0.0030]	37
Sup-t 90% simultaneous band (country bootstrap, $B = 2000$)	[−0.0004, +0.0042]	[−0.0006, +0.0059]	37
Balanced subsample (≥80 usable quarters)	0.0019* (0.0010)	0.0029** (0.0014)	35
Excl. low labor-share countries (mean labor share <0.35)	0.0010 (0.0011)	0.0024 (0.0015)	34

Notes: High-minus-low Mean Group gap in the cumulative markup response at quarterly horizons, with country-dispersion standard errors in parentheses. The baseline sorter is the WID top-10% income share and the baseline sample has 37 countries. The production-approach row uses annual data. Leave-one-out and sup-t rows report ranges or bands and carry no stars. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Summary: Signs by Shock and the Overall Reading

Table B.9 collects the qualitative record: the model-predicted and estimated signs of the markup and consumption-inequality responses, shock by shock. Conditional on the technology shock, both signs match: markups rise and consumption inequality rises, as the sticky-price mechanism and the cyclicality relation in Section 2.5 predict, though the inequality response is estimated on a short sample and the US markup response is concentrated pre-1975. Conditional on the demand shock, the markup sign disagrees—markups rise where the sticky-price core predicts a fall—while the inequality response is an uninformative null: the model's level prediction is a fall, with a negligible incremental TE effect, and the estimate is too imprecise to detect the predicted fall. Conditional on the oil shock, the markup sign matches, but the test of the TE amplification prediction in the cross-country gap is inconclusive—the gap is positive throughout, robust to the second factor-share-based proxy and to dropping any single country, but pointwise rather than simultaneously significant, dependent on the pre-tax top-income sorter, and not corroborated by the production-approach markup—while the flat inequality response neither supports nor contradicts the predicted rise on the available sample.

The prediction that the markup rises after a TFP shock is robustly confirmed (joint bootstrap $p = 0.041$; sup-t band excludes zero at the impact horizon), on a cost-margin measure that does not reduce to the labor-share identity, and it strengthens rather than weakens when hours are controlled for; an energy-inclusive marginal-cost markup confirms the oil-shock rise is not the mechanical labor-share/energy wedge either. The inequality-amplification prediction is directionally consistent in both the US state-dependent test and the cross-country oil-shock design ($\Delta_h > 0$ throughout): the US differential is statistically imprecise (joint $p = 0.33$), while the cross-country gap is significant at the 10 percent level at the headline horizons, at the 5 percent level in the balanced subsample, sharper under the GOS profit-share proxy and under Driscoll–Kraay inference, and positive under every single-country omission. The evidence is fragile in stated ways: the gap is pointwise but not simultaneously significant, it weakens at $L = 1$ and under a conservative time bootstrap, and it requires a top-concentration sorter rather than the broad SWIID Gini; the annual production-approach proxy is additionally uninformative on the gap and reverses the sign of the unconditional response. The US-TFP cross-country amplification split is wrong-signed and not informative. The sorter battery, the leave-one-out exercise, the alternative markup proxies and the US-TFP split behind these qualifications are collected in Table B.8. We therefore read the amplification test as inconclusive: the point estimates are directionally positive throughout, concentrated in the cleanest identification—the common global oil shock—and survive a confounder horse race and an extension through the post-2021

surge, but the statistical confirmation is marginal, sorter-dependent, and not corroborated across markup concepts.

TABLE B.9: Model Predictions versus Empirical Estimates: Signs by Shock

	Model prediction		Empirics		Aligned?	
	Markup	Inequality	Markup	Inequality	Markup	Inequality
TFP (expansionary)						
TANK	$\uparrow$	$\uparrow$	$\uparrow^{a}$	$\uparrow^{d}$	consistent	consistent^{d}
TE-TANK	$\Uparrow$	$\Uparrow$	+, imprecise^{b}	—	tested–inconcl.^{b}	not sep. tested
Demand (expansionary)^{g}						
TANK	$\downarrow$	$\downarrow$	$\uparrow^{c}$	0^{d}	inconsistent	tested–inconcl.^{d}
TE-TANK	≈ 0	≈ 0	—	—^{i}	not identified^{h}	not sep. tested^{i}
Cost-push (oil)						
TANK	$\uparrow$	$\uparrow$	$\uparrow^{e}$	0^{d}	consistent^{e}	tested–inconcl.^{d}
TE-TANK	$\Uparrow$	$\Uparrow$	+, fragile^{f}	—	tested–inconcl.^{f}	not sep. tested

Notes: $\uparrow$ / $\downarrow$ gives the response sign; $\Uparrow$ / $\Downarrow$ adds amplification relative to homogeneous-elasticity TANK; "—" means no dedicated estimate. "Consistent" and "inconsistent" require an informative estimate with the predicted or opposite sign. "Tested–inconcl." denotes an imprecise or design-sensitive estimate, "not sep. tested" no estimate of the incremental TE effect, and "not identified" a design that cannot isolate that effect. [a] The US TFP result is concentrated before 1975. [b] The US amplification estimate is positive but imprecise; the cross-country TFP split is wrong-signed. [c] The demand-shock markup rises, against the sticky-price prediction. [d] Consumption inequality is the CEX P90/P10 ratio; joint p-values are 0.018 (TFP), 0.66 (oil), and 0.95 (demand). [e] An energy-inclusive marginal-cost markup also rises after the oil shock. [f] The oil-shock markup gap is positive but not simultaneously significant; the time bootstrap gives $p = 0.254$ at $h = 4$ and 0.103 at $h = 8$, and the result depends on the sorter and markup proxy. [g] The markup and inequality outcomes use different monetary shocks. [h] The design cannot separate κ_h from κ_y. [i] A level null cannot identify the incremental TE effect.

C Data, Calibration, and Compact Benchmarks

This appendix combines the markup-estimation method, the data dictionary for the US time-series data, and the full model calibration. Series-construction detail beyond what is needed to interpret the tables below (raw source files, vintage handling, and code) is documented in the replication materials that accompany the paper. Variable definitions and sources are collected in Table C.1.

C.1 Data Dictionary: US Time Series

Table C.1 presents summary statistics for the quarterly US series (1960–2023). Of the series listed, the pre-tax top-10% income share is the inequality measure used in the sorter batteries of Appendix B; the inflation and output-gap series document the US time-series sample.

TABLE C.1: Summary Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max	Source
Dependent Variables						
Core PCE Inflation (%)	256	3.248	2.126	0.700	10.100	FRED
Headline PCE Inflation (%)	256	6.656	3.547	-14.281	29.947	FRED
Control Variables						
Output Gap (%)	256	-0.421	2.211	-9.024	5.684	FRED
Pre-tax Top 10% Share	256	0.395	0.044	0.336	0.469	Piketty et al. (2018)
SD Log Weekly Earnings (CPS ORG)	117	0.758	0.029	0.695	0.816	CPS ORG
BPT Earnings Inequality (CPS ORG)	117	0.586	0.019	0.549	0.621	CPS ORG

Notes: This table presents summary statistics for the quarterly US data. The sample period is 1960Q1–2023Q4 for FRED and WID series; 1994Q1–2023Q1 for CPS ORG series. Inflation and the output gap are expressed in percentage points. Income shares are expressed as fractions.

C.2 Markup Estimation: Production Approach

Markups in Figure A.1 are estimated using the production approach of De Loecker et al. (2020), which derives firm-level markups from cost-minimization behavior:

$$\mu_{it} = \theta_{it}^{v} \times \frac{P_{it}Q_{it}}{P_{it}^{V}V_{it}}, \tag{C.1}$$

where $\theta_{it}^{v} \equiv \frac{\partial Q(\cdot)}{\partial V_{it}} \frac{V_{it}}{Q_{it}}$ is the output elasticity of the variable input V_{it}, estimated from production data; $P_{it}Q_{it}$ is revenue and $P_{it}^{V}V_{it}$ is variable input cost. Under perfect competition with constant returns, $\mu_{it} = 1$. We estimate θ_{it}^{v} with the proxy-variable approach of De Ridder et al. (2024), building on Ackerberg et al. (2015), using intermediate inputs as the productivity proxy. This is the standard approach in the literature but is not uncontroversial: known concerns include sensitivity to functional-form assumptions and to the treatment of fixed costs in the cost of goods sold (Traina, 2018; Syverson, 2019). The patterns we document hold across the alternative markup measures summarized in Appendix B.1, with the production-approach markup the documented exception (Appendix B.4).

C.3 Calibration

The common parameters follow standard New Keynesian values: $\beta = 0.995$, $\sigma = \varphi = 1$, $\alpha = 1/3$, $\zeta = 105.63$, and a determinacy-consistent Taylor rule (Debortoli and Galí, 2018, 2024). The parameters that differ across TE-TANK, TANK, and RANK are the hand-to-mouth share $\lambda = 0.40$, the dividend and transfer parameters δ and τ, and the

demand elasticities $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ from Auer et al. (2024). Tables C.2 and C.3 report the two groups.

TABLE C.2: Common Model Parameters

Description	Symbol	Value
Discount factor	β	0.995
Risk aversion	σ	1.0
Inverse Frisch elasticity	φ	1.0
Capital share	α	1/3
Price adjustment cost	ζ	105.63
Taylor rule on inflation	ϕ_π	1.5
Taylor rule on output	ϕ_y	0.125
AR(1) preference shock	ρ_z	0.5
AR(1) technology shock	ρ_a	0.9
AR(1) monetary shock	ρ_ν	0.5
Cost-push persistence	ρ_u	0.9

TABLE C.3: Model Calibration Differences

Description	Symbol	TE-TANK	TANK	RANK
Share of hand-to-mouth	λ	0.40	0.40	0.00
Dividend tax	δ	1.00	1.00	1.00
Transfer share	τ	0.70	0.70	0.00
Rich elasticity	ϵ_R	3.00	3.933	3.933
Poor elasticity	ϵ_P	6.60	3.933	3.933

Steady states. Rich households consume about 23 percent more than the average household in both TANK specifications, where $\bar{H} = 1.2347$. The markup is $\bar{\mathcal{M}} = 1.3409$ in all three economies. The composite Phillips-curve slope is 0.0671 in TE-TANK, 0.0701 in TANK, and 0.0833 in RANK. The homogeneous economies use $\varepsilon^* = 3.933$, which reproduces TE-TANK's markup. At this benchmark, TANK and TE-TANK differ only in the inequality loading κ_h within the analytical first-order policy block. Table C.4 reports the full steady states.

TABLE C.4: Steady-State Values

Variable	Symbol	TE-TANK	TANK	RANK
Heterogeneity Index	H	1.2347	1.2347	1.0000
Markup	M	1.3409	1.3409	1.3409
Output	Y	0.7384	0.7384	0.7922
Marginal Cost	MC	0.7458	0.7458	0.7458
NKPC Slope wrt Output	κ_y	0.0833	0.0833	0.0833
NKPC Slope wrt Inequality	κ_h	0.0342	0.0278	0.0278
DIS GE-term	χ_y	0.4746	0.4746	0.0000
Composite NKPC Slope	κ	0.0671	0.0701	0.0833

C.4 Disciplining the Distributional Calibration

The distributional block uses $\lambda = 0.40$, at the upper end of evidence that places the hand-to-mouth share near one-third (Kaplan et al., 2014), and $R \equiv \delta(1-\tau) = 0.30$, following the literature calibration. The demand elasticities $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ come from Auer et al. (2024). These inputs imply $\bar{H} = 1.2347$ and a rich-to-poor consumption ratio of 1.905. The CEX consumption ratio is held out as a validation moment rather than targeted. Table C.5 records each source.

TABLE C.5: Calibration Ledger for the Distributional Block

Object	Value	Status
Hand-to-mouth share	$\lambda = 0.40$	Externally calibrated (hand-to-mouth share)
Demand elasticities	$(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	Externally calibrated (Auer et al.: gradient estimated, top-income level normalized; import-margin to variety-CES mapping assumed)
Steady-state markup	$\bar{\mathcal{M}} = 1.34$	Implied outcome, untargeted; equals the production-approach formula $(1-\alpha)/s_L$ in model data
Fiscal parameter	$R = 0.30$ ($\delta = 1$, $\tau = 0.70$)	Externally selected following the literature (CEX-implied value -0.042 infeasible)
CEX two-type ratio	2.93	Non-targeted validation; model reaches at most 2.759 (at $R = 0$), baseline value 1.905

The CEX moment exposes a limitation of the calibration. The moments in Coibion et al. (2017) imply a ratio of 2.93 at $\lambda = 0.40$ under a lognormal approximation. The model reaches at most 2.759 at the feasible boundary $R = 0$, so matching 2.93 would require $R = -0.042$. The baseline therefore understates consumption dispersion, and the fiscal block cannot close the gap. The raw CEX ratio includes durables and is not equivalized, but it would have to fall below 1.905 to reverse this conclusion.

The markup level is an implied outcome rather than a target because markup estimates measure different objects. The fixed-ratio path holds $\varepsilon_P/\varepsilon_R = 2.2$ while the implied markup spans 1.21–1.61, the range of published estimates (Edmond et al., 2023; Burstein et al., 2025). Along this path, the dispersion-pure Taylor-rule gap moves from $+2.36$ to $+3.82$ percent around the baseline $+3.11$ percent. This limited range is conditional on that path. Holding the 3.6-point elasticity gap fixed instead moves the statistic from $+1.62$ to $+5.67$ percent over the same markup range, while the elasticity ratio drifts from 1.75 to 3.01. The cost-push result is not tied to one markup level, but broader reparameterizations produce more variation. Table C.6 reports both paths and the three bounds that delimit them.

TABLE C.6: Sensitivity of the Dispersion-Pure Cost-Push Gap along Three Elasticity Paths

	$\bar{\mathcal{M}}$	ε_R	ε_P	$\varepsilon_P/\varepsilon_R$	$\Delta\varepsilon$	Gap (%)
Panel A: fixed-ratio path ($\varepsilon_P/\varepsilon_R = 2.2$ held, level varies)						
DLEU 1980 aggregate / French firm median	1.21	4.34	9.54	2.20	5.20	+2.36
EMX cost-weighted aggregate	1.25	3.78	8.31	2.20	4.53	+2.63
Baseline (implied, untargeted)*	1.34	3.00	6.60	2.20	3.60	+3.11
French firm-level mean	1.39	2.73	6.01	2.20	3.28	+3.30
DLEU 2016 aggregate / EMX sales-weighted	1.61	2.06	4.53	2.20	2.47	+3.82
Panel B: fixed-gap path ($\Delta\varepsilon = 3.6$ held, level varies)						
Low end of markup span	1.21	4.77	8.37	1.75	3.60	+1.62
High end of markup span	1.61	1.79	5.39	3.01	3.60	+5.67
Panel C: bounds						
(a) 10% gap, fixed-mean path	1.45	1.23	9.26	7.54	8.03	+10.00
(b) Ceiling at ratio 2.2	1.93	1.65	3.63	2.20	1.98	+3.99
(c) Supremum over λ, $R^\dagger$	1.31	3.00	6.60	2.20	3.60	+7.03

Notes: Each row reports the dispersion-pure cost-push gap of TE-TANK against its own markup-matched TANK(ε^*), $\Psi_\pi/\Psi_\pi^* - 1$ in percent, at the baseline $\lambda = 0.40$ and $R = 0.30$ unless stated otherwise. Panel A holds the elasticity ratio at its calibrated $\varepsilon_P/\varepsilon_R = 2.2$ and moves the level, so the implied markup varies and $\Delta\varepsilon$ adjusts with it. Panel B holds the elasticity gap at its calibrated $\Delta\varepsilon = 3.6$ and moves the level, so the ratio instead drifts from 1.75 to 3.01, away from the estimated gradient. The markup span 1.21–1.61 common to both panels is the range of published aggregate estimates (Edmond et al., 2023; Burstein et al., 2025). The starred row ($*$) is the baseline calibration $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$, whose markup is an untargeted outcome; all three paths intersect there, so it is common to both panels and to row (a). Panel C reports three bounds. Row (a) holds the *mean* elasticity at its calibrated $\bar{\varepsilon} = 4.44$ and varies $\Delta\varepsilon$ alone: it is a third path, and the parameterization it reports lies on neither the fixed-ratio path of Panel A nor the fixed-gap path of Panel B. Row (b) is the maximum over all elasticity levels at the fixed ratio 2.2, and it is attained at $\bar{\mathcal{M}} = 1.93$, above the published markup span; within that span the largest attainable value is the +3.82 of the last row of Panel A. The daggered row ($\dagger$) maximizes over the distributional parameters at the baseline elasticities: the gap is strictly decreasing in R, so $R = 0.0$ is attained, but it is strictly increasing in λ and is bounded only by the strict admissibility constraint $\chi_y < 1$. The entry is therefore a supremum, approached as $\lambda \to 0.656$ and $\chi_y \to 1.000$ and attained at no admissible point; every admissible λ delivers strictly less than +7.03. At $\alpha = 1/3$ and $\sigma = \varphi = 1$ that boundary is exact in rationals: $\chi_y = 1$ forces $\Gamma = 1/2$, hence $(1-\lambda)\bar{H} = 2/3$ independently of λ, so $\varepsilon_{\text{eff}} = 21/5$, $\bar{\mathcal{M}} = 21/16 = 1.3125$, and $\lambda^* = 21/32 = 0.65625$; the numerical root agrees with the closed form to 2.2×10^{-16}. Figure D.2 maps the same statistic over the full $(\bar{\varepsilon}, \Delta\varepsilon)$ plane, with the paths of Panels A and B drawn on it.

The model-matched distributional object is consumption dispersion. The top-10% income share is used only as the cross-country sorter in Section 4.3, because it is widely available. The top-10% wealth share is a robustness sorter. These objects are not interchangeable and the income or wealth sorters do not discipline the consumption block.

C.5 Homogeneous-Elasticity Benchmarks

The primary homogeneous benchmark is TANK($\varepsilon^* = 3.933$), whose steady-state markup equals TE-TANK's $\bar{\mathcal{M}} = 1.3409$. The two economies then share $\bar{H}$, $\bar{\mathcal{M}}$, Γ,

and χ_y and differ only in κ_h within the first-order policy block. The Taylor-rule pass-through and output-contraction ratios are both 1.031, so the dispersion-pure gap is 3.1 percent. Table C.7 reports this comparison and three alternative homogeneous conventions.

The population-mean-preserving benchmark $\bar{\varepsilon} = 4.44$ also includes the markup-level effect of introducing elasticity dispersion. Relative to this benchmark, pass-through rises by 15.74 percent and the output contraction by 14.53 percent. The lower panel separates each total into the dispersion-pure component through ε^* and the remaining markup-level component.

TABLE C.7: Benchmark Dependence of the TE-TANK/TANK Comparison

	κ	χ_y	TE-TANK / TANK ratio	
TANK benchmark $\bar{\varepsilon}$			Pass-through	Output
3.933 (ε^*, markup-matched)*	0.0701	0.4746	1.031	1.031
4.44 (population-mean-preserving)	0.0816	0.4930	1.157	1.145
4.00 (standard TANK)	0.0716	0.4773	1.048	1.046
4.80 (pair mean)	0.0898	0.5038	1.247	1.227
TE-TANK $(\varepsilon_R, \varepsilon_P)=(3, 6.6)$	0.0671	0.4746	1.000	1.000
RANK (ε^*, ref.)	0.0833	0.0000	–	–

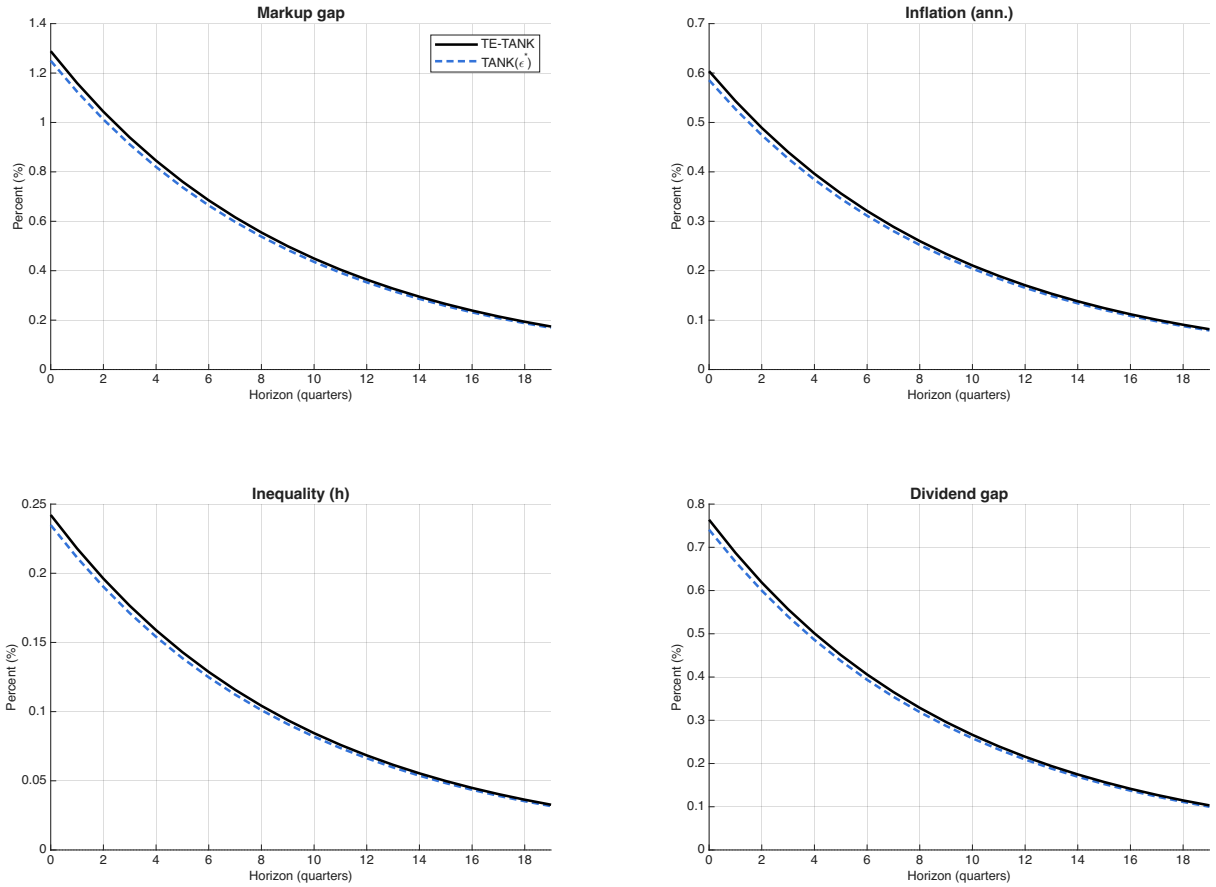
Decomposition of the TE-TANK/TANK(4.44) gap: dispersion-pure × markup-level

Object	Total TE / TANK(4.44)	Dispersion-pure TE / interm.	Markup-level interm. / TANK(4.44)
Taylor pass-through Ψ_π	+15.74%	+3.11%	+12.25%
Output contraction $\lvert\Psi_y\rvert$	+14.53%	+3.11%	+11.08%

Notes: Each upper-panel row is a homogeneous-elasticity TANK benchmark. The starred markup-matched row is the main-text benchmark. The next rows preserve the population mean, use the standard TANK elasticity, and use the unweighted pair mean. The κ and χ_y columns report each benchmark's own coefficients; the last two columns report TE-TANK relative to that benchmark. TE-TANK and markup-matched RANK are shown for reference. The lower panel decomposes the total against the population-mean-preserving benchmark through the exactly markup-matched ε^* intermediate.

Figure C.1 shows the same markup-matched comparison as impulse responses to the cost-push shock of Section 3.2. Because the two economies differ only in κ_h, the vertical distance between the two lines is the dispersion-pure effect of the elasticity gap.

FIGURE C.1: Cost-Push Shock: TE-TANK vs. TANK(ε^*)



Notes: Impulse responses to a one-standard-deviation cost-push shock, the AR(1) wedge u_t in equation (14) ($\rho_u = 0.9$, $\sigma_u = 0.05$). The **solid black line** is TE-TANK ($\varepsilon_R = 3.0$, $\varepsilon_P = 6.6$); the **dashed blue line** is TANK($\varepsilon^* = 3.933$), the homogeneous economy at the elasticity that reproduces TE-TANK's steady-state markup. Panel layout and units are as in Section 3.2. The sequential three-band decomposition of the same response, from a RANK baseline through hand-to-mouth aggregation to the elasticity gap, is Figure 2.

D Model Derivations and Core Robustness

This appendix derives the model's equilibrium objects and proves the results referenced in Section 2. It also reports the targeted comparative statics and the transmission of the monetary policy shock.

D.1 Household Problem

This subsection states the fiscal transfer rule, the household budget constraints, and the Ricardian first-order conditions summarized in Section 2; they are the starting point for the heterogeneity-index and Euler-equation derivations that follow.

Fiscal policy. The government collects a fraction δ of firm profits and redistributes them according to:

$$\begin{aligned} T_t^R &= \left(1 + \frac{\tau\lambda}{1-\lambda}\right)\delta D_t, \\ T_t^P &= (1-\tau)\,\delta D_t, \end{aligned} \tag{D.1}$$

where $\tau \in [0,1]$ governs the tilt of transfers toward rich households ($\tau = 0$ is the most progressive scheme, equal per-capita transfers). The government budget balances each period: $(1-\lambda)T_t^R + \lambda T_t^P = \delta D_t$.

Rich households. Rich households maximize expected lifetime utility (1) subject to a budget constraint:

$$C_t^R + \frac{B_t^R}{P_t^R} + Q_t F_t^R = \frac{B_{t-1}^R(1+i_{t-1})}{P_t^R} + \frac{W_t}{P_t^R}N_t^R + \big((1-\delta)D_t + Q_t\big)F_{t-1}^R + T_t^R, \tag{D.2}$$

where B_t^R are nominal bond holdings, F_t^R are equity shares, Q_t is the share price, D_t are real dividends, W_t is the nominal wage, and T_t^R are real transfers. A fraction δ of firm profits is collected by the government and redistributed as described above.

Rich household's optimization. The rich household maximizes $\mathbb{E}_0 \sum_t \beta^t \big[\frac{(C_t^R)^{1-\sigma}}{1-\sigma} Z_t - \frac{(N_t^R)^{1+\varphi}}{1+\varphi} Z_t\big]$ subject to the budget constraint (D.2) (multiplier ξ_t), with consumption expenditure $\frac{1}{P_t^R}\int_0^1 P_t(i)C_t^R(i)\,di$, labor income $\frac{W_t}{P_t^R}N_t^R$, bond and equity holdings, and transfers T_t^R. This is a standard Lagrangian problem; the first-order conditions, after using the CES demand aggregator of Appendix D.2 and the price-index identity to simplify the consumption FOC, are the marginal utility of wealth,

$$\xi_t = (C_t^R)^{-\sigma} Z_t, \tag{D.3}$$

the intratemporal labor-supply condition,

$$\frac{W_t}{P_t^R} = (C_t^R)^{\sigma}(N_t^R)^{\varphi}, \tag{D.4}$$

the consumption Euler equation,

$$1 = \mathbb{E}_t\left[\Lambda_{t+1,t}\frac{1+i_t}{\Pi_{t+1}^R}\right], \qquad \Lambda_{t+1,t} \equiv \beta\left(\frac{C_{t+1}^R}{C_t^R}\right)^{-\sigma}\frac{Z_{t+1}}{Z_t}, \quad \Pi_{t+1}^R \equiv \frac{P_{t+1}^R}{P_t^R}, \tag{D.5}$$

and the equity Euler equation,

$$Q_t = \mathbb{E}_t \left[\Lambda_{t+1,t} \left((1-\delta) D_{t+1} + Q_{t+1} \right) \right].$$

These are the Euler equation, labor-supply condition, and equity-pricing equation referenced in Section 2.

Poor households. Poor households cannot save or borrow, so they spend all labor income plus transfers each period:

$$C_t^P = \frac{W_t}{P_t^P} N_t^P + T_t^P. \tag{D.6}$$

Throughout we impose a common equilibrium employment level, $N_t^P = N_t^R \equiv N_t$, and price labor off the Ricardian condition (D.4); equation (D.6) then prices the common N_t at W_t / P_t^P and residually determines C_t^P. Free choice of hours at the common wage would instead imply $W_t / P_t^P = (C_t^P)^\sigma (N_t^P)^\varphi$ and hence $N_t^P > N_t^R$ whenever $C_t^P < C_t^R$, so common hours rations hand-to-mouth households below their desired hours.

D.2 CES Aggregation

Each household type $h \in \{R, P\}$ minimizes expenditure $\int_0^1 P_t(i) C_t^h(i)\, di$ subject to the CES aggregator $C_t^h = \left(\int_0^1 [C_t^h(i)]^{1-1/\varepsilon_h} di \right)^{\varepsilon_h/(\varepsilon_h - 1)}$. This is the standard Dixit–Stiglitz expenditure-minimization problem (e.g., Galí, 2015, ch. 3); standard steps (first-order conditions across varieties, substitution into the aggregator, and the price-index definition $P_t^h \equiv (\int_0^1 P_t(j)^{1-\varepsilon_h} dj)^{1/(1-\varepsilon_h)}$) yield the CES demand function

$$C_t^h(i) = \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} C_t^h. \tag{D.7}$$

From variety demand to aggregate output. Type h's CES demand for variety i is (D.7), and output of variety i sold to type h satisfies $Y_t^h(i) = C_t^h(i) + X_t^h(i)$, where $X_t^h(i)$ is the Rotemberg price-adjustment cost, $C_t^h(i) = Y_t^h(i)\left(1 - \frac{\zeta}{2}(P_t(i)/P_{t-1}(i) - 1)^2\right)$, aggregating to $C_t^h = Y_t^h\left(1 - \frac{\zeta}{2}(P_t/P_{t-1} - 1)^2\right)$. Substituting and imposing the symmetric equilibrium $P_t(i) = P_t$, $P_{t-1}(i) = P_{t-1}$ cancels the adjustment-cost terms on both sides, leaving the quantity supplied of variety i to type h as $Y_t^h(i) = (P_t(i)/P_t^h)^{-\varepsilon_h} Y_t^h$. Aggregate output of variety i, summed across household types with population share s_h, is

$$Y_t(i) = \sum_{h \in \{R,P\}} s_h Y_t^h(i) = \sum_{h \in \{R,P\}} s_h \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} Y_t^h. \tag{D.8}$$

D.3 Heterogeneity Index

We derive the heterogeneity index $H_t^h = C_t^h/C_t = Y_t^h/Y_t$ for rich households. Market clearing implies that consumption equals output minus price adjustment costs for each good i:

$$C_t^h(i) = Y_t^h(i) - X_t^h(i).$$

Aggregating across all goods $i \in [0,1]$ yields

$$C_t^h = Y_t^h - X_t^h.$$

The total price adjustment cost is given by

$$X_t^h = Y_t^h \left[\frac{\zeta}{2}(\Pi_t - 1)^2\right],$$

which allows us to write consumption as

$$C_t^h = Y_t^h \left[1 - \frac{\zeta}{2}(\Pi_t - 1)^2\right] = Y_t^h \Delta^p(\Pi_t), \tag{D.9}$$

where $\Delta^p(\Pi_t) = 1 - \frac{\zeta}{2}(\Pi_t - 1)^2$ captures the resource loss from price adjustment.

Starting from the rich household's budget constraint (D.2) and applying the CES aggregator for consumption, then using that bonds are in zero net supply and rich households' equity holdings are constant at $F_t^R = 1/(1-\lambda)$ (total shares equal one and the rich population share $1-\lambda$ is constant), the budget constraint simplifies to $C_t^R = \frac{W_t}{P_t^R} N_t^R + (1-\delta) D_t F_{t-1}^R + T_t^R$. Substituting C_t^R from (D.9), normalizing by aggregate output Y_t, and using the real wage $\frac{W_t}{P_t^R} = \frac{(1-\alpha)A_t}{\mathcal{M}_t} N_t^{-\alpha}$, labor-market clearing $N_t = N_t^R = N_t^P$, the transfer rule $T_t^R = \left(1 + \frac{\tau\lambda}{1-\lambda}\right)\delta D_t$, and $Y_t = A_t N_t^{1-\alpha}$ gives, after combining the dividend terms,

$$H_t = \frac{1}{\Delta^p(\Pi_t)} \left[\frac{(1-\alpha)}{\mathcal{M}_t} + \frac{D_t}{Y_t}\left(\frac{1 + \delta\lambda(\tau - 1)}{1-\lambda}\right)\right]. \tag{D.10}$$

The aggregate accounting identity $C_t = Y_t \Delta^p(\Pi_t) = W_t N_t / P_t + D_t$ states that output net of adjustment costs equals real labor income plus dividends; together with $W_t N_t / P_t = \frac{1-\alpha}{\mathcal{M}_t} Y_t$, this gives $D_t/Y_t = \Delta^p(\Pi_t) - \frac{1-\alpha}{\mathcal{M}_t}$. Substituting into (D.10) and collecting terms yields

$$H_t = \frac{1 + \delta\lambda(\tau - 1)}{1-\lambda} - \frac{1}{\Delta^p(\Pi_t)} \frac{(1-\alpha)}{\mathcal{M}_t} \left(\frac{\lambda(1 + \delta(\tau - 1))}{1-\lambda}\right).$$

In the steady state, where $\Delta^p(\Pi) = 1$, this simplifies to

$$H = \frac{1 + \delta\lambda(\tau - 1)}{1 - \lambda} - \frac{(1 - \alpha)}{\mathcal{M}} \left(\frac{\lambda(1 + \delta(\tau - 1))}{1 - \lambda} \right).$$

Using $R = \delta(1 - \tau)$ and the composite parameters defined in Section 2.4 gives equation (7). In contrast to the standard TANK model, the markup also depends on heterogeneity:

$$\mathcal{M} = \frac{H(1 - \lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P}{H(1 - \lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}.$$

This is equation (8). Substituting it into (7) gives the quadratic equation (9), whose economically admissible root is characterized in Appendix D.9.2.

Log-Linearization of the Heterogeneity Index. Write $H_t = C_1 - C_2/[\Delta^p(\Pi_t)\mathcal{M}_t]$ with constants $C_1 = \frac{1+\delta\lambda(\tau-1)}{1-\lambda}$ and $C_2 = (1 - \alpha)\frac{\lambda(1+\delta(\tau-1))}{1-\lambda}$. Since $1/\Delta^p(\Pi_t) \approx 1$ to first order, H_t varies only through $\mathcal{M}_t$; a first-order Taylor expansion around the steady state, using $\partial H_t/\partial \mathcal{M}_t|_{ss} = C_2/\mathcal{M}^2$, gives the proportional deviation $\hat{h}_t \equiv (H_t - H)/H \approx (C_2/\mathcal{M}H)\hat{\mu}_t$, i.e.,

$$\hat{h}_t = \frac{(1 - \alpha)}{\mathcal{M}H} \left(\frac{\lambda(1 + \delta(\tau - 1))}{1 - \lambda} \right) \hat{\mu}_t \equiv \Gamma\, \hat{\mu}_t, \qquad \Gamma \equiv \frac{(1 - \alpha)}{\mathcal{M}H} \left(\frac{\lambda(1 + \delta(\tau - 1))}{1 - \lambda} \right).$$

Substituting $\hat{\mu}_t = -\sigma\hat{h}_t - \left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right) \tilde{y}_t$ from (D.21), collecting $\hat{h}_t$ on the left-hand side, and solving yields

$$\hat{h}_t = -\underbrace{\frac{\Gamma\left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right)}{1 + \Gamma\sigma}}_{\chi_y} \tilde{y}_t,$$

where χ_y is the composite elasticity of heterogeneity with respect to the output gap. The denominator $1 + \Gamma\sigma > 1$ reflects a dampening feedback: higher heterogeneity raises Ricardian consumption relative to aggregate consumption, which through the intratemporal condition raises the real wage and compresses the markup, partially offsetting the original change in heterogeneity.

Note that when $\delta = 1$ (all profits are redistributed) and $\tau = 0$ (equal per-capita transfers), the term $\lambda(1 + \delta(\tau - 1))/(1 - \lambda) = 0$, and heterogeneity becomes acyclical in the log-linearized system.

D.4 Firm Problem and Flexible-Price Markup

The firm chooses $P_t(i)$ to maximize discounted real profits $\max_{P_t(i)} \sum_{s=t}^{\infty} \frac{\Lambda_{s,t}}{P_s} \left[P_s(i) Y_s(i) - P_s \mathcal{C}(Y_s(i)) - \frac{\zeta}{2} \left(\frac{P_s(i)}{P_{s-1}(i)} - 1 \right)^2 Y_s P_s \right]$, where $\Lambda_{s,t} \equiv \beta^{s-t} U'_{c,s}/U'_{c,t}$ is the stochastic discount factor, subject to the demand constraint

$Y_t(i) = \sum_h s_h (P_t(i)/P_t^h)^{-\varepsilon_h} Y_t^h$ aggregated across household types. Differentiating with respect to $P_t(i)$—the revenue and cost terms at $s = t$, and the Rotemberg adjustment-cost term at both $s = t$ (current-period cost) and $s = t+1$ (next period's cost, since $P_t(i)$ enters $P_{t+1}(i)/P_t(i)$), each evaluated under the symmetric equilibrium $P_t(i) = P_t$—and denoting real marginal cost $MC_t \equiv \mathcal{C}'(Y_t(i))$, gives the first-order condition

$$\sum_h s_h \frac{Y_t^h}{P_t} (1-\varepsilon_h) \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} + MC_t \sum_h s_h \frac{\varepsilon_h}{P_t(i)} \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h$$
$$- \zeta (\Pi_t - 1)\Pi_t \frac{Y_t}{P_t} + \mathbb{E}_t \left[\Lambda_{t+1,t} \zeta (\Pi_{t+1} - 1)\Pi_{t+1} \frac{Y_{t+1}}{P_t}\right] = 0. \tag{D.11}$$

Imposing the symmetric equilibrium $P_t(i) = P_t^h = P_t$, dividing by $\zeta Y_t / P_t$, and defining $H_t^h \equiv Y_t^h / Y_t$ as type h's output relative to the aggregate average gives

$$\sum_h \frac{s_h}{\zeta} H_t^h (1-\varepsilon_h) + MC_t \sum_h \frac{s_h}{\zeta} \varepsilon_h H_t^h + \mathbb{E}_t \left[\Lambda_{t+1,t} (\Pi_{t+1} - 1)\Pi_{t+1} \frac{Y_{t+1}}{Y_t}\right] = (\Pi_t - 1)\Pi_t. \tag{D.12}$$

At the zero-inflation steady state ($\Pi_{t+1} = \Pi_t = \Pi = 1$) the forward-looking term and the right-hand side vanish, giving steady-state marginal cost

$$MC = \frac{\sum_h s_h H^h (\varepsilon_h - 1)}{\sum_h s_h \varepsilon_h H^h} = \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h}. \tag{D.13}$$

D.4.1 Flexible-Price Markup

Flexible prices. With $\zeta = 0$, the symmetric-equilibrium first-order condition (D.11) reduces to $\sum_h s_h (1-\varepsilon_h) H_t^h - \frac{MC_t^{nom}}{P_t} \sum_h s_h (-\varepsilon_h) H_t^h = 0$, where $MC_t^{nom} \equiv P_t MC_t$ is nominal marginal cost, so real marginal cost is $MC_t^{nom}/P_t = \sum_h s_h (\varepsilon_h - 1) H_t^h / \sum_h s_h \varepsilon_h H_t^h$, and the flexible-price markup, its inverse, is

$$\mathcal{M}_t^n \equiv \frac{P_t}{MC_t^{nom}} = \frac{\sum_h s_h \varepsilon_h H_t^h}{\sum_h s_h (\varepsilon_h - 1) \cdot H_t^h}. \tag{D.14}$$

Relating heterogeneity shares. Goods-market clearing aggregates the output sold to each type with its population share $s_h \in \{1-\lambda, \lambda\}$:

$$Y_t = \sum_h s_h Y_t^h = (1-\lambda) Y_t^R + \lambda Y_t^P. \tag{D.15}$$

Dividing through by aggregate output Y_t and using the relative demand indices $H_t^h \equiv Y_t^h / Y_t$ defined above, each type's demand measured against the economy-wide

average, turns this into the adding-up condition

$$1 = (1-\lambda)H_t^R + \lambda H_t^P. \tag{D.16}$$

Solving for the poor household's share then expresses it in terms of the rich household's:

$$H_t^P = \frac{1-(1-\lambda)H_t^R}{\lambda}. \tag{D.17}$$

Effective elasticity. The demand curve (3) aggregates two CES schedules with different exponents, so the price elasticity of total demand for variety i is not a primitive parameter but an endogenous average of ε_R and ε_P. Define it as the (negative) elasticity of $Y_t(i)$ with respect to the firm's own price. Since each CES term satisfies $\partial(P_t(i)/P_t^h)^{-\varepsilon_h}/\partial \ln P_t(i) = -\varepsilon_h(P_t(i)/P_t^h)^{-\varepsilon_h}$, differentiating $Y_t(i) = \sum_h s_h(P_t(i)/P_t^h)^{-\varepsilon_h}Y_t^h$ gives

$$-\frac{\partial \ln Y_t(i)}{\partial \ln P_t(i)} = \frac{\sum_h s_h \varepsilon_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h}{\sum_h s_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h}.$$

Evaluating at the symmetric allocation ($P_t(i) = P_t^R = P_t^P = P_t$), every price ratio equals one and the denominator becomes $\sum_h s_h Y_t^h = Y_t$ by goods-market clearing, so

$$\varepsilon_{\text{eff},t} \equiv -\left.\frac{\partial \ln Y_t(i)}{\partial \ln P_t(i)}\right|_{P_t(i)=P_t^h=P_t} = \sum_h s_h \varepsilon_h \frac{Y_t^h}{Y_t} = \sum_h s_h \varepsilon_h H_t^h = (1-\lambda)H_t\,\varepsilon_R + \lambda H_t^P \varepsilon_P,$$

where the last equality uses $s_h \in \{1-\lambda, \lambda\}$ and $H_t^R = H_t$. Each type's elasticity is therefore weighted by its demand share $s_h H_t^h = s_h Y_t^h / Y_t$—equivalently its expenditure share, since both types pay the same price in the symmetric allocation—rather than by its population share s_h alone. Substituting the market-clearing identity (D.17) eliminates H_t^P:

$$\begin{aligned}
\varepsilon_{\text{eff},t} &= (1-\lambda)H_t\,\varepsilon_R + \lambda\,\frac{1-(1-\lambda)H_t}{\lambda}\,\varepsilon_P \\
&= (1-\lambda)H_t\,\varepsilon_R + \varepsilon_P - (1-\lambda)H_t\,\varepsilon_P \\
&= H_t(1-\lambda)\,(\varepsilon_R - \varepsilon_P) + \varepsilon_P.
\end{aligned} \tag{D.18}$$

Markup in terms of the effective elasticity. The flexible-price markup (D.14) is a function of the same share-weighted sums: its numerator is $\sum_h s_h \varepsilon_h H_t^h = \varepsilon_{\text{eff},t}$, and its denominator is $\sum_h s_h(\varepsilon_h - 1)H_t^h = \varepsilon_{\text{eff},t} - \sum_h s_h H_t^h = \varepsilon_{\text{eff},t} - 1$ by the adding-up condition (D.16). Hence $\mathcal{M}_t^n = \varepsilon_{\text{eff},t}/(\varepsilon_{\text{eff},t} - 1)$, and substituting (D.18) yields the flexible-price

markup as a function of H_t alone:

$$\mathcal{M}_t^n = \frac{H_t(1-\lambda)\left(\varepsilon_R - \varepsilon_P\right) + \varepsilon_P}{H_t(1-\lambda)\left(\varepsilon_R - \varepsilon_P\right) + \varepsilon_P - 1}.$$

D.5 Steady State

From the markup definition in (D.20), derived below in Appendix D.7, evaluated at the steady state with $A = 1$, $C^R = HC$, and market clearing $C = Y$, the steady-state markup is

$$\mathcal{M} = (1-\alpha)(HY)^{-\sigma} N^{-(\alpha+\varphi)}.$$

Substituting the steady-state production function $N = Y^{1/(1-\alpha)}$ and solving for Y yields

$$Y = \left(\frac{1-\alpha}{\mathcal{M}H^{\sigma}}\right)^{\frac{1}{\frac{\alpha+\varphi}{1-\alpha}+\sigma}} = \left(\frac{1-\alpha}{\mathcal{M}H^{\sigma}}\right)^{\frac{1-\alpha}{\alpha+\varphi+\sigma(1-\alpha)}}.$$

This is equation (10) in the main text.

D.6 Dynamic IS Curve

Start from the rich household's Euler equation (D.5), substitute $C_t^R = H_t C_t$, and use that the price index is common to both household types ($\Pi_{t+1}^R = \Pi_{t+1}$):

$$1 = \mathbb{E}_t\left[\beta\left(\frac{H_{t+1}C_{t+1}}{H_t C_t}\right)^{-\sigma}\frac{Z_{t+1}}{Z_t}\frac{1+i_t}{\Pi_{t+1}}\right].$$

Taking logs and log-linearizing around the deterministic steady state (a standard first-order Taylor expansion of the Euler equation, as in the textbook derivation of the New Keynesian IS curve; Galí, 2015), using the preference-shock process $z_t = \rho_z z_{t-1}$, and expressing variables as deviations from steady state ($\hat{i}_t = i_t - \bar{i}$, $\hat{\pi}_t = \pi_t - \bar{\pi}$) gives

$$y_t = \mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\right) + \mathbb{E}_t[\Delta\hat{h}_{t+1}] + \frac{1}{\sigma}(1-\rho_z)z_t.$$

Subtracting natural output y_t^n from both sides and defining the output gap $\tilde{y}_t \equiv y_t - y_t^n$ yields the pre-substitution dynamic IS curve,

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\right) + \mathbb{E}_t[\Delta\hat{h}_{t+1}] + \frac{1}{\sigma}(1-\rho_z)z_t + \mathbb{E}_t[\Delta\hat{y}_{t+1}^n], \tag{D.19}$$

which contains $\mathbb{E}_t[\Delta\hat{h}_{t+1}]$ explicitly, as referenced in Section 2.5; substituting (11) gives the reduced form (12).

D.7 New Keynesian Phillips Curve

We start with the first-order condition (D.12) rewritten as:

$$(\Pi_t - 1)\,\Pi_t = \sum_h \frac{s_h}{\zeta} H_t^h \left(1 - \varepsilon_h + MC_t \varepsilon_h\right) + \mathbb{E}_t\left[\Lambda_{t+1,t}\,(\Pi_{t+1} - 1)\,\Pi_{t+1}\frac{Y_{t+1}}{Y_t}\right]$$

and log-linearize around the zero-inflation steady state. A standard first-order expansion (dropping products of deviations; cf. Galí, 2015, ch. 3–4) gives $(\Pi_t - 1)\Pi_t \approx \pi_t \equiv \ln \Pi_t$ on the left, collapses the forward-looking term to $\beta \mathbb{E}_t\{\pi_{t+1}\}$, and, using $H_t^h \approx H^h(1 + \hat{h}_t^h)$ and $MC_t \approx MC(1 + \hat{mc}_t)$, linearizes the heterogeneity-and-marginal-cost term to the model-specific form below.

Combining terms. Putting all linearized terms together:

$$\pi_t \approx \sum_h \frac{s_h}{\zeta}\varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC\right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h MC \hat{mc}_t + \beta \mathbb{E}_t\left\{\pi_{t+1}\right\}$$

Using the steady-state result $MC = \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h}$ from (D.13), we can simplify:

$$\begin{aligned}
\pi_t &= \sum_h \frac{s_h}{\zeta}\varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC\right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h} \hat{mc}_t + \beta \mathbb{E}_t\left\{\pi_{t+1}\right\} \\
&= \sum_h \frac{s_h}{\zeta}\varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC\right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right) \hat{mc}_t + \beta \mathbb{E}_t\left\{\pi_{t+1}\right\}
\end{aligned}$$

Defining the coefficients:

$$\delta_1^P = \frac{s_P}{\zeta}\varepsilon_P \left(\frac{1}{\varepsilon_P} - 1 + MC\right) H^P, \quad \delta_1^R = \frac{s_R}{\zeta}\varepsilon_R \left(\frac{1}{\varepsilon_R} - 1 + MC\right) H^R,$$
$$\delta_2 = \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)$$

The linearized Phillips Curve becomes:

$$\pi_t = \beta \mathbb{E}_t\left\{\pi_{t+1}\right\} + \delta_1^P \hat{h}_t^P + \delta_1^R \hat{h}_t^R + \delta_2 \hat{mc}_t$$

Relating markup to marginal cost. From the markup definition $\mathcal{M}_t = \frac{P_t}{MC_t^{nom}} = \frac{P_t}{MC_t \cdot P_t} = \frac{1}{MC_t}$, we have $\mu_t = -mc_t$ in logs. Since $\hat{mc}_t = mc_t - mc$ and $\hat{\mu}_t = \mu_t - \mu$, and given that $\mu = -mc$, it follows that $\hat{\mu}_t = -\hat{mc}_t$. Substituting:

$$\pi_t = \beta \mathbb{E}_t\left\{\pi_{t+1}\right\} + \delta_1^P \hat{h}_t^P + \delta_1^R \hat{h}_t^R - \delta_2 \hat{\mu}_t$$

Expressing markup in terms of output gap. To arrive at the NKPC, we replace $\hat{\mu}_t$ with the output gap $\tilde{y}_t$. Starting with the definition of the gross nominal markup:

$$\mathcal{M}_t = \frac{P_t}{MC_t^{nom}} = \frac{P_t}{\frac{W_t}{\partial Y_t/\partial N_t}} = P_t \frac{\partial Y_t/\partial N_t}{W_t} = \frac{(1-\alpha)A_t N_t^{-\alpha} P_t}{W_t}.$$

From the Ricardian household's intratemporal condition (D.4), using $P_t^R = P_t$ in symmetric equilibrium and $N_t^R = N_t$ from labor market clearing, $W_t/P_t = (C_t^R)^\sigma N_t^\varphi$, so

$$\begin{aligned} \mathcal{M}_t &= \frac{(1-\alpha)A_t}{(C_t^R)^\sigma N_t^\varphi} N_t^{-\alpha} = (1-\alpha)A_t (C_t^R)^{-\sigma} N_t^{-\alpha-\varphi}, \\ \mu_t &= \ln(1-\alpha) + a_t - \sigma c_t^R - (\alpha+\varphi) n_t. \end{aligned} \tag{D.20}$$

Using $C_t^R = H_t C_t$, market clearing $C_t = Y_t$ (Rotemberg costs are second-order), and the production function $Y_t = A_t N_t^{1-\alpha}$ (so $n_t = (y_t - a_t)/(1-\alpha)$) to substitute out c_t^R and n_t, and taking log-deviations from steady state, gives

$$\hat{\mu}_t = \left(1 + \frac{\alpha+\varphi}{1-\alpha}\right) a_t - \sigma \hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \hat{y}_t.$$

Deriving the natural rate of output. With flexible prices, $\mathcal{M}_t = \mathcal{M}_t^n(H_t)$ for all t. Note that $\mathcal{M}_t^n \neq \mathcal{M}_t^e = 1$. This is also the case in the standard TANK without heterogeneous elasticities. However, in the TANK the difference $\mathcal{M}^n - \mathcal{M}^e = \frac{\varepsilon}{\varepsilon-1} - 1$ is constant, whereas in our model $\mathcal{M}_t^n(H_t)$ is in principle state-dependent. Crucially, however, the heterogeneity index in the flex-price equilibrium does not respond to technology or preference shocks: from equation (D.10), H_t^n is determined by an implicit equation involving only $\mathcal{M}_t^n(H_t^n)$ and the fiscal parameters (δ, τ), with no dependence on A_t or Z_t. Since $H_t^n = H$ for all t, the flex-price markup remains at its steady-state value ($\hat{\mu}_t^n = 0$), and divine coincidence is preserved. The time-varying markup channel would, however, generate a cost-push term and break divine coincidence if the model included shocks that directly affect the demand composition (e.g., stochastic redistribution or heterogeneous preference shocks).

In the flex-price equilibrium, the markup equals the desired markup $\hat{\mu}_t = \hat{\mu}_t^n$, where $\hat{\mu}_t^n$ denotes the log-deviation of the flexible-price markup (D.14) from its steady-state value. In general, log-linearizing the flex-price markup around H with $\Phi \equiv (1-\lambda)(\varepsilon_R - \varepsilon_P)$ gives:

$$\hat{\mu}_t^n = \frac{-\Phi H}{(\Phi H + \varepsilon_P)(\Phi H + \varepsilon_P - 1)} \hat{h}_t$$

Since $H_t^n = H$ for all t (as argued above), we have $\hat{h}_t^n = 0$ and therefore $\hat{\mu}_t^n = 0$. Setting

$\hat{\mu}_t = \hat{\mu}_t^n = 0$ in the markup expression:

$$0 = \left(1 + \frac{\alpha+\varphi}{1-\alpha}\right) a_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \hat{y}_t^n$$

Subtracting this from the sticky-price markup expression:

$$\begin{aligned} \hat{\mu}_t &= -\sigma \hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right)(\hat{y}_t - \hat{y}_t^n) \\ &= -\sigma \hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \tilde{y}_t \end{aligned} \tag{D.21}$$

Since $\hat{mc}_t = -\hat{\mu}_t$, $\hat{mc}_t = \sigma \hat{h}_t + \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \tilde{y}_t$, so the NKPC is $\pi_t = \beta \mathbb{E}_t\{\pi_{t+1}\} + \delta_1^P \hat{h}_t^P + \delta_1^R \hat{h}_t^R + \delta_2[\sigma \hat{h}_t + (\frac{\alpha+\varphi}{1-\alpha} + \sigma)\tilde{y}_t]$.

Expressing in terms of aggregate heterogeneity. We relate $\hat{h}_t^P$ to $\hat{h}_t^R$ using market clearing, $\lambda H_t^P + (1-\lambda) H_t^R = 1$: log-linearizing around the steady state (where $\lambda H^P + (1-\lambda) H^R = 1$) gives $\lambda H^P \hat{h}_t^P + (1-\lambda) H^R \hat{h}_t^R = 0$, so, using $\lambda H^P = 1 - (1-\lambda) H^R$,

$$\hat{h}_t^P = \frac{(\lambda - 1) H^R}{1 + (\lambda - 1) H^R} \hat{h}_t^R.$$

Define $\hat{h}_t^R \equiv \hat{h}_t$ and $H^R \equiv H$ as the aggregate heterogeneity index. Substituting, and using $\delta_1^P = \frac{1}{\zeta} \varepsilon_P (\frac{1}{\varepsilon_P} - 1 + MC)(1 - (1-\lambda)H)$ and $\delta_1^R = \frac{1-\lambda}{\zeta} \varepsilon_R (\frac{1}{\varepsilon_R} - 1 + MC) H$ (both immediate from the definitions of δ_1^P, δ_1^R and $H^P = [1 - (1-\lambda)H]/\lambda$), the coefficient on $\hat{h}_t$ collapses, after simplification, to $\delta_1^P \frac{(\lambda-1)H}{1+(\lambda-1)H} + \delta_1^R = \frac{(1-\lambda)H}{\zeta}(1 - MC)(\varepsilon_P - \varepsilon_R)$. Adding the $\delta_2 \sigma$ term from marginal cost, the NKPC with heterogeneity is

$$\pi_t = \beta \mathbb{E}_t \{\pi_{t+1}\} + \kappa_h \hat{h}_t + \kappa_y \tilde{y}_t + u_t,$$

where

$$\kappa_h = \underbrace{\frac{(1-\lambda)H}{\zeta}\left((1 - MC)(\varepsilon_P - \varepsilon_R)\right)}_{\text{from firm FOC heterogeneity}} + \underbrace{\delta_2 \sigma}_{\text{from Ricardian consumption in markup}},$$

$$\kappa_y = \delta_2 \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right),$$

and, substituting $H^P = [1 - (1-\lambda)H]/\lambda$ into the definition of δ_2 and simplifying,

$$\delta_2 = \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right) = \frac{1}{\zeta}\left[H(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1\right].$$

The cost-push disturbance u_t is appended as an additive wedge to this linearized opti-

mality condition, normalized to enter with a unit coefficient: a disturbance that raises real marginal cost by $\hat{u}_t$ independently of the output gap appears as $u_t = \delta_2 \hat{u}_t$. This is equation (14) in the main text.

D.8 Dividends

Linearization of Dividends. The aggregate accounting identity $Y_t \Delta^p(\Pi_t) = \frac{W_t}{P_t} N_t + D_t$ (output net of price dispersion equals real labor income plus dividends), together with $W_t N_t / P_t = \frac{1-\alpha}{\mathcal{M}_t} Y_t$, gives $D_t = Y_t[\Delta^p(\Pi_t) - \frac{1-\alpha}{\mathcal{M}_t}]$, so in steady state ($\Pi = \Delta^p(1) = 1$), $D = Y(\mathcal{M} - (1-\alpha))/\mathcal{M}$. Since price dispersion is minimized at zero inflation, $\partial D / \partial \Pi|_{\Pi=1} = 0$, so a first-order Taylor expansion of the deviation $D_t - D$ around the steady state, normalized by D and using $1 - \frac{1-\alpha}{\mathcal{M}} = D/Y$, simplifies to

$$\hat{d}_t = \hat{y}_t + \frac{1-\alpha}{\mathcal{M} - (1-\alpha)} \hat{\mu}_t. \tag{D.22}$$

D.9 Proofs of the Lemma and Propositions

The proofs below establish Lemma 1 and Propositions 1, 2, and 3; Corollary 4, which builds on all four, is stated and proved in Appendix D.10. They use the derivations in Appendices D.3, D.6, D.7, and D.4.1. Throughout, $\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P)$, $\mathsf{B} = \varepsilon_P$, $\mathsf{C} = \frac{1+\delta\lambda(\tau-1)}{1-\lambda}$, and $\mathsf{D} = \frac{(1-\alpha)\lambda(1+\delta(\tau-1))}{1-\lambda}$, as in Proposition 1.

D.9.1 Fiscal Sufficient Statistic (Lemma 1)

Proof of Lemma 1. After eliminating gross transfer flows and equity values, the fiscal parameters enter the reduced equilibrium through the following combinations, each a function of $R = \delta(1-\tau)$ through $1 + \delta(\tau - 1) = 1 - R$:

$$\begin{aligned}
\mathsf{C} &= \frac{1+\delta\lambda(\tau-1)}{1-\lambda} = \frac{1-\lambda R}{1-\lambda}, \\
\mathsf{D} &= \frac{(1-\alpha)\lambda\big(1+\delta(\tau-1)\big)}{1-\lambda} = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}, \\
\bar{H} - 1 &= \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right), \\
\Gamma &= \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}}.
\end{aligned}$$

To see why the eliminated objects do not add a second fiscal sufficient statistic, first consider equity. Total equity supply is one and only rich households hold shares, so

market clearing implies

$$F_t^R = F_{t-1}^R = \frac{1}{1-\lambda}.$$

The purchase and resale-value terms in (D.2) therefore cancel period by period:

$$Q_t F_{t-1}^R - Q_t F_t^R = Q_t\left(F_{t-1}^R - F_t^R\right) = 0.$$

Although the equity Euler equation implies

$$\bar{Q} = \frac{\beta(1-\delta)}{1-\beta}\bar{D},$$

so that the equity valuation depends separately on δ, this valuation does not enter the household's flow resources or the reduced equilibrium.

Second, the transfer rule (D.1) can be written as

$$\begin{aligned} T_t^P &= (1-\tau)\delta D_t = RD_t, \\ T_t^R &= \frac{\delta(1-\lambda+\tau\lambda)}{1-\lambda}D_t = \frac{\delta-\lambda R}{1-\lambda}D_t. \end{aligned}$$

Thus T_t^R itself depends on δ separately at fixed R, but rich households receive it together with after-tax dividend income. Their combined profit-related resources are

$$\begin{aligned} (1-\delta)D_t F_{t-1}^R + T_t^R &= \frac{(1-\delta)+\delta-\lambda R}{1-\lambda}D_t \\ &= \frac{1-\lambda R}{1-\lambda}D_t = \mathsf{C}D_t. \end{aligned}$$

Hence changing (δ, τ) while holding R fixed changes the composition of rich households' income between dividends and transfers, as well as the equity valuation, but not either household type's disposable resources.

The quadratic primitives A and B contain no fiscal parameters; hence $\bar{H}$, the root of (9), depends on (δ, τ) only through (C, D), that is, through R. Then $\bar{\mathcal{M}} = \varepsilon_{\text{eff}}(\bar{H})/(\varepsilon_{\text{eff}}(\bar{H})-1)$; the coefficients δ_2, κ_y, and κ_h (functions of $\bar{H}$ and $\bar{\mathcal{M}}$ only); Γ (an explicit factor $1-R$ times $1/(\bar{\mathcal{M}}\bar{H})$); $\chi_y = \Gamma\left(\frac{\varphi+\alpha}{1-\alpha}+\sigma\right)/(1+\Gamma\sigma)$; and $\kappa = \kappa_y - \chi_y\kappa_h$ all inherit dependence on R alone. No other occurrence of δ or τ appears in the derivations of Appendix D.3, Appendix D.7, or the dynamic IS block of Appendix D.6. Therefore fiscal pairs with the same R yield the same steady-state allocation and first-order dynamics. Finally, $\bar{H} = 1 \iff 1-R = 0 \iff R = 1$, and $(\delta, \tau) = (1, 0)$ gives $R = 1$; the cyclicality derivation below establishes $\operatorname{sign}(\chi_y) = \operatorname{sign}(\Gamma) = \operatorname{sign}(1-R)$. □

D.9.2 Existence, Uniqueness, and Fiscal Comparative Statics (Proposition 1)

Proof of Proposition 1. (i) The fixed point $\bar{H} = \mathsf{C} - \mathsf{D}/\bar{\mathcal{M}}$ is the steady-state heterogeneity index of Appendix D.3 evaluated at $\Delta^p(\bar{\Pi}) = 1$, that is, equation (7); the flex-price markup $\bar{\mathcal{M}} = (\mathsf{A}\bar{H} + \mathsf{B})/(\mathsf{A}\bar{H} + \mathsf{B} - 1)$ is equation (8), derived in Appendix D.4.1, written with (A, B). Substituting $1/\bar{\mathcal{M}} = (\mathsf{A}\bar{H} + \mathsf{B} - 1)/(\mathsf{A}\bar{H} + \mathsf{B})$ and clearing the denominator:

$$
\begin{aligned}
& & (\bar{H} - \mathsf{C})(\mathsf{A}\bar{H} + \mathsf{B}) &= -\mathsf{D}(\mathsf{A}\bar{H} + \mathsf{B} - 1) \\
\Rightarrow \quad & \mathsf{A}\bar{H}^2 + \big[\mathsf{B} - \mathsf{A}(\mathsf{C} - \mathsf{D})\big]\bar{H} - \big[\mathsf{B}(\mathsf{C} - \mathsf{D}) + \mathsf{D}\big] &&= 0,
\end{aligned}
$$

which is the quadratic (9). Existence and uniqueness on the fiscal square. Let $e(H) \equiv \mathsf{A}H + \mathsf{B} = \varepsilon_{\text{eff}}(H)$ and

$$
H_{\max} \equiv \frac{\varepsilon_P - 1}{(1-\lambda)(\varepsilon_P - \varepsilon_R)},
$$

so that $e(H) > 1$, equivalently $\mathcal{M}(H) > 1$, exactly on $H < H_{\max}$. The steady state is a zero of

$$
q(H; R) \equiv \mathsf{C}(R) - \mathsf{D}(R) + \frac{\mathsf{D}(R)}{e(H)} - H.
$$

Using $\mathsf{C} - 1 = \frac{\lambda(1-R)}{1-\lambda}$ and $\mathsf{D} = (1-\alpha)(\mathsf{C} - 1)$, for $R < 1$,

$$
q(1) = (\mathsf{C} - 1)\left[\alpha + \frac{1-\alpha}{\bar{\varepsilon}}\right] > 0,
$$

since $e(1) = \bar{\varepsilon} > 1$. Moreover $H_{\max} - \frac{1}{1-\lambda} = \frac{\varepsilon_R - 1}{(1-\lambda)(\varepsilon_P - \varepsilon_R)} > 0$ while $\mathsf{C} \leq \frac{1}{1-\lambda}$, so $\lim_{H \uparrow H_{\max}} q(H; R) = \mathsf{C} - H_{\max} < 0$. Finally $q_{HH}(H; R) = 2\mathsf{D}\mathsf{A}^2/e(H)^3 > 0$: q is convex, positive at $H = 1$, and negative at the upper boundary, so it crosses zero exactly once on $(1, H_{\max})$. The crossing is strict: if $q_H(\bar{H}; R) \geq 0$, convexity would imply $q(H; R) \geq 0$ for every $H > \bar{H}$, contradicting its negative limit at $H_{\max}$. Thus $q_H(\bar{H}; R) < 0$. The quadratic's second root lies beyond $H_{\max}$, where $\mathcal{M} < 1$ and, because $H_{\max} > \frac{1}{1-\lambda}$, poor-household consumption is negative (the quadratic equals $-e(H)q(H; R)$; it is positive at $H_{\max}$, where $q < 0$ and $e = 1$, and has negative leading coefficient A, so its remaining root exceeds $H_{\max}$). At the root, $\bar{H} = \mathsf{C} - \mathsf{D} + \mathsf{D}/e(\bar{H}) < \mathsf{C} \leq \frac{1}{1-\lambda}$, so the poor household's consumption relative to the average, $\bar{H}^p = [1 - (1-\lambda)\bar{H}]/\lambda$, is strictly positive. Hence for every $R \in [0, 1)$ there is exactly one admissible steady state, with $1 < \bar{H} < \mathsf{C}$. At $R = 1$, $\mathsf{C} = 1$ and $\mathsf{D} = 0$, so the quadratic factors as $(H - 1)(\mathsf{A}H + \mathsf{B}) = 0$. Under $\varepsilon_P > \varepsilon_R$, its second root is

$$
-\frac{\mathsf{B}}{\mathsf{A}} = \frac{\varepsilon_P}{(1-\lambda)(\varepsilon_P - \varepsilon_R)} > \frac{1}{1-\lambda},
$$

which implies negative poor-household consumption; hence $\bar{H} = 1$ is the unique feasible root. In the homogeneous TANK case $\mathsf{A} = 0$, the system is linear and trivially unique. Existence and uniqueness therefore hold on the entire fiscal square $(\delta, \tau) \in [0,1]^2$.

(ii) Differentiate the fixed-point condition $q(\bar{H}; R) = 0$. Its partial derivatives are

$$\begin{aligned} q_R(\bar{H}; R) &= -\frac{\lambda}{1-\lambda} + \frac{(1-\alpha)\lambda}{1-\lambda} - \frac{(1-\alpha)\lambda/(1-\lambda)}{e(\bar{H})} \\ &= -\frac{\lambda}{1-\lambda}\left[\alpha + \frac{1-\alpha}{e(\bar{H})}\right] < 0, \\ q_H(\bar{H}; R) &= -\left[1 + \frac{\mathsf{DA}}{e(\bar{H})^2}\right] < 0, \end{aligned}$$

where the second inequality follows from the strict downcrossing established in part (i). The implicit-function theorem therefore gives

$$\frac{\mathrm{d}\bar{H}}{\mathrm{d}R} = -\frac{q_R}{q_H} = -\frac{\lambda}{1-\lambda} \cdot \frac{\alpha + (1-\alpha)/e(\bar{H})}{1 + \mathsf{DA}/e(\bar{H})^2} < 0.$$

This derivative extends continuously to $R = 1$ from the left. For the markup, $\bar{\mathcal{M}} = e/(e-1)$ implies

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}\bar{H}} = -\frac{\mathsf{A}}{[e(\bar{H})-1]^2} = \frac{(1-\lambda)(\varepsilon_P - \varepsilon_R)}{[e(\bar{H})-1]^2} > 0.$$

Combining the two derivatives yields

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}R} = -\frac{\lambda(\varepsilon_P - \varepsilon_R)\,[\alpha + (1-\alpha)/e(\bar{H})]}{[e(\bar{H})-1]^2\;[1 + \mathsf{DA}/e(\bar{H})^2]} < 0.$$

Thus redistribution lowers both steady-state inequality and the desired markup. Using $\mathsf{D} = \Gamma\bar{\mathcal{M}}\bar{H}$ and the definition of η_μ in (15), the feedback denominator satisfies

$$1 + \frac{\mathsf{DA}}{e(\bar{H})^2} = 1 - \Gamma\eta_\mu > 0.$$

Hence the endogenous-markup feedback amplifies the direct redistribution effect by $(1 - \Gamma\eta_\mu)^{-1}$. □

Benchmark nesting. Subtracting one from (7) gives

$$\bar{H} - 1 = \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right).$$

Because $\bar{\mathcal{M}} > 1 > 1-\alpha$, the second factor is strictly positive, so $\bar{H} = 1 \iff R = 1$. On the fiscal square, $R = 1$ requires $(\delta, \tau) = (1, 0)$. At this point $\varepsilon_{\text{eff}}(1) = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P = \bar{\varepsilon}$ and $\bar{\mathcal{M}} = \bar{\varepsilon}/(\bar{\varepsilon}-1)$. Moreover,

$$\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\bar{\mathcal{M}}\bar{H}}, \qquad \chi_y = \frac{\Gamma\left(\frac{\varphi+\alpha}{1-\alpha}+\sigma\right)}{1+\Gamma\sigma},$$

so $\bar{H} = 1 \iff R = 1 \iff \Gamma = 0 \iff \chi_y = 0 \iff \hat{h}_t \equiv 0$ along every first-order equilibrium path, establishing (13). The elasticity-gap term in the NKPC then multiplies $\hat{h}_t = 0$, while the DIS curve loses its heterogeneity term; every structural equation reduces to its RANK($\bar{\varepsilon}$) counterpart to first order. In the alternative homogeneous-elasticity benchmark $\varepsilon_R = \varepsilon_P = \varepsilon$, $\mathsf{A} = 0$ and the steady-state system is linear, with $\bar{H} = \mathsf{C} - \mathsf{D}(\varepsilon-1)/\varepsilon$ and fixed $\bar{\mathcal{M}} = \varepsilon/(\varepsilon-1)$. Thus standard TANK can retain consumption inequality, but its desired markup does not respond to it.

D.9.3 Elasticity Dispersion and IS Amplification (Proposition 2)

IS semi-elasticity and admissible region. The DIS coefficient $1/[\sigma(1-\chi_y)]$ follows from substituting (11) into the dynamic IS curve of Appendix D.6: its $\mathbb{E}_t\Delta\hat{h}_{t+1}$ term contributes $-\chi_y\mathbb{E}_t\Delta\tilde{y}_{t+1}$, whose contemporaneous piece scales $\tilde{y}_t$ by $(1-\chi_y)^{-1}$, yielding (12). The amplification factor relative to RANK is $1/(1-\chi_y)$ since RANK has $\chi_y = 0$. Finiteness and positivity: $\chi_y < 1 \iff \Gamma\left(\frac{\varphi+\alpha}{1-\alpha}+\sigma\right) < 1+\Gamma\sigma \iff \Gamma\frac{\varphi+\alpha}{1-\alpha} < 1$; with $\chi_y > 0$ under the baseline, this gives $\chi_y \in (0,1)$.

Proof of Proposition 2. The TANK/TE ordering: both economies share (C, D), since (λ, δ, τ) are common, and each steady state solves $H = \mathsf{C} - \mathsf{D}/\mathcal{M}(H)$. For $R < 1$ the TANK fixed point is $\bar{H}^{\text{TANK}} = \mathsf{C} - \mathsf{D}/\mathcal{M}(\bar{\varepsilon}) > 1$. For any $H > 1$, the TE effective elasticity is $\varepsilon_{\text{eff}}^{\text{TE}}(H) = \bar{\varepsilon} + \mathsf{A}(H-1) < \bar{\varepsilon}$ because $\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P) < 0$, and $\mathcal{M}(\cdot)$ is decreasing in the elasticity, so $\mathcal{M}^{\text{TE}}(H) > \mathcal{M}(\bar{\varepsilon})$. Hence the TE fixed-point residual $q(H) = \mathsf{C} - \mathsf{D}/\mathcal{M}^{\text{TE}}(H) - H$ of the proof of Proposition 1(i)—the same q, since $1/\mathcal{M}(H) = 1 - 1/e(H)$—is strictly positive at $\bar{H}^{\text{TANK}}$:

$$q\left(\bar{H}^{\text{TANK}}\right) = \mathsf{D}\left[\frac{1}{\mathcal{M}(\bar{\varepsilon})} - \frac{1}{\mathcal{M}^{\text{TE}}(\bar{H}^{\text{TANK}})}\right] > 0.$$

Since q is convex, positive at $H = 1$, and crosses zero exactly once on the admissible band (proof of Proposition 1(i)), and since $\bar{H}^{\text{TANK}} < \mathsf{C} \le \frac{1}{1-\lambda} < H_{\max}$ places $\bar{H}^{\text{TANK}}$ inside that band, $q > 0$ exactly to the left of the TE root, so $\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}$. With $\bar{H}^{\text{TE}} > 1$, also $\bar{\mathcal{M}}^{\text{TE}} = \mathcal{M}^{\text{TE}}(\bar{H}^{\text{TE}}) > \mathcal{M}(\bar{\varepsilon}) = \bar{\mathcal{M}}^{\text{TANK}}$. Both factors of $\bar{\mathcal{M}}\bar{H}$ are

therefore strictly larger in TE-TANK, so

$$\bar{\mathcal{M}}^{\text{TE}}\bar{H}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}\bar{H}^{\text{TANK}} \;\Rightarrow\; \Gamma^{\text{TE}} < \Gamma^{\text{TANK}} \;\Rightarrow\; \chi_y^{\text{TE}} < \chi_y^{\text{TANK}}.$$

□

This is the one comparison in the paper that is made at the mean-matched benchmark rather than at ε^*, and it has to be. At the baseline calibration, $1/(1-\chi_y)$ orders 1.0000 (RANK) $<$ 1.9032 (TE-TANK) $<$ 1.9724 (mean-matched TANK, $\bar{\varepsilon} = 4.44$): elasticity dispersion dampens the demand amplification by 3.5 percent, and it does so through the markup level. The gap raises $\bar{\mathcal{M}}$ from 1.291 to 1.341, which is more than it raises $\bar{H}$, so $\bar{\mathcal{M}}\bar{H}$ rises, Γ falls, and χ_y falls with it, from 0.4930 to 0.4746. At ε^* the two economies share χ_y, so the difference is exactly zero. The proposition compares markup levels; it has no dispersion-pure content.

D.9.4 Cost-Push Pass-Through (Proposition 3)

Cyclicality and flattening. The relation $\hat{h}_t = -\chi_y \tilde{y}_t$ is derived in Appendix D.3. Write $\Gamma = k \cdot \big(1 + \delta(\tau - 1)\big)$ with

$$k = \frac{(1-\alpha)\lambda}{\bar{\mathcal{M}}\bar{H}(1-\lambda)} > 0$$

on the admissible region ($\bar{\mathcal{M}} > 1$, $\bar{H} > 0$), so $\text{sign}(\Gamma) = \text{sign}\big(1 + \delta(\tau - 1)\big)$. Since $\chi_y = \Gamma\big(\frac{\varphi+\alpha}{1-\alpha} + \sigma\big)/(1 + \Gamma\sigma)$ with a strictly positive numerator factor and $1 + \Gamma\sigma > 0$, $\text{sign}(\chi_y) = \text{sign}(\Gamma)$. At $\delta = 1$, $1 + \delta(\tau - 1) = \tau$.

The reduced form $\kappa = \kappa_y - \chi_y \kappa_h$ and the decomposition of κ_h are derived in Appendix D.7. Signs: $\overline{MC} = 1/\bar{\mathcal{M}} < 1$ since $\bar{\mathcal{M}} > 1$, so $1 - \overline{MC} > 0$; $\varepsilon_P - \varepsilon_R > 0$ by assumption; and the second piece $\delta_2 \sigma > 0$ because $\delta_2 = \big[(1-\lambda)\bar{H}(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1\big]/\zeta = (\varepsilon_{\text{eff}}(\bar{H}) - 1)/\zeta > 0$ (Appendix D.7). Therefore $\kappa_h > 0$. Since $\kappa_y - \kappa = \chi_y \kappa_h$ and $\kappa_h > 0$, $\kappa < \kappa_y \iff \chi_y > 0 \iff R < 1$.

Benchmark. κ_y is the slope obtained by holding $(\bar{H}, \bar{\mathcal{M}}, \delta_2)$ at their TE-TANK values and switching off the $-\chi_y \kappa_h$ feedback, equivalently by imposing $\hat{h}_t \equiv 0$ in the structural NKPC (14). This is the same-steady-state, feedback-off comparison. It is not the slope of the homogeneous-elasticity economy, which additionally resets δ_2, κ_y, and κ_h to their $\bar{\varepsilon}$ values.

Proof of Proposition 3. With u_t the only disturbance, $z_t = a_t = v_t = 0$ implies $r_t^n = 0$ and $\hat{y}_t = \tilde{y}_t$, so the equilibrium reduces to (16), (12), and (6) with forcing process $u_t = \rho_u u_{t-1} + \varepsilon_t^u$. Guess $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$, so $\mathbb{E}_t[\pi_{t+1}] = \rho_u \Psi_\pi u_t$ and $\mathbb{E}_t[\tilde{y}_{t+1}] = \rho_u \Psi_y u_t$. Matching coefficients in (16) gives $(1 - \beta\rho_u)\Psi_\pi = \kappa \Psi_y + 1$; substituting the Taylor rule into (12) gives $\Psi_y\big[\tilde{\sigma}(1-\rho_u) + \phi_y\big] = -(\phi_\pi - \rho_u)\Psi_\pi$, that is, $\Psi_y = -\Theta\Psi_\pi$

with $\Theta = (\phi_\pi - \rho_u)/\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right] > 0$. Combining the two conditions yields $\Psi_\pi = \left[(1-\beta\rho_u) + \kappa\Theta\right]^{-1} \equiv 1/D_\pi > 0$. The standard determinacy condition is $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$, which holds under the maintained restrictions; hence this is the unique bounded solution. Differentiating,

$$\frac{\partial \Psi_\pi}{\partial \kappa} = -\Theta\,\Psi_\pi^2 < 0, \qquad \frac{\partial \Psi_\pi}{\partial \tilde{\sigma}} = \frac{(1-\rho_u)\,\kappa\,(\phi_\pi - \rho_u)}{\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right]^2}\,\Psi_\pi^2 > 0.$$

For the ratio, $\Psi_\pi^{\mathrm{TE}}/\Psi_\pi^{\mathrm{TANK}} = D_\pi^{\mathrm{TANK}}/D_\pi^{\mathrm{TE}} > 1$ if and only if $\kappa^{\mathrm{TE}}\Theta^{\mathrm{TE}} < \kappa^{\mathrm{TANK}}\Theta^{\mathrm{TANK}}$. The sufficient conditions stated after the proposition follow because Θ is increasing in χ_y through $\tilde{\sigma} = \sigma(1-\chi_y)$. Finally, $\Psi_y = -\Theta\Psi_\pi < 0$, and dividing the TE and TANK output responses gives the stated contraction ratio. □

Quantitative magnitudes. At the baseline calibration, the feedback removes 19.5 percent of κ_y. Moving from κ_y to κ at fixed $\tilde{\sigma}$ raises cost-push pass-through by 16.6 percent. Table C.7 reports the pass-through and output ratios across four homogeneous-elasticity benchmarks.

D.10 Comparative Statics

The markup-matched comparison isolates the elasticity-gap channel. Table D.1 shows that RANK, TANK, and TE-TANK share the steady-state markup at $\varepsilon^* = 3.933$. Moving from RANK to TANK raises χ_y from zero to 0.4746 and lowers κ from 0.0833 to 0.0701. Moving from TANK to TE-TANK changes only κ_h within the first-order policy block and lowers the slope further to 0.0671.

TABLE D.1: Structural Objects Across RANK, TANK, and TE-TANK

	RANK	TANK	TE-TANK
ε_R	3.93	3.93	3.00
ε_P	3.93	3.93	6.60
λ	0.00	0.40	0.40
τ	0.00	0.70	0.70
$\bar{H}$	1.0000	1.2347	1.2347
$\bar{\mathcal{M}}$	1.3409	1.3409	1.3409
Γ	0.0000	0.1879	0.1879
χ_y	0.0000	0.4746	0.4746
δ_2	0.0278	0.0278	0.0278
κ_y	0.0833	0.0833	0.0833
κ_h	0.0278	0.0278	0.0342
$\kappa = \kappa_y - \chi_y \kappa_h$	0.0833	0.0701	0.0671
$1/(1-\chi_y)$	1.0000	1.9032	1.9032

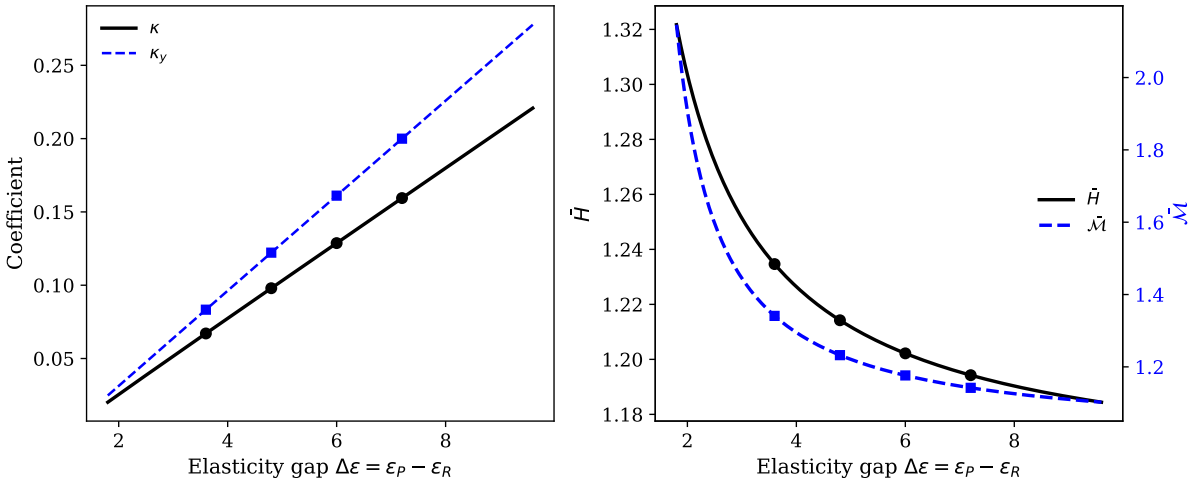
Notes: Structural steady-state and NKPC objects across three calibrations. $\kappa_h = \frac{(1-\lambda)\bar{H}}{\zeta}(1-\overline{MC})(\varepsilon_P - \varepsilon_R) + \delta_2\sigma$ and $\chi_y = \Gamma \cdot \text{slope_factor}/(1+\Gamma\sigma)$. Common parameters: $\sigma = \varphi = 1$, $\alpha = 1/3$, $\zeta = 105.63$, $\delta = 1$. $\bar{\varepsilon} = \varepsilon^* = 3.93$ matches TE-TANK's markup $\bar{\mathcal{M}} = 1.3409$ (Table C.7): all three columns share $\bar{\mathcal{M}}$, δ_2, κ_y; TANK and TE-TANK also share $\bar{H}$, Γ, χ_y, differing only in κ_h (hence κ).

The fixed-ratio path varies the elasticity level while holding $\varepsilon_P/\varepsilon_R = 2.2$. Both κ_y and κ rise along this path, while the feedback remains a drag of about 20 percent on κ_y. The elasticity gap and level rise together, so the sweep does not isolate either dimension. The heterogeneity index changes little, and the movement comes mainly from the pricing coefficients rather than from a large change in the consumption distribution. Figure D.1 reports the path.

Figure D.2 places this path and the fixed-gap path of Table C.6 in the full $(\bar{\varepsilon}, \Delta\varepsilon)$ plane, where the dispersion-pure cost-push gap rises with the elasticity gap and falls with the level, and where the two paths cross at the baseline calibration; the right panel repeats the exercise for the gain from commitment over discretion, whose TE-TANK versus TANK(ε^*) ordering reverses along the $\bar{\mathcal{M}} \approx 1.52$ iso-markup contour rather than along any contour of $\Delta\varepsilon$.

Steady-state inequality and cost-push amplification. Higher steady-state inequality amplifies the markup response to a cost-push shock. The mechanism is compositional. With more steady-state inequality, a larger share of spending comes from rich households, whose demand is less price-elastic, so the demand curve firms face is less elastic

FIGURE D.1: Comparative Statics along the Fixed-Ratio Elasticity Path



Notes: Phillips-curve slopes and steady-state objects as functions of $\Delta\varepsilon = \varepsilon_P - \varepsilon_R$, holding $\varepsilon_P/\varepsilon_R = 2.2$ fixed and all other parameters at baseline. Left: reduced-form slope κ (solid black) and feedback-off slope κ_y (dashed blue). Their distance is $\chi_y\kappa_h$. Right: heterogeneity index $\bar{H}$ (solid black, left axis) and markup $\bar{\mathcal{M}}$ (dashed blue, right axis). Markers trace $\Delta\varepsilon = 3.6$, 4.8, 6.0, and 7.2 along the same path. The leftmost marker is the baseline $(\varepsilon_R, \varepsilon_P) = (3, 6.6)$.

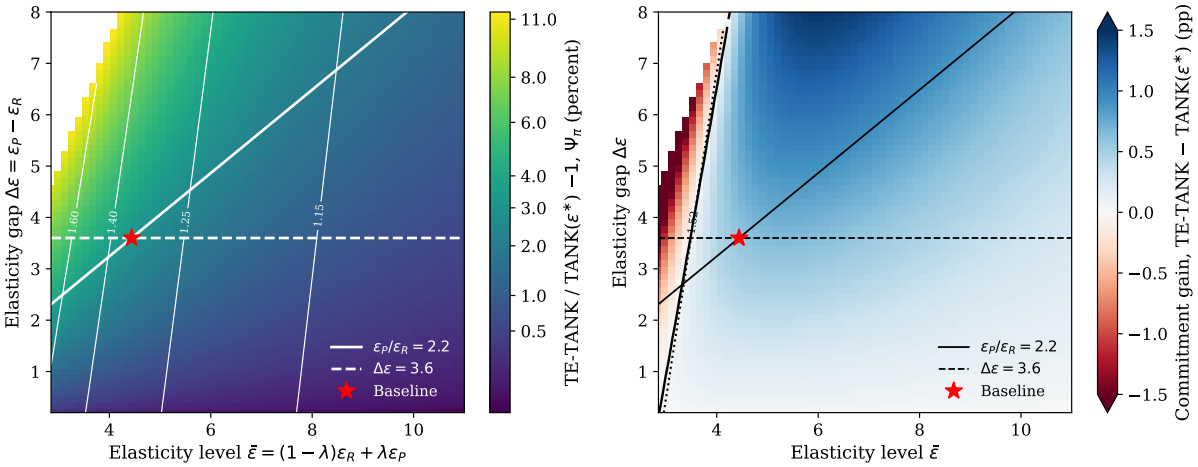
and the desired markup is more sensitive to any further shift in the composition of demand. A cost-push shock produces exactly such a shift: it opens a negative output gap, inequality rises, spending tilts further toward the rich, and desired markups rise. Corollary 4 makes this comparative static exact, and it is the model's sharpest cross-sectional prediction: after the same shock, a more unequal economy shows a larger markup response. The cross-country local projections of Section 4.3 test this prediction.

The comparative static runs along the admissible fiscal branch: the family of steady states traced out as the rebate parameter $R = \delta(1-\tau)$ moves over $[0,1]$, with all non-fiscal parameters held at baseline. At $R = 1$ profits are fully rebated as equal transfers and steady-state inequality vanishes; at $R = 0$ poor households receive no profit income and inequality is largest. The branch can be indexed by steady-state inequality itself, the heterogeneity index $\bar{H}$ (rich households' consumption relative to average consumption, equal to one under full equality), rather than by R. The *Benchmark nesting* display above, obtained by subtracting one from (7), gives $\bar{H} - 1 = \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right)$, whose second factor is strictly positive because $\bar{\mathcal{M}} > 1 > 1-\alpha$ on the branch (Proposition 1(i)). Inverting is therefore legitimate:

$$1 - R = \frac{(1-\lambda)(\bar{H}-1)}{\lambda\left[1 - (1-\alpha)/\bar{\mathcal{M}}(\bar{H})\right]}. \tag{D.23}$$

Since $\mathrm{d}\bar{H}/\mathrm{d}R < 0$ (Proposition 1(ii)), $R \mapsto \bar{H}$ is a strictly decreasing bijection from $[0,1]$

FIGURE D.2: Comparative Statics over the Elasticity Plane



Notes: Comparative statics over the elasticity plane, spanned by the level $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$ and the gap $\Delta\varepsilon = \varepsilon_P - \varepsilon_R$, all other parameters at baseline; at each grid point the homogeneous comparison economy sits at that point's own markup-matched ε^*. The grid is 60×60 over $\bar{\varepsilon} \in [2.85, 11]$ and $\Delta\varepsilon \in [0.2, 8]$; cells in which $\varepsilon_R \leq 1.01$, or in which the commitment root leaves the unit interval, are left blank. Left panel: the dispersion-pure cost-push gap of TE-TANK against the markup-matched TANK(ε^*) under the Taylor rule, $\Psi_\pi / \Psi^*_\pi - 1$ in percent (color scale); thin white lines are iso-markup contours of the implied steady-state markup $\bar{\mathcal{M}}$, labeled at 1.15, 1.25, 1.40, and 1.60. The color scale is a power norm with exponent one half, which compresses the high-gap corner without clipping; the colorbar reports untransformed values. Right panel: the difference between TE-TANK and TANK(ε^*) in each economy's own commitment gain over discretion, $(L^{\text{disc}} - L^{\text{com}})/L^{\text{disc}}$ computed from the linear-quadratic loss, in percentage points; blue means TE-TANK gains more from commitment, red means less. The thick solid black curve is the zero contour of the difference and the dotted black curve, labeled numerically, is the $\bar{\mathcal{M}} \approx 1.52$ iso-markup contour; the two nearly coincide, which is the content of the panel. The color scale is saturated at ± 1.5 percentage points, indicated by the arrows on the colorbar. In both panels the straight solid line is the fixed-ratio path $\varepsilon_P/\varepsilon_R = 2.2$ of Figure D.1 and Panel A of Table C.6, the straight dashed line is the fixed-gap path $\Delta\varepsilon = 3.6$ of Panel B, and the red star marks the baseline calibration $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$, where the two paths cross.

onto $[1, \bar{H}(R{=}0)]$, with $\bar{H}(R{=}0) = 1.342$ at the baseline elasticities, and every object below is a function of $\bar{H}$ alone along the branch. "Increasing in $\bar{H}$" therefore means increasing as redistribution weakens and steady-state inequality rises.

The corollary tracks the composite objects of Sections 2.5 and 2.6 along the branch. In words: Γ is the strength of the demand-composition feedback, zero in the RANK limit; χ_y is the loading of inequality on the output gap in the distributional relation $\hat{h}_t = -\chi_y \tilde{y}_t$, so it measures how much inequality rises when a shock opens a negative gap; η_μ is the elasticity of the desired markup with respect to the heterogeneity index, that is, how strongly desired markups react to a shift in demand composition; δ_2 is the marginal-cost loading in the Phillips curve; κ is the reduced-form Phillips-curve slope; Θ converts inflation into the output contraction generated by the Taylor rule and the IS curve, through $\Psi_y = -\Theta\Psi_\pi$; and Ψ_π is cost-push pass-through, the impact response of inflation to a unit cost-push shock. Throughout, write $G = \frac{\varphi+\alpha}{1-\alpha} + \sigma$ for the coefficient linking real marginal cost to the output gap and $b = G - \sigma = \frac{\varphi+\alpha}{1-\alpha}$ for its labor-market part.

Two conditions on primitives organize the comparative statics:

$$(\varepsilon_R - 1)\big(1 - \alpha + \alpha\varepsilon_R\big) \;\geq\; \frac{\lambda}{1-\lambda}\big(\varepsilon_P - \varepsilon_R\big), \tag{D.24}$$

$$(1 - \rho_u)\,\frac{\varphi + \alpha}{1 - \alpha} \;\geq\; \phi_y. \tag{D.25}$$

Condition (D.24) keeps the rich elasticity far enough above one relative to the elasticity gap. Its left side rises with ε_R and vanishes as $\varepsilon_R \to 1$, the point where the desired markup diverges; its right side is the elasticity gap weighted by the relative population share of the poor. The condition rules out configurations in which the effective elasticity firms face comes close to one on the admissible branch; the remark after the proof makes this boundary explicit. Condition (D.25) caps the Taylor rule's output response ϕ_y at $(1-\rho_u)b$, the marginal-cost coefficient scaled by one minus the persistence of the cost-push shock; it rules out rules that respond so strongly to output that the demand-side term in the proof of part (iii) would work against the amplification. Both hold strictly at the baseline calibration, with room to spare: (D.24) reads $3.333 \geq 2.400$ and (D.25) reads $0.200 \geq 0.125$.

Corollary 4 (Steady-State Inequality and Cost-Push Amplification)**.** *Index the admissible fiscal branch by $\bar{H} \in \big[1, \bar{H}(R{=}0)\big]$ through (D.23), and maintain the cost-push conditions of Proposition 3. Then:*

(i) Under (D.24), Γ, χ_y, η_μ, and Θ are strictly increasing in $\bar{H}$, while δ_2 and κ are strictly decreasing.

(ii) *Under* (D.24) *alone, the impact output contraction* $|\Psi_y| = \Theta\Psi_\pi$ *is strictly increasing in* $\bar{H}$.

(iii) *Under* (D.24) *and* (D.25)*, the impact markup response to the cost-push wedge,*

$$\frac{\partial \hat{\mu}_t}{\partial u_t} = \big[G - \sigma\chi_y\big]\Theta\Psi_\pi = \frac{G}{1+\Gamma\sigma}\,\Theta\Psi_\pi,$$

is strictly increasing in $\bar{H}$*. Higher steady-state inequality therefore raises the markup response to a cost-push shock.*

(iv) Numerical, at the baseline calibration only; not implied by (i)–(iii). *Inflation pass-through* Ψ_π *is strictly increasing in* $\bar{H}$ *along the branch, rising from* 2.739 *at* $R = 1$ *to* 3.212 *at* $R = 0$*, while* κ *falls from* 0.0977 *to* 0.0574 *and the markup response of (iii) rises by 27.5 percent.*

Proof of Corollary 4. Throughout, write $e \equiv e(\bar{H}) = \mathsf{A}\bar{H} + \mathsf{B} = \varepsilon_{\text{eff}}(\bar{H})$ for the effective demand elasticity firms face, the expenditure-weighted average of the two types' elasticities. Since $\mathsf{A} = -|\mathsf{A}| < 0$ with $|\mathsf{A}| = (1-\lambda)(\varepsilon_P - \varepsilon_R)$, e is strictly decreasing in $\bar{H}$: as inequality rises, spending shifts toward the low-elasticity rich and the average elasticity falls. Proposition 1(i) gives $1 \le \bar{H} < \mathsf{C} \le \frac{1}{1-\lambda}$ and $e > 1$, equivalently $\bar{\mathcal{M}} = e/(e-1) > 1$, everywhere on the branch.

Step 1: the feedback strength Γ *is strictly increasing in* $\bar{H}$*; this is where* (D.24) *does its work.* Substituting (D.23) into $\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\bar{\mathcal{M}}\bar{H}}$ and using $\bar{\mathcal{M}} - (1-\alpha) = \frac{\alpha e + 1 - \alpha}{e-1}$ expresses Γ as a function of $\bar{H}$ alone,

$$\Gamma(\bar{H}) = \frac{(1-\alpha)(\bar{H}-1)}{\bar{H}\big[\bar{\mathcal{M}} - (1-\alpha)\big]} = \frac{(1-\alpha)(\bar{H}-1)(e-1)}{\bar{H}\big[\alpha e + 1 - \alpha\big]}.$$

Two forces compete: the factor $\bar{H} - 1$ rises with inequality, while the factor $e - 1$ falls with it. Log-differentiating and collecting the two e terms separates them,

$$\frac{\mathrm{d}\log\Gamma}{\mathrm{d}\bar{H}} = \frac{1}{\bar{H}(\bar{H}-1)} - \frac{|\mathsf{A}|}{(e-1)(\alpha e + 1 - \alpha)}, \qquad \text{so} \qquad \Gamma' > 0 \iff (e-1)(\alpha e + 1 - \alpha) > |\mathsf{A}|\,\bar{H}(\bar{H} -$$

Because e is strictly decreasing in $\bar{H}$, the left side of this inequality is strictly decreasing and the right side strictly increasing in $\bar{H}$; it therefore suffices to check it weakly at the right endpoint of the branch's closure. That endpoint is $\bar{H} = \frac{1}{1-\lambda}$, where $e\big(\frac{1}{1-\lambda}\big) = \frac{\mathsf{A}}{1-\lambda} + \mathsf{B} = \varepsilon_R$ exactly and $|\mathsf{A}|\frac{1}{1-\lambda}\big(\frac{1}{1-\lambda} - 1\big) = \frac{\lambda}{1-\lambda}(\varepsilon_P - \varepsilon_R)$, so the inequality evaluated there is precisely (D.24). Since $\bar{H} < \frac{1}{1-\lambda}$ strictly on the branch, the weak condition (D.24) delivers the strict inequality at every admissible $\bar{H}$, hence $\Gamma' > 0$. At the left endpoint $\Gamma(1) = 0$ and $\Gamma'(1) = \frac{(1-\alpha)(e(1)-1)}{\alpha e(1) + 1 - \alpha} > 0$.

Step 2: the remaining objects of part (i) follow by inspection once e falls and Γ rises with $\bar{H}$. By Appendix D.7, $\delta_2 = (e-1)/\zeta$, so $\delta_2 > 0$ and $\delta_2' < 0$. By (15), $\eta_\mu = |\mathsf{A}|\bar{H}/\big[e(e-1)\big]$ has a strictly increasing positive numerator and a strictly decreasing positive denominator, so $\eta_\mu > 0$ is strictly increasing: desired markups become more sensitive to demand composition. Since $\mathrm{d}\chi_y/\mathrm{d}\Gamma = G/(1+\Gamma\sigma)^2 > 0$, Step 1 makes χ_y strictly increasing; hence $\tilde{\sigma} = \sigma(1-\chi_y)$ is strictly decreasing and $\Theta = (\phi_\pi - \rho_u)/\big[\tilde{\sigma}(1-\rho_u) + \phi_y\big]$ strictly increasing: the demand side contracts more per unit of inflation. Finally, by (17), $\kappa = \delta_2 G\frac{1-\Gamma\eta_\mu}{1+\Gamma\sigma}$. The proof of Proposition 1(ii) already establishes $1-\Gamma\eta_\mu = 1 + \mathsf{DA}/e^2 > 0$ analytically, from the strict downcrossing of q. So κ is G times a product of three strictly positive factors, δ_2, $1-\Gamma\eta_\mu$, and $1/(1+\Gamma\sigma)$, of which the first is strictly decreasing and the other two are decreasing; therefore $\kappa' < 0$: higher steady-state inequality flattens the Phillips curve.

Step 3: part (ii), the output contraction deepens. By Proposition 3, $|\Psi_y| = \Theta\Psi_\pi = \Theta/\big[(1-\beta\rho_u) + \kappa\Theta\big] = \big[(1-\beta\rho_u)/\Theta + \kappa\big]^{-1}$. By part (i), Θ is strictly increasing and κ strictly decreasing, so both terms in the bracket fall, the bracket is strictly decreasing, and $|\Psi_y|$ is strictly increasing. Only (D.24) is used.

Step 4: part (iii), the markup response rises; the strategy is to write the response as a fixed positive numerator over a denominator $N(\bar{H})$ and show that N falls. Combining (D.21) with the distributional relation (11) gives $\hat{\mu}_t = -\sigma\hat{h}_t - G\tilde{y}_t = -\big[G - \sigma\chi_y\big]\tilde{y}_t$, and Proposition 3 gives $\tilde{y}_t = -\Theta\Psi_\pi u_t$, which yields the response displayed in the statement. The second equality in that display is the exact identity

$$G - \sigma\chi_y = G - \frac{\sigma\Gamma G}{1+\Gamma\sigma} = \frac{G\big[(1+\Gamma\sigma) - \Gamma\sigma\big]}{1+\Gamma\sigma} = \frac{G}{1+\Gamma\sigma},$$

so the $\Gamma\sigma$ terms cancel exactly, as they do in (17). Substituting Θ and Ψ_π and clearing $1+\Gamma\sigma$ from numerator and denominator, using $(1+\Gamma\sigma)\kappa = \delta_2 G(1-\Gamma\eta_\mu)$ and $(1+\Gamma\sigma)\sigma(1-\chi_y) = \sigma(1-\Gamma b)$, the denominator collects all the $\bar{H}$ dependence:

$$\frac{\partial\hat{\mu}_t}{\partial u_t} = \frac{G(\phi_\pi - \rho_u)}{N(\bar{H})}, \qquad N = (1-\beta\rho_u)\big[\sigma(1-\rho_u)(1-\Gamma b) + \phi_y(1+\Gamma\sigma)\big] + (\phi_\pi - \rho_u)G\delta_2(1-\Gamma\eta_\mu) > 0$$

with $N > 0$ because $D_\pi > 0$, $\tilde{\sigma}(1-\rho_u) + \phi_y > 0$, and $1+\Gamma\sigma > 0$, and with $\phi_\pi - \rho_u > 0$ under the maintained restrictions $\phi_\pi > 1 > \rho_u$. It remains to show $N' < 0$. Differentiating in $\bar{H}$, and noting that b, σ, ϕ_y, ϕ_π, ρ_u, and β are fixed along the branch,

$$N' = (1-\beta\rho_u)\,\sigma\,\Gamma'\big[\phi_y - (1-\rho_u)b\big] + (\phi_\pi - \rho_u)G\big[\delta_2'(1-\Gamma\eta_\mu) - \delta_2(\Gamma\eta_\mu)'\big].$$

The second bracket is the supply side, and under (D.24) it is strictly negative: $\delta_2' < 0$ and $1-\Gamma\eta_\mu > 0$ by Step 2, while $\Gamma \geq 0$ and $\eta_\mu > 0$ are both increasing with at least one

strictly so, hence $(\Gamma\eta_\mu)' > 0$ and $\delta_2 > 0$. The first term is the demand side, and (D.25) is exactly what signs it: it gives $\phi_y - (1-\rho_u)b \leq 0$, and $\Gamma' > 0$ by Step 1, so the term is non-positive. Therefore $N' < 0$ and $G(\phi_\pi - \rho_u)/N$ is strictly increasing in $\bar{H}$.

Step 5: part (iv) is a numerical statement, not a theorem. By Proposition 3, pass-through is $\Psi_\pi = \left[(1-\beta\rho_u) + \kappa\Theta\right]^{-1}$, so it depends on $\bar{H}$ only through the product $\kappa\Theta$, and along the branch the two factors move in opposite directions. Part (i) gives $\mathrm{d}\kappa/\mathrm{d}\bar{H} < 0$: the Phillips curve flattens, each unit of output contraction buys less disinflation, and this force alone would raise Ψ_π through $\partial\Psi_\pi/\partial\kappa < 0$. But part (i) also gives $\mathrm{d}\tilde{\sigma}/\mathrm{d}\bar{H} < 0$, so Θ rises: the same policy rule generates a larger output contraction per unit of inflation, and this force alone would lower Ψ_π through $\partial\Psi_\pi/\partial\tilde{\sigma} > 0$. Conditions (D.24)–(D.25) sign the factors but not the product $\kappa\Theta$, so neither force dominates on paper; in particular, $\mathrm{d}\kappa/\mathrm{d}\bar{H} < 0$ does *not* by itself imply that pass-through rises. Part (iv) is accordingly verified numerically rather than proved: at the baseline calibration we evaluate the branch on a 201-point grid in $R \in [0,1]$ and find Ψ_π strictly increasing in $\bar{H}$ at every grid step, with the endpoint values reported in the statement. Parts (i)–(iii) are unaffected, since none of them uses the monotonicity of Ψ_π. □

Parts (i)–(iii) are comparative statics across steady states on the fiscal branch, conditional on (D.24)–(D.25); they are not a global ordering over all parameterizations. The inequality in Step 1 genuinely fails as $e(\bar{H}) \downarrow 1$, where the desired markup $e/(e-1)$ diverges, and (D.24) is exactly the requirement that the admissible branch stay clear of that boundary. The magnitudes in part (iv) are calibration-specific.

D.11 Monetary Policy Shock Transmission

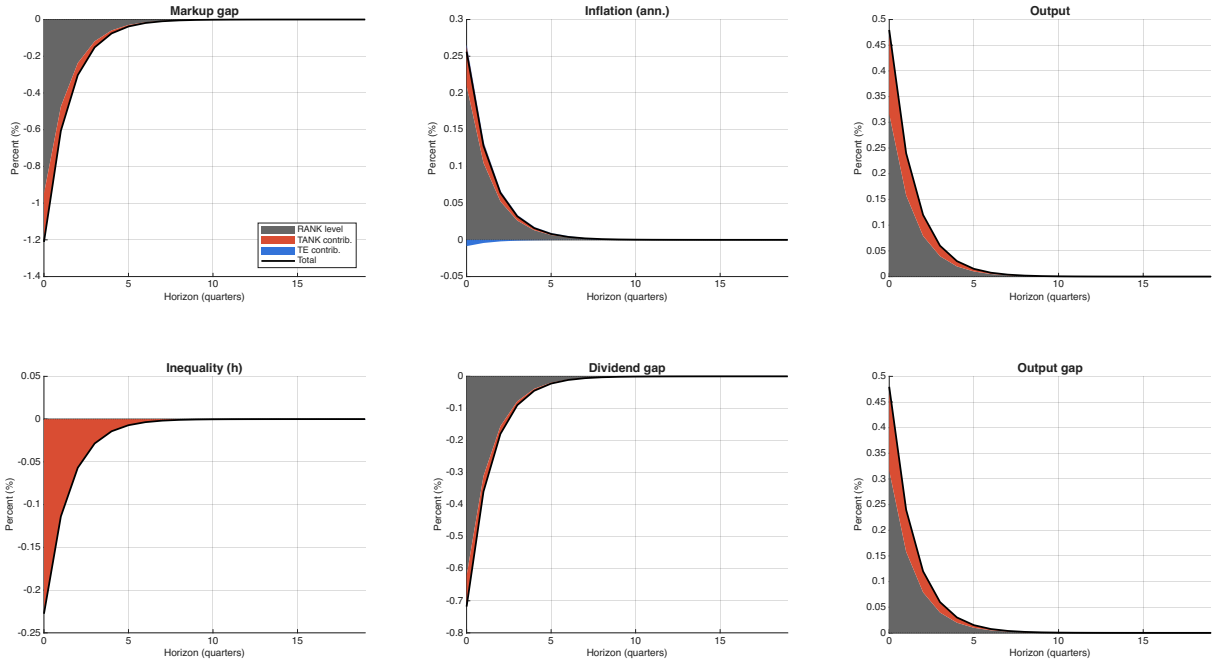
Section 3.2 reports the band decomposition for the technology and cost-push shocks. Figure D.3 applies the same decomposition to an expansionary monetary policy shock, and this subsection traces its transmission.[17] TE-TANK produces dynamics nearly identical to the markup-matched TANK($\varepsilon^* = 3.933$), so the blue elasticity-gap band is thin; both economies deviate substantially from the RANK corner, consistent with the literature documenting amplification from hand-to-mouth households.

The similarity between TE-TANK and TANK for demand shocks reflects a property of the transmission mechanism. Following the interest rate cut, rich households raise consumption. Because prices are sticky, this raises labor demand and the real wage for all households, including the poor, who consume their additional income. Firms accommodate the additional demand by hiring more labor.

As output rises, marginal cost rises under decreasing returns, which lowers markups through (D.21). The fall in markups dominates the offsetting rise in out-

[17] The preference shock case, not shown, yields no additional insights.

FIGURE D.3: Monetary Policy Shock: Channel Decomposition



Notes: Impulse responses to a one-standard-deviation reduction in the nominal interest rate (expansionary shock). Bands, homogeneous corners, panel layout, and units are as in Section 3.2. The red band amplifies output and inflation relative to RANK through the high marginal propensity to consume of hand-to-mouth households; inequality is zero in RANK by construction, so its response is carried entirely by the red and blue bands, and it falls because the expansion raises labor income while dividends fall. The blue band is negative in the inflation panel, the elasticity-gap damping discussed in the text.

put, so dividends fall in (D.22) and transfers to both types decline. Poor households nevertheless gain, because their income is disproportionately labor income; firms must therefore produce still more, hiring more labor and pushing the real wage up further. This channel is absent in RANK, where the real effects of a demand shock are muted relative to TANK and TE-TANK.

The fall in the heterogeneity index leaves desired markups unchanged in TANK. In TE-TANK it lowers the desired markup, because the composition of demand shifts toward the more price-elastic poor households. Firms therefore cut prices by more in TE-TANK than in TANK, which damps inflation in TE-TANK relative to TANK. This is the negative blue band in the inflation panel of Figure D.3; the output gap rises by essentially the same amount in the two economies.

D.12 Fixed versus Consumption-Dependent Elasticities

The model treats the elasticity gap as a fixed preference difference: rich households always have $\varepsilon_R = 3.0$ and poor households always $\varepsilon_P = 6.6$. The micro-evidence we calibrate to (Auer et al., 2024) and the closest structural model (Mongey and Waugh, 2025) instead generate the gap from non-homothetic preferences, so that each type's

elasticity is a function of its own consumption, $\varepsilon_h = \varepsilon_h(C_h)$, and drifts over the business cycle. What matters for the comparisons in this paper is not how steeply elasticities move with consumption, but whether the two types' gradients differ. If the elasticity is linear in log consumption, fixing ε_h is exactly innocuous for every comparison in the paper. Only curvature of $\varepsilon(\cdot)$ can weaken the channel, and the calibrations below bound it.

Let $\eta_h \equiv \mathrm{d}\varepsilon_h / \mathrm{d}\ln C_h < 0$ be the within-type gradient, measured in elasticity points per log point of consumption (equivalently, per 100 percent), so that $\varepsilon_{h,t} = \bar{\varepsilon}_h + \eta_h \hat{c}_t^h + O(\hat{c}^2)$ around the steady state. The effective elasticity firms face is the expenditure-weighted average $\varepsilon_{\text{eff},t} = \sum_h s_h H_t^h \varepsilon_{h,t}$, and it now moves for two reasons,

$$\mathrm{d}\varepsilon_{\text{eff},t} = \underbrace{(\varepsilon_R - \varepsilon_P)(1-\lambda)\bar{H}\,\hat{h}_t}_{\text{composition}} + \underbrace{\omega_R \eta_R\, \hat{c}_t^R + \omega_P \eta_P\, \hat{c}_t^P}_{\text{within-type}}, \quad \omega_R = (1-\lambda)\bar{H}, \quad \omega_P = \lambda \bar{H}^P,$$

where ω_R and ω_P are the steady-state expenditure shares of the two types, $\bar{H}^P = [1 - (1-\lambda)\bar{H}]/\lambda$, and $\omega_R + \omega_P = 1$. The composition term is the paper's mechanism: expenditure shifts between rich and poor, the weights in the average tilt, and each ε_h stays fixed. The within-type term is what the fixed-elasticity assumption switches off. Substituting $\hat{c}_t^R = \hat{y}_t + \hat{h}_t$ and $\hat{c}_t^P = \hat{y}_t - (\omega_R/\omega_P)\hat{h}_t$, which follow from $C_t^R = H_t C_t$, the market-clearing restriction $\lambda H_t^P + (1-\lambda) H_t^R = 1$, and $C_t = Y_t$, gives

$$\mathrm{d}\varepsilon_{\text{eff},t} = (1-\lambda)\bar{H}\big[(\varepsilon_R - \varepsilon_P) + (\eta_R - \eta_P)\big]\hat{h}_t + \bar{\eta}\,\hat{y}_t, \qquad \bar{\eta} \equiv \omega_R \eta_R + \omega_P \eta_P.$$

The levels of the gradients collect entirely in the $\hat{y}_t$ term; only their difference reaches the $\hat{h}_t$ term.

Linearizing the firm first-order condition (D.12) in ε_h adds $-[(1-\overline{MC})/\zeta]\,\mathrm{d}\varepsilon_{\text{eff},t}$ to the Phillips curve, the same route through which the composition term already enters. The $\bar{\eta}\hat{y}_t$ part therefore adds a constant $\Delta\kappa_y \equiv -(1-\overline{MC})\bar{\eta}/\zeta > 0$ to κ_y and leaves κ_h untouched. Two consequences follow. The flattening of Section 2.5 is the drag $\chi_y \kappa_h$, which is unchanged, so only the drag expressed as a percentage of a now larger κ_y shrinks. And the markup-matched TANK(ε^*) inherits the same $\varepsilon(\cdot)$ and, having $\varepsilon^* = \varepsilon_{\text{eff}}(\bar{H})$, the same $1-\overline{MC}$ and ζ, so it picks up an identical $\Delta\kappa_y$ and the dispersion-pure difference $\kappa^{\text{TE}} - \kappa^{\text{TANK}^*} = -\chi_y \delta_2 \eta_\mu$ is unaffected. How steeply elasticities move with consumption is therefore not the relevant statistic.

The composition loading keeps its sign if and only if $\eta_R - \eta_P < \varepsilon_P - \varepsilon_R = 3.6$ points

per log point. Writing this as a curvature index gives the exact scaling,

$$\gamma \equiv \frac{\eta_R - \eta_P}{\varepsilon_P - \varepsilon_R} \geq 0, \qquad \frac{\mathrm{d}\varepsilon_{\text{eff}}}{\mathrm{d}\hat{h}} = (1-\lambda)\bar{H}(\varepsilon_R - \varepsilon_P)(1-\gamma),$$
$$\kappa^{\text{TE}} = \kappa^{\text{TANK}^*}\left[1 - \Gamma\eta_\mu(1-\gamma)\right].$$

Here $\gamma = 0$ exactly when the two types share a gradient, that is, when ε is linear in log consumption. The desired-markup route of κ_h scales by exactly $1-\gamma$, while the marginal-cost route $\delta_2\sigma$ is untouched. The dispersion-pure slope gap $\Gamma\eta_\mu = 4.34$ percent therefore becomes $4.34(1-\gamma)$ percent.

Three benchmarks bracket γ. The first is ε linear in log consumption, the form implicit in summarizing Auer et al. (2024) by a single cross-sectional slope of about -2.0 points per log point ($3.6/\ln 6$); then $\gamma = 0$ and nothing in the paper changes. The second is a constant proportional gradient, $\varepsilon = AC^{-\gamma}$, calibrated to the same cross-section, that is, to an elasticity ratio of 2.2 across a factor-of-six range of permanent income; then $\gamma = \ln 2.2/\ln 6 = 0.44$, the desired-markup route retains 56 percent of its baseline strength, the slope gap falls from 4.34 to 2.43 percent, and the dispersion-pure pass-through gap falls from $+3.1$ percent to roughly $+1.7$ percent—smaller, same sign. The third forces the same power law through the model's own steady-state consumption ratio $\bar{C}^R/\bar{C}^P = \bar{H}/\bar{H}^P = 1.91$; then $\gamma = \ln 2.2/\ln 1.91 = 1.22 > 1$ and the loading reverses sign. This third case is the adverse bound, and it overstates curvature: it requires one structural $\varepsilon(\cdot)$ to deliver the full endpoint gap of 3.6 points across a consumption range far narrower than the income range over which that gap is estimated. The steady-state calibration is unaffected in all three cases, since $\bar{\varepsilon}_R$ and $\bar{\varepsilon}_P$ stay at the baseline quantification of Auer et al. (2024).

The $\hat{y}_t$ term cancels from the comparisons but not from the natural allocation. Appendix D.7 obtains $\hat{\mu}_t^n = 0$ because $H_t^n = \bar{H}$; with $\bar{\eta} \neq 0$ the effective elasticity moves with natural output even at constant H, so

$$\hat{\mu}_t^n = -\frac{\bar{\eta}}{\varepsilon_{\text{eff}}(\bar{H})\left[\varepsilon_{\text{eff}}(\bar{H}) - 1\right]}\,\hat{y}_t^n,$$

and disturbances that move natural output enter the gap-form Phillips curve as an endogenous cost-push term. Fixed elasticities therefore do buy divine coincidence; the term vanishes as $\bar{\eta} \to 0$ and is orthogonal to the dispersion-pure comparisons above.